The Inflation Expectations of Savers and Hand-to-Mouth Households

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Abstract

Whose inflation expectations drive aggregate demand? Using the Survey of Consumer Expectations, I show that unconstrained households (Savers), whose expectations enter intertemporal consumption decisions, hold lower, more accurate, less volatile, and less dispersed inflation expectations than Hand-to-Mouth (HtM) households. Estimating a noisy-information model separately for the two groups, I find that during the 2021–2023 inflation surge all households underreact to inflation news and over-perceive the persistence of inflation. Compared with HtM households, Savers underreact more to news but over-perceive inflation persistence less. Introducing the estimated expectation parameters into a tractable heterogeneous-agent New Keynesian (THANK) model, I show that the expectations of Savers make the cumulative output contraction after a cost-push shock 45% larger than under full-information rational expectations. With the expectations of all households instead, the contraction is 98% larger than under full-information rational expectations. This result suggests that the choice of survey expectations used to discipline macroeconomic models matters for understanding the propagation of inflationary shocks.

Keywords: inflation expectations, household heterogeneity, hand-to-mouth, noisy information, perceived persistence, underreaction, tractable HANK.

JEL-Codes: D84, E21, E31, E52, E71.

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1. Introduction

Inflation expectations are central to understanding macroeconomic dynamics. They affect perceived real interest rates, and therefore intertemporal consumption choices and the transmission of monetary policy. Survey data show that households disagree substantially about future inflation, make persistent forecast errors, and differ in how strongly their expectations respond to inflation news (Coibion and Gorodnichenko, 2012, 2015; D'Acunto et al., 2022, 2021; Weber et al., 2022). At the same time, the heterogeneous-agent literature suggests that not all households' inflation expectations play the same role. Inflation expectations matter directly for households that are unconstrained and able to adjust consumption intertemporally, because their consumption-saving decision is affected by the perceived real rate. By contrast, for constrained households, this channel is muted, as their consumption is tied to current resources (Kaplan et al., 2018; Kaplan and Violante, 2014).

A natural question is therefore whether inflation expectations differ systematically between unconstrained households, henceforth *Savers*, and constrained households, henceforth *Hand-to-Mouth* (HtM). This distinction matters because, if the two groups form expectations differently, the inflation expectations of Savers, which enter the intertemporal consumption-saving decision, may differ from a simple average expectation observed in the population. In that case, aggregate survey expectations may not be the best proxy for the expectation that matters for forward-looking aggregate demand.

In this paper, I show that the distinction between the expectations of Savers and HtM households matters quantitatively. In a heterogeneous-agent model with subjective expectations, using the expectations of all households instead of those of Savers alone makes the expectations channel of a cost-push shock about twice as strong.

In the first part of the paper, I decompose household inflation expectations along the Saver-HtM margin in the Survey of Consumer Expectations (SCE), and document four stylized empirical facts. First, Savers report lower inflation expectations than HtM households. Second, Savers display smaller forecast errors and smaller absolute forecast errors. Third, Savers' expectations are less volatile. Fourth, Savers' expectations are less dispersed in the cross section. These differences are persistent over the sample period, August 2015 to August 2025, including the recent inflation surge, and remain after controlling for demographic and socioeconomic characteristics. The facts therefore suggest that the Saver-HtM margin is informative not only about intertemporal consumption smoothing, but also about how households forecast inflation. The classification of households in the SCE as Savers or HtM relies on their reported ability to meet an unexpected expense. The resulting classification is consistent with the existing literature on HtM households: it assigns approximately 40% of households to the HtM group and reproduces a distribution of HtM households across wealth that is close to existing estimates for the US population (Aguiar et al., 2020). This makes the SCE particularly appealing for the purpose of this paper, because it combines information on households' constrained status with information on their inflation expectations.

To rationalize the four empirical facts, I propose a noisy-information model of household inflation expectations. In the model, households form forecasts from a perceived inflation

signal, which can differ from the actual signal through three parameters: the perceived bias in the inflation signal, the perceived persistence of inflation, and the perceived signal-noise variance. Estimating the model separately for Savers and HtM households shows that expectation formation differs across these two groups. During the low-inflation period, 2015–2020, neither Savers nor HtM households react systematically to current inflation news, so differences in expectations are interpreted as differences in inflation priors. During the inflation surge, 2021–2023, instead, both household groups update their expectations in response to inflation news, but display underreaction: their Kalman gains are below one. In particular, underreaction is more pronounced among Savers than among HtM households.

The estimates also reveal how persistent households perceive inflation to be. During the surge, Savers perceive a persistence of 0.84 at quarterly frequency and HtM households of 1.02, while the actual persistence of headline CPI inflation is about 0.5. All households over-perceive the persistence of inflation, and HtM households do so the most. Since HtM households make up about 40% of the population, expectation parameters estimated on all households therefore overstate the persistence perceived by unconstrained households.

In the second part of the paper, I ask whether these differences in expectations matter for aggregate demand. I derive the consumption functions of Savers and HtM households in the tractable heterogeneous-agent New Keynesian (THANK) model of Bilbiie (2025) under subjective expectations at all horizons. HtM households consume their current income, so that their inflation expectations do not enter the intertemporal margin of aggregate demand. The consumption of Savers, on the other hand, depends on the path of real interest rates they expect. The inflation expectations of Savers therefore enter aggregate demand directly, and their effect depends strongly on perceived inflation persistence.

I then introduce the estimated expectation parameters into THANK and study a cost-push shock. With the expectation parameters of Savers, the cumulative output contraction is 45% larger than under full-information rational expectations (FIRE). With those of all households instead, which a model disciplined by population-wide survey data would use, the contraction is almost twice as large as under FIRE, and the expectations channel is 2.2 times as strong as with those of Savers alone. The amplification comes from the over-perception of persistence, not from underreaction, which on its own dampens the effect of the shock. Since Savers expect the central bank to respond to inflation, the more persistent they perceive inflation to be, the higher the path of real rates they expect, and the lower their demand.

Related literature. This paper contributes to the empirical literature on household inflation expectations in surveys. This literature documents systematic deviations from FIRE and substantial heterogeneity in beliefs across households (Coibion and Gorodnichenko, 2012, 2015; Vellekoop and Wiederholt, 2019; Weber et al., 2022). A large part of this work relates heterogeneity in inflation expectations to demographic and socioeconomic characteristics, such as income, gender, education, cognitive ability, or personal inflation experiences (D'Acunto et al., 2022, 2021; Malmendier and Nagel, 2016). More recent work emphasizes that households' attention to inflation, and therefore the responsiveness of expectations to inflation news, also varies with the inflation environment: households are often poorly informed about inflation

in normal times, but attention rises when inflation becomes salient (Bracha and Tang, 2025; Korenok et al., 2022; Link et al., 2026; Pfäuti, 2024; Weber et al., 2024). This paper shows that the Saver-HtM distinction yields sizeable differences not only in the level and dispersion of inflation expectations, but also in how expectations react to inflation news. This response is muted in the low-inflation period, and during the inflation surge it becomes positive but remains weak, especially for Savers.

The paper also relates to the literature on underreaction and overreaction in macroeconomic expectations. Consensus forecasts underreact to news (Coibion and Gorodnichenko, 2015; Coibion et al., 2018), while individual forecasts often overreact (Bordalo et al., 2020; Kohlhas and Walther, 2021), and forecasters tend to perceive economic variables as more persistent than they are (Afrouzi et al., 2023; Fuster et al., 2010). Angeletos et al. (2021) show that both patterns can coexist in the same expectations. In the estimated model of this paper, households underreact to inflation news but over-perceive the persistence of inflation. In general equilibrium the two forces work in opposite directions, and the amplification of a cost-push shock comes from over-perceived persistence.

Finally, this paper speaks to the heterogeneous-agent literature that incorporates departures from FIRE into models with household heterogeneity (Broer et al., 2021; Pfäuti and Seyrich, 2026). This literature studies how expectation frictions interact with wealth, liquidity, and MPC heterogeneity. For the purposes of connecting these models to survey data, an important question is which measured expectations should be mapped into which household decision. This paper provides direct survey evidence on this question by estimating expectation-formation parameters separately for Savers and HtM households. It then derives consumption functions under subjective expectations at all horizons, in the spirit of Preston (2005), in the THANK model of Bilbiie (2025), and feeds them with the measured beliefs, so that the expectations of each group enter only the margin they affect. Approaches that introduce expectation frictions through a single behavioral parameter, such as cognitive discounting in the Euler equation (Gabaix, 2020) or level-k thinking (Farhi and Werning, 2019), determine how expectations at different horizons enter demand through that parameter. Here, instead, the consumption of Savers depends on the expected path of real interest rates at all horizons, so that the beliefs estimated from survey data determine directly how expectations at each horizon affect aggregate demand.

Outline. The rest of the paper is structured as follows. Section 2 describes the SCE data, the Saver and HtM classification, and the four empirical facts. Section 3 presents the noisy-information model and estimates it separately for Savers and HtM households. Section 4 introduces subjective expectations into THANK. Section 5 quantifies whose expectations matter for a cost-push shock. Section 6 concludes.

2. Four Facts about Inflation Expectations of Households

This section documents systematic differences between the inflation expectations of Saver and HtM households. The empirical analysis relies on microdata from the Survey of Consumer

Expectations (henceforth, SCE) of the Federal Reserve Bank of New York (FRBNY, 2026).[1] The SCE elicits households' expectations about inflation over the next twelve months in its main module, while complementary modules record household characteristics that can be used to classify respondents as constrained or unconstrained.

In this study, I refer to constrained households as *Hand-to-Mouth* (HtM) households and to unconstrained households as *Savers*. This terminology is adopted to map the empirical classification directly into the distinction between constrained and unconstrained households that is standard in the heterogeneous-agent literature.

2.1. The SCE Microdata

Inflation Expectations in the SCE Households' inflation expectations are elicited in the main module of the SCE. That module circulates on a monthly basis, in a rotating panel of approximately 1,300 household heads every month, each of whom stays in the panel for up to twelve consecutive months. The relevant question has two parts, the first one asks:

> *Over the next 12 months, do you think that there will be inflation or deflation?*

Given the answer to this question, the second part asks:

> *What do you expect the rate of [inflation/deflation] to be over the next 12 months?*

From these answers, I follow the convention described in Armantier et al. (2017) and compute, in each cross section, the linearly interpolated median based on point predictions across all respondents, treated as grouped data and weighted with the survey weights.[2] I refer to the resulting number as the average inflation expectation in the cross section. Figure 6 in the Appendix reports the average inflation expectation over the sample period, together with the series published by the Federal Reserve Bank of New York and realized inflation.

The period covered in this analysis spans from August 2015 to August 2025, covering both the low-inflation environment before the recent inflation surge and the high-inflation period that followed. The start of the sample is determined by the assignment rule described below, which allows to classify households as Savers or HtM from August 2015 onwards. The end of the sample, instead, is determined by the need to observe realized CPI inflation. As the next sections will describe, the same SCE responses are used to compute average forecast errors, average absolute forecast errors, the volatility of average inflation expectations, and their cross-sectional dispersion.

Identifying Savers and HtM Households The empirical classification of HtM and Saver households relies on the following question, which is asked in the complementary "Credit Access" module of the SCE from October 2015 onwards:

[1] For a thorough description of the Survey of Consumer Expectations, see Armantier et al. (2017).

[2] This procedure is needed because almost all answers to this question are integers, so that simply taking the cross-sectional median would result in an extremely stable time series. The resulting series tracks the median point prediction published by the Federal Reserve Bank of New York closely, with a mean absolute difference of about 10 basis points. Answers below −100%, which are logically impossible, are excluded, and no other answers are trimmed.

What do you think is the percent chance that you could come up with $2,000 if an unexpected need arose within the next month?

Accordingly, I classify a responding household as *Hand-to-Mouth* if the reported probability is below 80%. Conversely, households at or above this threshold are classified as *Savers*. The definition is therefore directly linked to households' perceived distance from their liquidity constraint, and identifies households for whom an unexpected expense is likely to be binding. The Credit Access module is fielded every four months, in February, June and October, to the respondents of the main module in those months. In the remaining months, each household is assigned the answer given in the nearest wave of the module.[3]

In the literature, HtM households are usually identified according to their liquid balance over a pay period (Kaplan and Violante, 2014; Kaplan et al., 2014). This identification is only partially possible in the SCE, where a proxy for households' liquid balances is publicly available only from 2015 to 2019, in the annual "Household Finance" module. The threshold of 80% is chosen so that, for the households that answer both the credit and the finance SCE modules, the two identification methods align as closely as possible: the agreement between the two classifications is highest for thresholds between 70% and 85% where, as Figure 7 in the Appendix shows, the two classifications also deliver similar distributions of households across wealth quintiles. In addition, while the exact threshold value of 80% affects the share of households classified as HtM, the empirical facts documented below hold for any threshold between 75% and 99%, as shown in the robustness exercises.

The summary statistics on liquid wealth and net worth in the SCE are reported in Table 8 in the Appendix, and they are comparable to those from other surveys, such as those reported in Kaplan and Violante (2014). In particular, using the identification described above, I find that the share of HtM households is approximately 42%, a number that is consistent with other estimates for the US population (Aguiar et al., 2020; Kaplan and Violante, 2014). Importantly, HtM households are not only located at the lower end of the wealth distribution, and in fact a significant fraction of HtM households are *wealthy* HtM. For example, Figure 7 shows that in the middle quintile of the wealth distribution, approximately 39% of households are classified as HtM. Under this identification, these households are not certain that they could meet an unexpected expense of $\$2,000$ within the next month, even though this amount represents only a small fraction of their net worth.[4] Hence, despite holding a sizeable wealth stock, their perceived ability to meet short-run liquidity needs is limited.

Thus, the Saver-HtM classification adopted here has a natural interpretation through the lens of a standard heterogeneous-agent model. It captures perceived ability to meet short-run liquidity needs, which is directly related to households' ability to smooth consumption intertemporally at the time of observation.

[3]Ties are resolved in favor of the earlier wave. The June 2022 wave of the module has not been released, and around it the window is therefore extended to four months.

[4]Using data from the 2019 Survey of Consumer Finances, Bhutta et al. (2020) find that the median net worth of US families is $\$121,700$ in 2019 dollars, or about $\$123,000$ in 2020 dollars. In the Household Finance module of the SCE, which covers the years 2015 to 2019, I find that median net worth is approximately $\$107,000$ in 2020 dollars (see Table 8).

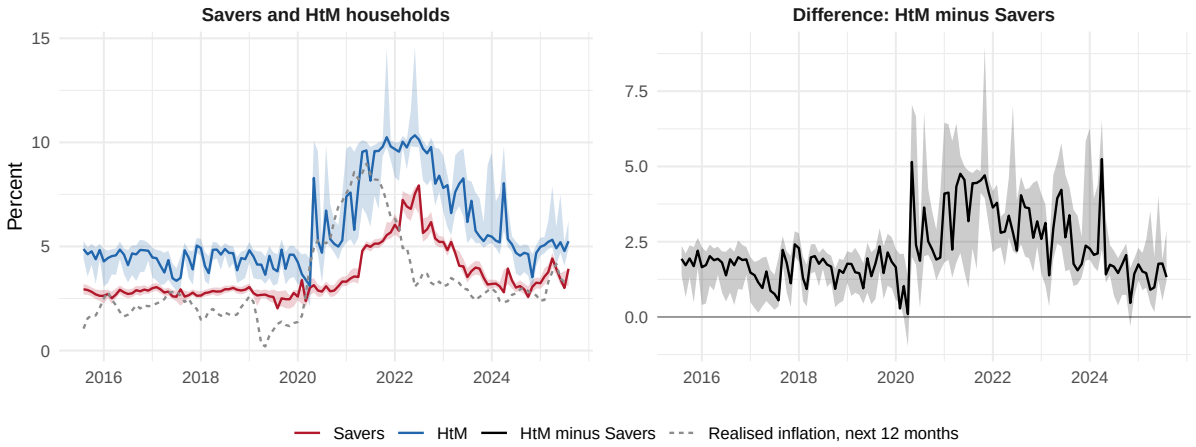


Figure 1: Inflation expectations of Savers and HtM households.

Notes: In the left panel, the Savers' expectations are in red, HtM expectations are in blue, and realized inflation over the following twelve months is dashed in grey. The right panel shows the difference between HtM households and Savers. The bands represent the 90% confidence interval.

2.2. The Four Facts

The facts are organized around four aspects of household inflation expectations: the *level* of average expectations, their *forecast errors*, their *volatility*, and their cross-sectional *dispersion*. In each case, HtM households and Savers display systematic differences.

Each figure below shows these differences by reporting the corresponding time series separately for the two groups in the left panel, and the difference between them in the right panel, both over the full sample. For every figure, the estimates are computed at a monthly frequency and using the SCE weights. Confidence intervals are obtained from a bootstrap that resamples respondents, together with all their answers.

Fact 1: Savers Have Lower Inflation Expectations than HtM Households The first fact is that Savers tend to have lower inflation expectations than HtM households. Figure 1 plots one-year-ahead average inflation expectations, separately for the two groups.

HtM inflation expectations are higher than those of Savers in every month of the sample, and the difference is statistically significant in almost all of them. Before the inflation surge, the gap averages 170 basis points, and it falls below 50 basis points only in two months in early 2020. During the surge, the gap widens to 330 basis points on average, reaching almost 480 basis points at its peak in 2021. After the surge, it returns to about 180 basis points. That inflation expectations tend, overall, to be higher than realized inflation is well established in the literature: Weber et al. (2022) refer to it as the "systematic upward bias" of inflation expectations. The decomposition between Savers and HtM households shows that this upward bias is stronger among HtM households.

Fact 2: Savers Have Smaller Forecast Errors The second fact is directly related to the first and concerns forecast errors. I define the individual forecast error as the difference between household i's one-year-ahead inflation expectation and realized inflation over the corresponding

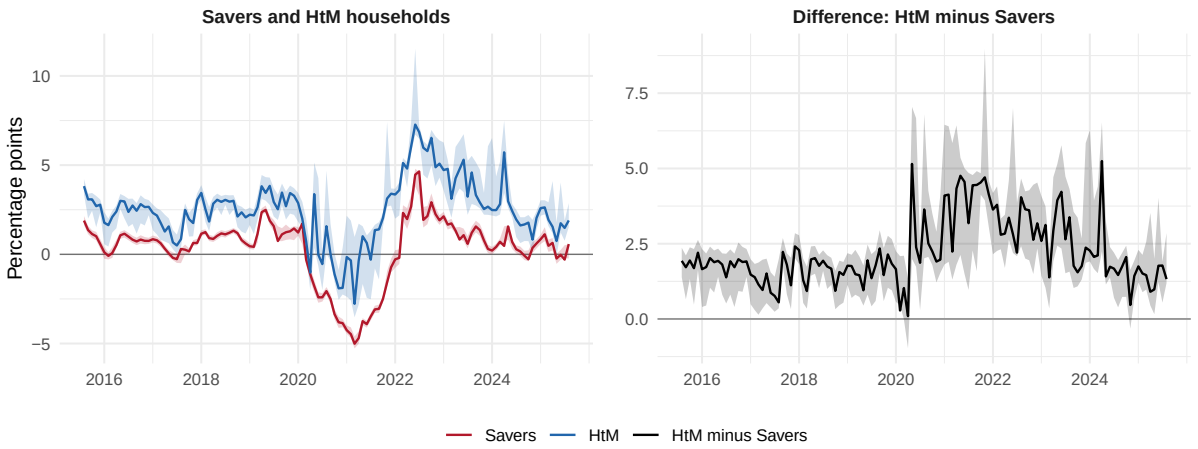


Figure 2: Inflation expectations forecast errors of Savers and HtM households.

Notes: In the left panel, Saver forecast errors are in red and HtM forecast errors are in blue. The right panel shows the difference between HtM households and Savers. The bands represent the 90% confidence interval.

horizon:

$$\mathbb{E}_t^i [\Pi_{t+12}] - \Pi_{t+12} \ .$$

Realized inflation is measured as the growth rate of the seasonally adjusted Consumer Price Index over the twelve months that follow the survey month, using data from the U.S. Bureau of Labor Statistics (2026). After constructing individual forecast errors, I compute group-level forecast-error time series for HtM households and Savers separately, following the same aggregation procedure used to construct average inflation expectations.

Figure 2 shows that, in line with the first fact, Savers consistently display smaller forecast errors than HtM households. On average over the sample, the forecast error of Savers is close to zero, at about 30 basis points, while HtM households overexpect inflation by about 250 basis points. At the beginning of the inflation surge, forecast errors turn negative, as realized inflation rises above expectations: for Savers from the spring of 2020 until early 2022, for HtM households only in some months between the spring of 2020 and mid-2021. Once inflation peaks, both groups' forecast errors become positive again.

Because positive and negative forecast errors can offset each other, I also consider *absolute* forecast errors. The corresponding time series are presented in Figure 8 in the Appendix and show that the persistent difference in (signed) forecast errors translates into a difference in absolute forecast errors of similar magnitude: about 290 basis points over the full sample, 285 during the surge, and as much as 470 after the surge. The gap narrows only temporarily, to about 50 basis points in October and November 2020, when realized inflation rises above the expectations of both groups, and remains statistically significant in every month. The difference in absolute forecast errors hence indicates that HtM forecasts tend to be further away from realized inflation.

Fact 3: Savers' Expectations Are Less Volatile The third fact is that Savers' inflation expectations are less volatile than those of HtM households. In the sample, the unconditional variance

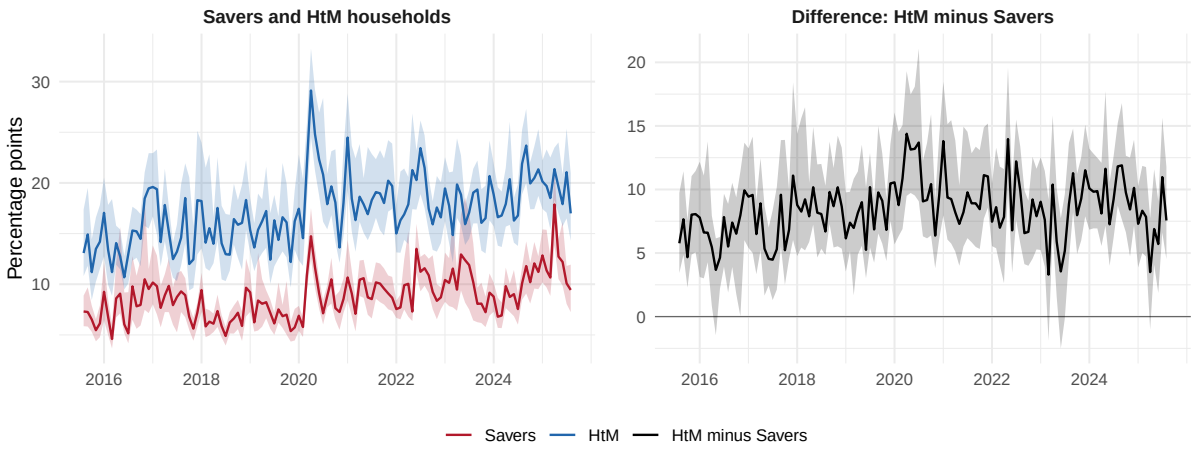


Figure 3: Dispersion of inflation expectations for Savers and HtM households.

Notes: In the left panel, Saver dispersion is in red and HtM dispersion is in blue. The right panel shows the difference between HtM households and Savers. Dispersion is the root-mean-square deviation from the group mean, after winsorizing expectations at the 1st and 99th percentiles of each month. The bands represent the 90% confidence interval.

of the Saver inflation-expectation time series is only about one third of the corresponding variance for HtM households: the ratio of the two variances is 0.37, with a 90% confidence interval between 0.27 and 0.41 (see Table 1). The ratio is lowest before the surge (0.08), when Savers' average expectations barely moved, and highest during the surge (0.62), when the expectations of both groups moved with realized inflation. Thus, the difference between the two groups is not only visible in the level of expectations and in corresponding forecast errors: Savers' expectations also fluctuate less over time.

Figure 9 in the Appendix illustrates this pattern further, by considering the average revision of inflation expectations by group. These are computed here as the absolute change, for households that answer the survey in two consecutive months, of the inflation forecast, subject to the same aggregation procedure used for the other facts. The average revision of Savers' inflation expectations is consistetly smaller than that of HtM households, and the difference is statistically significant in every month.

Overall, the third fact suggests that the average expectation of unconstrained Savers is more stable, without taking a stand on whether this reflects different information, different updating, or both.

Fact 4: Savers' Expectations Are Less Dispersed Finally, the fourth fact is that Savers' inflation expectations are less dispersed in the cross section than those of HtM households. For each group $j \in \{\text{Savers}, \text{HtM}\}$, dispersion in period t is defined as

$$\sqrt{\frac{1}{\sum_{i \in j} w_{i,t}} \sum_{i \in j} w_{i,t} \left(\mathbb{E}_t^i \left[\Pi_{t+12} \right] - \bar{\mathbb{E}}_t^j \left[\Pi_{t+12} \right] \right)^2},$$

where $w_{i,t}$ is the survey weight of respondent i at date t, $\mathbb{E}_t^i$ is household i's one-year-ahead inflation expectation, and $\bar{\mathbb{E}}_t^j \left[\Pi_{t+12} \right]$ is the weighted average expectation in the group j to

Table 1: Four facts: Savers and HtM households by period.

	Full sample 2015–2025	Pre-surge 2015–2020	Surge 2021–2023	Post-surge 2024–2025
Fact 1: average inflation expectation				
Savers	3.56	2.80	5.05	3.34
HtM	5.76	4.52	8.38	5.10
Difference, HtM − Savers	2.21	1.73	3.33	1.76
	[2.02, 2.42]	[1.52, 1.94]	[2.84, 3.78]	[1.45, 2.34]
Fact 2a: average forecast error				
Savers	0.29	0.40	0.02	0.41
HtM	2.50	2.12	3.35	2.17
Difference, HtM − Savers	2.21	1.73	3.33	1.76
	[2.02, 2.42]	[1.52, 1.94]	[2.84, 3.78]	[1.45, 2.34]
Fact 2b: average absolute forecast error				
Savers	2.37	1.75	3.62	2.17
HtM	5.26	4.10	6.47	6.84
Difference, HtM − Savers	2.88	2.35	2.85	4.68
	[2.77, 3.17]	[2.23, 2.81]	[2.52, 3.30]	[4.11, 4.97]
Fact 3: variance of the average expectation over time				
Savers	1.51	0.05	1.57	0.21
HtM	4.11	0.56	2.53	0.67
Ratio, Savers / HtM	0.37	0.08	0.62	0.31
	[0.27, 0.41]	[0.03, 0.15]	[0.33, 0.76]	[0.06, 0.52]
Fact 4: dispersion (root-mean-square deviation)				
Savers	8.81	7.77	9.73	10.51
HtM	17.23	15.93	18.40	19.31
Difference, HtM − Savers	8.42	8.17	8.67	8.81
	[7.87, 9.27]	[7.33, 9.27]	[7.63, 10.05]	[7.44, 10.52]

Notes: Facts 1, 2 and 4: averages over the months of each period of the monthly statistics (percent and percentage points). Fact 3: variance over the months of each period of the monthly average expectation. In each month, the average expectation of a group is the weighted interpolated median described in Section 2.1, and similarly for forecast errors. Survey weights. The full sample covers August 2015 to August 2025; pre-surge: August 2015 to December 2020; surge: January 2021 to December 2023; post-surge: January 2024 to August 2025. In brackets, 90% confidence intervals from a cluster bootstrap that resamples respondents (500 replications). Dispersion is computed after winsorizing point forecasts at the 1st and 99th percentiles of each month.

which household i belongs. Because the root-mean-square deviation is sensitive to a few extreme answers, individual expectations are winsorized at the 1st and 99th percentiles of each month's cross section, with the same cutoffs for both groups.

Figure 3 plots the time series of dispersion for Savers and HtM households. The dispersion of expectations among Savers is about half of that among HtM households throughout the sample, and the ratio between the two is similar before, during and after the surge. It rises above three quarters only in a few months of 2023 and 2025. The same pattern holds without winsorization and when dispersion is measured by the interquartile range instead (see Figures 10 and 11 in the Appendix). This means that Savers' expectations are not only lower and more stable over time, but tend also to be more clustered within the group. By contrast, HtM households display greater disagreement in their inflation forecasts.

Summary and Robustness Taken together, these four facts show that inflation expectations differ systematically and persistently along the Saver-HtM margin. Savers' expectations tend to be lower, to have smaller forecast errors, to be less volatile over time, and to be less dispersed in the cross section. Table 1 summarizes the four facts over the full sample and over three

sub-periods: before the inflation surge (August 2015 to December 2020), during the surge (2021 to 2023), and after the surge (January 2024 to August 2025).

The facts do not depend on the details of the classification. Figure 12 in the Appendix repeats the analysis for thresholds between 75% and 99%, for a classification that uses only the months in which the Credit Access module is fielded, for a classification that assigns each household its most frequent status, and for the liquid-balance classification over 2015–2019. In all cases, the four differences between Savers and HtM households remain statistically significant.

2.3. Liquidity Margin or Other Household Characteristics?

HtM households differ from Savers along several dimensions that are known to be associated with inflation expectations, such as gender, income, cognitive abilities, and financial literacy (D'Acunto et al., 2022, 2021; Weber et al., 2022). In the SCE, HtM households make up between 62% and 70% of the households with low numeracy, without basic inflation literacy, with a household income below $\$50,000$, or with net worth below the median, compared with between 20% and 36% in the Savers groups.[5] This naturally raises the question of whether the four facts reflect the liquidity constraints of households or other characteristics that are correlated with it. I address this question in three ways.

First, I repeat the analysis of the four facts within groups of households that share a given characteristic. Figure 13 in the Appendix shows that the differences between HtM households and Savers survive within every group for the four facts: among women and men, among households with and without a college degree, with high and low numeracy, with and without inflation literacy, with low and high income, and with low and high wealth.[6] The differences between HtM and Savers tend to be larger among those household groups whose four facts of inflation expectations align with those of the HtM, these are referred to as the "high-expectations" groups (consisting of women, households without a college degree, with low numeracy, without inflation literacy, with an income below $\$50,000$, below median wealth, and non-homeowners). In particular, the Saver-HtM gap in average inflation expectations is between 250 and 290 basis points among women, households without a college degree, households with low numeracy or without inflation literacy, and households with an income below $\$50,000$. In the other groups, it ranges from 60 basis points among households with net worth above the median to 190 basis points among homeowners. In every group, the variance of the Savers' average expectation remains below that of HtM households, and so does the dispersion of expectations.

Second, I regress individual expectations' outcomes on a HtM dummy, survey wave fixed effects, and all household characteristics jointly. The outcomes are individual measures related

[5]Numeracy and income groups follow the categories published by the Federal Reserve Bank of New York. Inflation literacy is measured by the answer to a question on the real return of savings when inflation exceeds the interest rate. Net worth is observed only in the Household Finance waves of 2015 to 2019, and it is assigned to the other survey months of the same respondents. The corresponding comparisons concerning net worth therefore cover the survey months from August 2015 to February 2020.

[6]Net worth is only observed between 2015 and 2019, when answers to the Household Finance module of the SCE are public.

Table 2: HtM status and other household characteristics.

	Inflation expectation	Absolute forecast error	Absolute revision	Absolute deviation from group median
(1) HtM only	4.720***	6.702***	5.543***	6.504***
	(0.278)	(0.247)	(0.233)	(0.243)
(2) + age, gender	4.407***	6.073***	4.971***	5.870***
	(0.264)	(0.237)	(0.227)	(0.233)
(3) + education, numeracy, inflation literacy	3.015***	3.647***	2.704***	3.509***
	(0.225)	(0.209)	(0.197)	(0.205)
(4) + income, region	2.510***	3.089***	2.399***	2.988***
	(0.223)	(0.208)	(0.203)	(0.205)
(5) + homeownership	2.485***	2.970***	2.296***	2.871***
	(0.223)	(0.206)	(0.202)	(0.203)
Observations	119,092	119,092	107,481	119,092

Notes: HtM coefficient in weighted least-squares regressions of individual outcomes on an HtM indicator, household characteristics added step by step, and month fixed effects. Outcomes: one-year-ahead inflation expectation; its absolute difference from realized inflation over the following twelve months; its absolute change from the previous month; its absolute deviation from the month's median expectation of its Saver-HtM group. Expectations are winsorized at the 1st and 99th percentiles of each month. All specifications use the same sample of respondent-months with all characteristics observed (the absolute revision requires an answer in the previous month). Age groups: under 20, 20 to 40, 40 to 60, and over 60. Table 11 in the Appendix reports all coefficients of specification (5). Standard errors clustered by respondent and month in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

to the four facts: the inflation expectation, its absolute forecast error, its absolute revision from the previous month (which accounts for the stability of individual forecasts) and its absolute deviation from the month's group expectation. Table 2 adds the household characteristics step by step. It shows that the HtM coefficient is roughly halved once household characteristics are controlled for, mostly because of education, numeracy, inflation literacy, and income while age and gender reduce it only slightly. The coefficient on the HtM dummy remains large and highly statistically significant for all four facts: conditional on all characteristics, HtM households' expectations are about 250 basis points higher, 300 basis points further away from realized inflation, revised by 230 basis points more from one month to the next, and 290 basis points further away from the median of their group.[7] The Saver-HtM margin therefore carries information about inflation expectations beyond the household demographic characteristics that are usually considered in the literature.

Third, I systematically consider the distinction between *poor* and *wealthy* HtM households. I define wealthy HtM households as those who report owning their primary residence, and hence poor HtM as those who do not. Table 9 in the Appendix shows that the share of poor HtM is 20% of all SCE respondents, while the share of wealthy HtM is 23%, both are broadly consistent with the shares estimated by Aguiar et al. (2020). Wealthy HtM households also report a higher income, have a higher share of college graduates, have a higher numeracy, and are more likely to be inflation literate compared to poor HtM households. Remarkably, over the period from 2015 to 2019, in which households' wealth and liquidity are observed,

[7]The regression estimates refer to means rather than medians, as a result the reported differences are larger than those reported in Table 1.

the wealthy HtM display a substantially higher mean net worth than the poor HtM ($111,720 vs −$10,190), but a very low liquid wealth: on average, liquid wealth amounts to $94,690 for Savers, −$4,090 for the poor HtM, and $6,890 for the wealthy HtM.[8] Despite these striking differences, the four facts documented above hold for *both* poor and wealthy HtM households: Figures 14 to 17 in the Appendix show that Facts 1, 2 and 4 survive among both homeowners and non-homeowners, and that the gaps between HtM households and Savers have the same sign in both groups and have a comparable magnitude. Finally, Table 10 in the Appendix shows that Fact 3 is also robust to the distinction between poor and wealthy HtM households: the volatility of inflation expectations is higher for both poor and wealthy HtM households than for Savers with the same homeownership status.

Taken together, the evidence shows that the Saver-HtM distinction is informative about households' inflation expectations beyond demographics, income, and wealth. While it is certainly not the only margin along which inflation expectations differ across households, it accounts for a substantial part of these persistent differences.

The next section presents a noisy-information model. The purpose of the model is to provide a parsimonious framework for understanding which parameters can account for these differences in inflation expectations.

3. A Noisy-Information Model of Inflation Expectations

The model of inflation expectations presented here provides a correspondence between observed inflation forecasts in the SCE and expectation-formation parameters that differ between HtM households and Savers. In the model, households form inflation forecasts from a *perceived* inflation signal rather than from the actual signal. Households might differ from the actual data-generating process along three dimensions: the perceived persistence of the inflation process, the perceived bias in the signal, and the perceived variance of the signal.

Apart from this feature, the model builds on the standard noisy-signal literature as in, for example, Woodford (2003), Coibion and Gorodnichenko (2012), and Coibion and Gorodnichenko (2015), while the framework presented here is closely related to Vellekoop and Wiederholt (2019). The central mechanism of the model is that households form their forecast through the Kalman filter, and the steady-state Kalman gain determines how strongly a household revises its inflation forecast after observing a new inflation signal. A household with a high gain treats the signal as informative and moves expectations strongly, while a household with a low gain puts more weight on its prior belief and updates only weakly.

3.1. The Model

I assume that a given household is either Hand-to-Mouth or Saver. In what follows, this distinction is referred to as the household *state*, and the superscript $j \in \{\text{Saver, HtM}\}$ denotes a parameter that is specific to whether the household is either in the Saver or the Hand-to-Mouth state. The model states correspond directly to the Saver and HtM groups observed in the SCE.

[8] Liquid wealth means are computed excluding the top 1% of the distribution, to avoid contamination by large outliers. Table 9 in the Appendix also reports the medians that are in line with this ordering.

The perceived law of motion for inflation is an AR(1) at quarterly frequency:[9]

$$\pi_t = \rho^j \pi_{t-1} + \eta_t, \tag{1}$$

where π_t is inflation at time t, $\rho^j \in (0,1)$ is the inflation persistence perceived by a household in state j, and $\eta_t \sim$ i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$ is the innovation to inflation.[10]

The *actual* inflation signal reads:

$$s_t^A = \pi_t + \varepsilon_t^A \tag{2}$$

where the signal noise satisfies $\varepsilon_t^A \sim \mathcal{N}(0, \sigma_A^2)$ and is drawn independently across households. Similarly, the *perceived* inflation signal for a household in state j is given by:

$$s_t^j = \pi_t - \xi^j + \varepsilon_t^j \tag{3}$$

where ξ^j is interpreted as the bias perceived by a household in state j and $\varepsilon_t^j \sim \mathcal{N}(0, \sigma_j^2)$, with σ_j^2 capturing the variance in the signal perceived by a household in state j. Hence, the parameter ξ^j shifts the perceived signal, while σ_j^2 is inversely related to the household's confidence in the perceived signal. In fact, a household that perceives a positive ξ^j believes that the true level of inflation π_t is higher than the perceived signal s_t^j. Similarly, a household for which $\sigma_j^2 < \sigma_A^2$ perceives that the inflation signal is less noisy than it actually is, meaning that the household is "overconfident" about the signal.

The forecast of inflation at $t+1$ generated by a household in state j in period t is denoted $\hat{\pi}^j_{t+1|t}$ such that:[11]

$$\hat{\pi}^j_{t+1|t} \equiv \hat{E}\left[\pi_{t+1} \mid \mathcal{I}_t^j\right] \tag{4}$$

where the information set $\mathcal{I}_t^j$ consists of the history of perceived signals and the perceived persistence parameter of the law of motion of inflation, that is:

$$\mathcal{I}_t^j = \left(\rho^j, s_t^j, s_{t-1}^j, ..., s_1^j\right) \tag{5}$$

Similarly, the forecast of the perceived inflation signal is denoted $\hat{s}^j_{t|t-1}$ and reads:

$$\hat{s}^j_{t|t-1} \equiv \hat{E}\left[s_t^j \mid \mathcal{I}_{t-1}^j\right] \tag{6}$$

and it then follows from the observation equation of the perceived signal (3) that the ξ^j parameter captures the discrepancy between the forecast of inflation and the forecast of the

[9] Section 3.3 makes explicit that I formulate the model at quarterly frequency in order to deliver an estimation that is meaningful with the SCE data.

[10] Hence the "rational expectations" error of the inflation forecast is, for a given time period, some linear combination of the innovation η_t.

[11] The notation here is borrowed from Hamilton (1994).

signal. In fact, given that $\hat{E}\left[\varepsilon_t^j \mid \mathcal{I}_{t-1}^j\right] = 0$, one has:

$$\hat{s}_{t|t-1}^j = \hat{\pi}_{t|t-1}^j - \xi^j \ . \tag{7}$$

From the standard derivation of the Kalman filter,[12] it follows that the nowcast of inflation for a household in state j simplifies to:

$$\hat{\pi}_{t|t}^j = \hat{\pi}_{t|t-1}^j + \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(s_t^A - \hat{s}_{t|t-1}^j\right) \tag{8}$$

where the term $s_t^A - \hat{s}_{t|t-1}^j$ captures the difference between the actual signal realization in period t and the expected signal realization in period t. where $\mathbf{P}_{t|t-1}^j$ is the mean-square error of the forecast, such that:

$$\mathbf{P}_{t|t-1}^j \equiv \mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j\right)^2\right] \ . \tag{9}$$

Substituting yields

$$\hat{\pi}_{t|t}^j = \hat{\pi}_{t|t-1}^j + \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(\pi_t - \hat{\pi}_{t|t-1}^j + \xi^j + \varepsilon_t^A\right) \tag{10}$$

Finally, from the perceived law of motion of inflation (1), the forecast at time t of inflation at time $t+1$ for a household in state j is given by $\hat{\pi}_{t+1|t}^j = \rho^j \hat{\pi}_{t|t}^j$. Combining this expression with the updating equation yields the inflation forecast

$$\hat{\pi}_{t+1|t}^j = \rho^j \hat{\pi}_{t|t-1}^j + \rho^j \mathbf{K}_t^j\left(\pi_t - \hat{\pi}_{t|t-1}^j + \xi^j + \varepsilon_t^A\right). \tag{11}$$

Using standard notation, $\mathbf{K}_t^j \equiv \mathbf{P}_{t|t-1}^j / \left(\mathbf{P}_{t|t-1}^j + \sigma_j^2\right)$ is the Kalman gain, and summarizes how strongly households update their forecast after observing inflation news. In what follows, households form their conditional inflation expectations using the steady-state Kalman filter implied by their perceived law of motion for inflation and their perceived signal, so the gain is denoted $\mathbf{K}^j$. It should also be noted that under full-information rational expectations, the signal is observed without noise (so $\mathbf{K}^j = 1$), the perceived signal is unbiased ($\xi^j = 0$), and the perceived persistence ρ^j coincides with the actual persistence of the inflation process.

The Saver or HtM state of the household in period t therefore affects the forecast of inflation in period $t+1$ via three parameters: the perceived signal bias ξ^j, the perceived inflation persistence ρ^j, and the Kalman gain $\mathbf{K}^j$, which is decreasing in the perceived signal noise σ_j^2. The next subsection shows how combinations of these parameters can rationalize the four empirical facts.

[12] See Appendix B.1.

3.2. The Model's Parameters and the Empirical Facts

The ξ^j parameter and level differences. The ξ^j parameter is a systematic bias that enters the household's perceived inflation signal (3). From the inflation forecast equation (11), this parameter shifts inflation expectations without changing the responsiveness of expectations to incoming inflation news: holding ρ^j and $\mathbf{K}^j$ fixed, a higher ξ^j raises the household's inflation forecast.

Differences in ξ^j across Savers and HtM households can therefore rationalize level differences in inflation expectations. In particular, a larger perceived signal bias for HtM households would raise their average inflation expectations relative to those of Savers, consistent with Fact 1. The same mechanism also affects forecast errors. The forecast error here is simply:

$$\hat{\pi}^j_{t+1|t} - \pi_{t+1} = \rho^j \hat{\pi}^j_{t|t-1} + \rho^j \, \mathbf{K}^j \left(\pi_t - \hat{\pi}^j_{t|t-1} + \xi^j + \varepsilon^A_t \right) - \pi_{t+1} \, , \tag{12}$$

and hence when the forecast error is positive, a higher ξ^j increases the forecast error even more. When instead the forecast error is negative, a higher ξ^j moves expectations closer to realized inflation. Figure 19 in the Appendix shows this mechanism in a numerical simulation where ξ^j varies while other parameters are held constant.

The $\mathbf{K}^j$ parameter and responsiveness to inflation news. The steady-state Kalman gain $\mathbf{K}^j$ determines how strongly households update their inflation forecast after observing the perceived inflation signal. Holding ξ^j and ρ^j fixed, a higher $\mathbf{K}^j$ places more weight on the incoming signal and less weight on the prior forecast of current inflation $\hat{\pi}^j_{t|t-1}$. In other words, the gain summarizes the extent to which households react to inflation news.

As a result, $\mathbf{K}^j$ can rationalize differences in the volatility of expectations: if Savers have a lower gain than HtM households, their expectations react less strongly to inflation signals, and therefore display smaller movements over time, consistent with Fact 3. Figure 20 in the Appendix shows this mechanism in a numerical simulation where a higher perceived signal-noise variance σ^2_j (and hence a lower $\mathbf{K}^j$), delivers a less volatile time series of inflation forecasts.

The same mechanism can also speak to cross-sectional dispersion, corresponding to Fact 4. For that purpose, the distinction between actual and perceived signal noise is important. Cross-sectional dispersion is generated by different realizations of the idiosyncratic noise attached to the actual inflation signal, which enters through ε^A_t in the forecast (12). The Kalman gain, however, is determined by the perceived signal-noise variance, σ^2_j. Therefore, even if the actual signal realizations are drawn from the same distribution for Savers and HtM households, a higher perceived signal-noise variance among Savers implies a lower Kalman gain. That, in turn, dampens the transmission from idiosyncratic signal realizations to forecasts, and can therefore rationalize lower dispersion among Savers. Figure 21 in the Appendix shows this mechanism in a numerical simulation where a lower $\mathbf{K}^j$ delivers lower cross-sectional dispersion.

Finally, $\mathbf{K}^j$ also affects the level of expectations, and therefore their forecast errors. A higher gain moves expectations in the direction of the current perceived signal, but this does not

necessarily imply a smaller forecast error with respect to future realized inflation. If forecast errors are initially positive and the signal raises expected inflation, a higher gain increases the forecast error. If forecast errors are negative, the same higher gain can instead move expectations closer to realized inflation. Thus, differences in $\mathbf{K}^j$ can contribute to differences in forecast errors, but the sign of this effect depends on the sign of the forecast error and on whether the current perceived signal moves expectations toward or away from future realized inflation.

The ρ^j parameter and the persistence of expectations. The parameter ρ^j governs the persistence of inflation perceived by the household. Holding ξ^j and $\mathbf{K}^j$ fixed, the relation

$$\hat{\pi}^j_{t+1|t} = \rho^j \, \hat{\pi}^j_{t|t}$$

shows that a higher ρ^j makes a given nowcast of current inflation translate into a higher forecast of future inflation. The parameter therefore determines how strongly the updated nowcast is carried forward into future forecasts.

Differences in ρ^j can affect several empirical facts at once. A higher ρ^j raises the level of inflation expectations for a given nowcast, amplifies the effect of incoming inflation news on future forecasts, and increases the persistence of expectation dynamics. It can therefore contribute to differences in average expectations, forecast errors, volatility, and dispersion. These effects operate by scaling the nowcast that has already been updated through the perceived bias ξ^j and Kalman gain $\mathbf{K}^j$. The estimation below uses this distinction to separate differences in responsiveness to inflation news from differences in perceived inflation persistence, and thereby addresses the empirical question of which of these parameters accounts to which extent.

3.3. From Quarterly Inflation to One-Year-Ahead Expectations

Inflation expectations in the SCE are elicited at monthly frequency, and each respondent remains in the panel for up to twelve months. I formulate the model at quarterly frequency, as is common in the literature on noisy-information models of expectations, and exploit the panel dimension of the survey to observe the same household one model period apart. Each observation pairs the one-year-ahead inflation expectation of a respondent in survey month t with the expectation reported by the same respondent three months earlier, and with realized CPI inflation over the three months up to month t. All survey months are used, so that consecutive observations of the same respondent overlap. Accordingly, in the noisy-information model, the signal is observed at quarterly frequency.

Since the SCE elicits expectations about inflation over the next twelve months, a mapping between the model frequency and the survey horizon is necessary for the estimation to be meaningful. Throughout, π_t denotes quarter-on-quarter, non-annualized CPI inflation, so that the four quarterly rates sum to the twelve-month rate elicited in the survey. The one-year-ahead

inflation expectation in state j is denoted $\hat{\Pi}^j_{t+4|t}$ and is then

$$\hat{\Pi}^j_{t+4|t} = \sum_{s=1}^{4} \hat{\pi}^j_{t+s|t}. \tag{13}$$

Using the perceived law of motion for inflation (1), this can be written as

$$\hat{\Pi}^j_{t+4|t} = \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \hat{\pi}^j_{t|t} \ . \tag{14}$$

Combining this definition with the inflation forecast equation derived from the Kalman filter (11), the year-ahead expectation can be written as the linear regression

$$\hat{\Pi}^j_{t+4|t} = \beta^j_0 + \beta^j_1 \hat{\Pi}^j_{t+3|t-1} + \beta^j_2 \pi_t + \nu^j_t \ , \tag{15}$$

with coefficients

$$\begin{cases} \beta^j_0 = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \xi^j \\ \beta^j_1 = \rho^j (1 - \mathbf{K}^j) \\ \beta^j_2 = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \ . \end{cases} \tag{16}$$

Appendix B.2 provides the details of the derivation and shows that this regression recovers the population coefficients.

Hence, this specification makes it possible to recover the parameters of the inflation-expectation model from the survey observations. The mapping is useful because the regression coefficients have direct economic content: the coefficient on lagged expectations, β^j_1, captures perceived persistence net of updating, the coefficient on current inflation, β^j_2, captures the response to inflation news, and the constant, β^j_0, captures the average level shift implied by the perceived signal bias and the strength of updating. The perceived inflation persistence ρ^j, the Kalman gain $\mathbf{K}^j$, and the perceived signal bias ξ^j are then obtained numerically from the system (16), so that the structural interpretation comes from the three coefficients jointly. In the estimation, ρ^j is not restricted to the unit interval.

3.4. Estimation

The regression (15) is estimated by weighted least squares, using survey weights, over the full sample and separately over the three periods considered in Section 2.2: before, during, and after the inflation surge. In each sample, HtM households and Savers are estimated jointly with group-specific coefficients, so that the differences between the two groups can be tested directly. Standard errors are clustered by respondent and by survey month, to account both for the repeated observations of the same household and for the common shocks that affect all households in a given month. The parameters ρ^j, $\mathbf{K}^j$, and ξ^j are recovered from the estimated coefficients through the mapping (16), and their standard errors are computed using the delta method.

Table 3: Noisy-information estimates by household group (quarterly model, OLS).

	Dependent variable: $\hat{\Pi}^i_{t+4\|t}$							
	Full sample 2015–2025		Pre-surge 2015–2020		Surge 2021–2023		Post-surge 2024–2025	
	HtM	Savers	HtM	Savers	HtM	Savers	HtM	Savers
$\hat{\beta}^j_1$: lagged expectation, $\hat{\Pi}^i_{t+3\|t-1}$	0.370***	0.344***	0.300***	0.251***	0.412***	0.438***	0.442***	0.376***
	(0.019)	(0.025)	(0.026)	(0.031)	(0.029)	(0.040)	(0.039)	(0.055)
$\hat{\beta}^j_2$: 3-month inflation, π_t	2.507***	1.173***	1.378	0.069	2.509***	1.277***	−0.502	0.351
	(0.391)	(0.181)	(0.881)	(0.156)	(0.538)	(0.203)	(1.707)	(0.756)
$\hat{\beta}^j_0$: constant	3.539***	1.866***	4.312***	2.552***	3.498***	1.613***	4.782***	1.959***
	(0.404)	(0.179)	(0.574)	(0.131)	(0.831)	(0.339)	(1.298)	(0.598)
Implied $\hat{\rho}^j$	0.998***	0.764***	0.781	0.299	1.020***	0.843***	0.442	0.549
	(0.051)	(0.040)			(0.068)	(0.042)		
Implied $\hat{\mathbf{K}}^j$	0.629***	0.549***	0.616	0.163	0.597***	0.480***	0.000	0.316
	(0.025)	(0.035)			(0.037)	(0.044)		
Implied $\hat{\xi}^j$	1.412***	1.591***	–	–	1.394**	1.263***	–	–
	(0.362)	(0.376)			(0.603)	(0.429)		
p-values, H_0: HtM = Savers								
β^j_2	0.000		0.125		0.031		0.536	
ρ^j	0.000		–		0.032		–	
$\mathbf{K}^j$	0.045		–		0.025		–	
ξ^j	0.663		–		0.864		–	
Observations	23,780	46,677	12,307	25,731	7,505	14,113	3,968	6,833
Respondents	5,585	9,092	3,134	5,318	1,782	2,905	987	1,543
R^2	0.182	0.181	0.111	0.092	0.212	0.262	0.242	0.199

Notes: Weighted OLS estimates of $\hat{\Pi}^i_{t+4|t} = \beta^j_0 + \beta^j_1 \hat{\Pi}^i_{t+3|t-1} + \beta^j_2 \pi_t + \nu^i_t$, where $\hat{\Pi}^i_{t+4|t}$ is respondent i's one-year-ahead point forecast in survey month t, $\hat{\Pi}^i_{t+3|t-1}$ the same respondent's forecast three months earlier and π_t CPI inflation over the three months up to month t (not annualized). All respondent-months with a forecast three months earlier and the same Saver-HtM status in both months are used. Forecasts are winsorized at the 1st and 99th weighted percentiles of each month; survey weights. HtM and Savers are estimated jointly with group-specific coefficients. Standard errors (in parentheses) are clustered two-way by respondent and survey month. ρ^j, $\mathbf{K}^j$ and ξ^j solve (16); standard errors by the delta method. Statistical significance at the 1%, 5% and 10% levels is reported with ***, ** and *, respectively.

From (16), the Kalman gain is zero exactly when β^j_2 is zero, so that the test on β^j_2 is a test on $\mathbf{K}^j$. When $\hat{\beta}^j_2$ is not statistically different from zero, the mapping is strongly nonlinear around the estimate, and the delta method can deliver misleadingly small standard errors for the implied parameters. In these cases, I report the point estimates of ρ^j and $\mathbf{K}^j$ without standard errors, and I do not report ξ^j, which is not identified. Table 3 reports the estimates.

Before the inflation surge, between 2015 and 2020, the coefficient $\hat{\beta}^j_2$ on quarterly inflation π_t is not statistically different from zero for either HtM households or Savers. Hence, neither group reacts systematically to current inflation news, and a Kalman gain equal to zero cannot be rejected. In that case, $\beta^j_1 = \rho^j$, so that the coefficient on lagged expectations directly measures the perceived persistence of inflation, which is estimated at 0.30 for HtM households and 0.25 for Savers. In addition, ξ^j cannot be identified, and the difference in the intercepts $\hat{\beta}^j_0$ is interpreted as a difference in inflation priors. The same holds after the surge, between 2024 and 2025.

Conversely, during the inflation surge, between 2021 and 2023, the $\hat{\beta}^j_2$ coefficient is positive and statistically significant for both groups of households, implying that they both

update their expectations in response to current inflation news. In particular, $\hat{\beta}_2^j$ is about twice as large for HtM households as for Savers, and the difference is statistically significant. Given the definition of β_2^j, this stronger reduced-form response can reflect either a higher Kalman gain or a more persistent perceived inflation process, and the estimates point to both. The coefficients on lagged expectations are similar across groups. Nevertheless, the numerical inversion of (16) implies that HtM households perceive inflation as more persistent than Savers, $\hat{\rho}^H = 1.02$ against $\hat{\rho}^S = 0.84$,[13] and that they react more strongly to inflation news: $\hat{\mathbf{K}}^H = 0.60 > 0.48 = \hat{\mathbf{K}}^S$. Both differences are statistically significant at the 5% level. Over the full sample, in which the parameters are identified by the same large movements in inflation, the ordering is the same: $\hat{\mathbf{K}}^H = 0.63$ against $\hat{\mathbf{K}}^S = 0.55$, and the difference is also statistically significant.

The larger HtM intercept should be read through the same mapping. Since $\beta_0^j = \beta_2^j \xi^j$, a higher intercept does not necessarily require a larger perceived signal bias ξ^j. Indeed, the implied perceived bias is very similar for the two groups, 1.39 for HtM households and 1.26 for Savers during the surge, and the difference is far from statistically significant. Given a similar perceived bias, the higher perceived persistence and Kalman gain of HtM households translate it more strongly into year-ahead expectations, which are therefore higher on average, consistent with Fact 1.

Figure 4 shows the estimation of the model in 36-month windows moved by one month, and shows the Kalman gains and the perceived persistence of the two groups.[14] Following the reporting rule described above, the estimates are shown only in windows in which the response to inflation news is statistically significant. For both groups, this is the case in 48 of the 83 windows, essentially all those centred from 2020 onwards, which include the fall in inflation of 2020 and the inflation surge. In all of them, the Kalman gain of HtM households exceeds that of Savers, significantly so in 36. The gap is stable, at around 0.1 to 0.15, in the windows covering 2020 to 2023, and shrinks as the windows move into the post-surge period. The same pattern holds for perceived persistence. In all these windows, the perceived persistence of HtM households exceeds that of Savers, significantly so in 41, with a median difference of 0.21. HtM households perceive inflation as close to a unit root in all windows centred from 2020 onwards, while the perceived persistence of Savers rises from about 0.55 to about 0.9 as the windows move through the surge.[15]

In other words, whenever inflation moves substantially, and the data are therefore informative about the Kalman gains, the gain of HtM households exceeds that of Savers. In periods of low and stable inflation, instead, households' responses to inflation news cannot be estimated precisely.

[13] The estimate for HtM households is not statistically different from one, and imposing $\rho^H = 1$ implies $\mathbf{K}^H = 1 - \beta_1^H = 0.59$, very close to the unrestricted estimate.

[14] I am grateful to Uwe Sunde for suggesting this estimation exercise.

[15] The same pattern is truw with 24-month windows, where the HtM gain exceeds the Saver gain in all of the 36 windows in which both can be estimated, significantly so in 25, and the same holds for perceived persistence, significantly so in 26 (see Figure 18 in the Appendix).

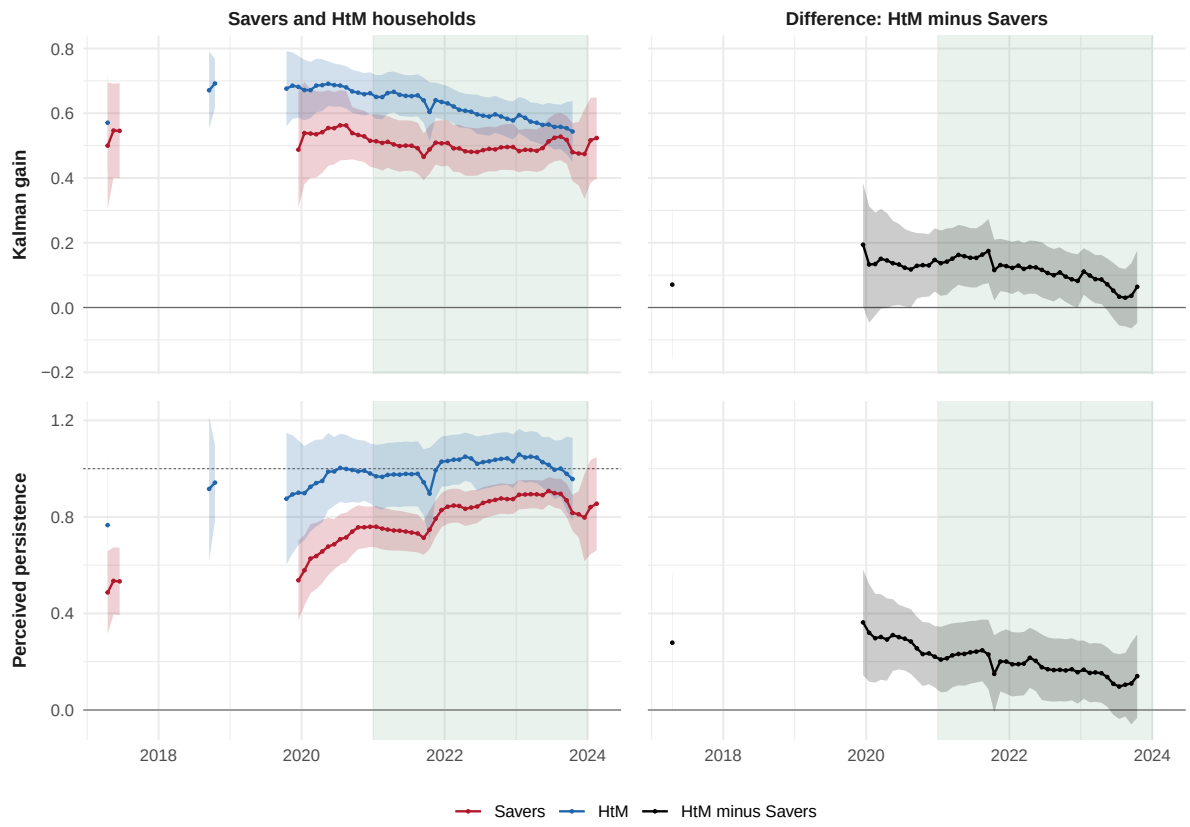


Figure 4: Kalman gains and perceived persistence of Savers and HtM households in rolling windows.

Notes: Estimates of the quarterly noisy-information model in 36-month windows moved by one month, plotted at the centre of each window. The top panels show the Kalman gains and the bottom panels the perceived persistence of inflation. In the left panels, Savers are in red and HtM households in blue, and the right panels show the difference between HtM households and Savers. The estimates are shown only in windows where the response to inflation, $\hat{\beta}_2^j$, is statistically significant at the 10% level (for both groups, in the case of the difference). The bands represent the 90% confidence interval, and the dashed line marks a perceived persistence of one. The shaded area marks the inflation surge, 2021 to 2023.

A different estimation of perceived inflation persistence. The difference in perceived persistence can also be assessed with information that the estimation does not use. The SCE asks respondents about inflation over the twelve months between 24 and 36 months ahead. In the noisy-information model, the forecast at any horizon is the current nowcast of inflation carried forward with the perceived persistence, so that this three-year expectation moves with the one-year expectation by a factor $(\rho^j)^8$. Households that perceive inflation as more persistent should therefore display a stronger co-movement between their three-year and one-year expectations.[16] Unlike the level of longer-horizon expectations, which also depends on the rate at which households expect inflation to settle in the long run, this co-movement depends only on perceived persistence. Table 12 in the Appendix reports the corresponding slopes. During the inflation surge, the average three-year expectation of HtM households moves with their one-year expectation with a slope of 0.38, against 0.12 for Savers, and the difference is statistically significant. In terms of quarterly persistence, these slopes imply values of 0.89 for HtM

[16] I am grateful to Giorgio Topa for suggesting this estimation of perceived persistence in the SCE.

households and 0.77 for Savers, which preserve the ordering of the estimates in Table 3. Hence, longer-horizon expectations confirm, independently of the estimation, that HtM households perceive inflation as more persistent than Savers during the inflation surge.

Summary. In sum, the estimation suggests that in periods of low and stable inflation, before and after the surge, neither Savers nor HtM households update their inflation expectations systematically in response to current inflation news. In these periods, their Kalman gains cannot be distinguished from zero, and the level differences in forecasts are interpreted as differences in inflation priors. During the inflation surge, instead, both groups react to inflation news, but they do so with Kalman gains well below the full-information benchmark. Underreaction is therefore present for both groups, and it is more pronounced for Savers: in the baseline estimates, their Kalman gain is lower than that of HtM households in every period and every window in which the gains can be estimated. At the same time, HtM households perceive inflation as more persistent than Savers, a difference that is also reflected in the co-movement of their longer-horizon expectations. The perceived bias in the inflation signal is instead similar for the two groups.

3.5. Savers, All Households, and the Actual Persistence of Inflation

The estimates above describe the beliefs of Savers and HtM households separately. Two further comparisons are needed for the general equilibrium analysis of Sections 4 and 5. The first compares the beliefs of Savers with those of all households, since surveys typically summarize the expectations of the whole population. The second compares the persistence that households perceive with the actual persistence of inflation. Table 4 reports both comparisons for the two periods in which perceived persistence is identified: the full sample and the inflation surge.

For all households, the regression (15) is estimated on all respondents of the baseline sample with common coefficients, using the same sample, survey weights, estimator, and standard errors as above. The Kalman gain of all households is close to that of Savers: 0.52 against 0.48 during the surge, and 0.57 against 0.55 over the full sample. Perceived persistence is not. All households perceive inflation as considerably more persistent than Savers do: 0.93 against 0.84 during the surge, and 0.89 against 0.76 over the full sample. The reason is the composition of the population. HtM households make up about 40 percent of the weighted sample, and they perceive inflation as close to a random walk. In other words, population-wide estimates describe how Savers react to inflation news well, but they overstate how persistent Savers perceive inflation to be.

The actual persistence of inflation is measured in the same way as the expectation regressions: three-month inflation, the series that households forecast in the model, is regressed on its value three months earlier, using every month. Actual persistence is about 0.5 for headline inflation, 0.48 over the full sample and 0.53 during the surge. Core inflation, which excludes food and energy, is about as persistent over the full sample, 0.42, and much less persistent during the surge, although this estimate is imprecise. Hence, all groups of households perceive inflation as considerably more persistent than it actually is, and this over-perception is smallest for Savers.

Table 4: Perceived and actual persistence of inflation.

	Full sample, 2015–2025	Surge, 2021–2023
Perceived persistence, $\hat{\rho}$		
Savers	0.76 (0.04)	0.84 (0.04)
HtM households	1.00 (0.05)	1.02 (0.07)
All households	0.89 (0.04)	0.93 (0.04)
Kalman gain, $\hat{\mathbf{K}}$		
Savers	0.55 (0.04)	0.48 (0.04)
HtM households	0.63 (0.03)	0.60 (0.04)
All households	0.57 (0.02)	0.52 (0.03)
Actual persistence of inflation		
Headline CPI	0.48 (0.18)	0.53 (0.18)
Core CPI	0.42 (0.15)	0.07 (0.24)

Notes: Perceived persistence and Kalman gains: baseline OLS estimates of the quarterly noisy-information model (Table 3), standard errors by the delta method. All households: the same regression estimated on all respondents of the baseline sample with common coefficients (survey-weighted share of HtM households: 41 percent in the full sample and 42 percent during the surge). Actual persistence: coefficient of a regression of three-month CPI inflation (not annualized, seasonally adjusted) on its value three months earlier, using every month (121 months in the full sample, 36 during the surge); Newey–West standard errors with six lags, which account for the overlap of the three-month windows. Core CPI: all items less food and energy (FRED series CPILFESL).

4. THANK with Subjective Inflation Expectations

This section presents the state-contingent consumption functions of Savers and HtM households in the Tractable HANK framework of Bilbiie (2025), and shows how they can be used with the subjective inflation expectations estimated in Section 3.

4.1. The THANK Model

Household States and Transitions. The economy is populated by a unit mass of households that are either unconstrained *Savers*, denoted with superscript S, or constrained *Hand-to-Mouth* (HtM), denoted with superscript H. A fraction λ of households are HtM, while the remaining fraction $1-\lambda$ are Savers. Household states evolve according to an exogenous two-state Markov chain, that generates the idiosyncratic uncertainty faced by households in this economy: conditional on being in state S in period t, a household remains a Saver in period $t+1$ with probability s and transitions to state H with probability $1-s$. Similarly, conditional on being in state H in period t, a household remains HtM with probability h and transitions to state S with probability $1-h$.[17]

The stationary distribution of household states is determined by the invariant distribution of the Markov transition matrix. Let $(\lambda,\ 1-\lambda)$ denote the stationary fraction of households in states (H, S). Solving for the stationary distribution yields that the unconditional probability

[17]This formulation nests the standard *Two-Agent New Keynesian* (TANK) framework (see Bilbiie, 2008, Bilbiie, 2020): when states are absorbing ($s = h = 1$), the fraction sizes are not pinned down by transition probabilities and λ is exogenous.

of being in the HtM state is given by

$$\frac{\lambda}{1-\lambda} = \frac{1-s}{1-h} \quad \Longleftrightarrow \quad \lambda = \frac{1-s}{2-s-h}\,. \tag{17}$$

Assets and Portfolio Choice. Households can hold two types of assets: *illiquid stocks*, which represent claims on firm profits, and *liquid bonds*, which correspond to public debt. In period t, households choose current consumption and the asset portfolio they carry into period $t+1$. Idiosyncratic uncertainty arises because households choose their asset positions in period t *before* knowing whether they will be in the Saver or HtM state in period $t+1$.

The timing of the household decision problem is as follows. At the beginning of period t, each household's current state, S or H, is revealed. Conditional on the realized state distribution, assets are pooled within each state group and redistributed equally among households in that group. Each household therefore receives the per-capita asset holdings of its current state group, together with the associated returns. Stock holdings and dividend payments are assumed to accrue only to households in the Saver state. Hence, under these pooling assumptions, the model features full insurance *within* each current state group, but incomplete insurance *across* states.

By the end of period t, households choose consumption and determine their asset holdings for period $t+1$. Savers can allocate their portfolio across both bonds and stocks, whereas HtM households can hold only bonds.

At the beginning of period $t+1$, the new states are revealed, and assets are again pooled and redistributed within each current state group. Assets available to Savers in period $t+1$ consist of the assets carried by households who were Savers in period t and remained Savers, together with the assets carried by households who were HtM in period t and became Savers. Similarly, assets available to HtM households in period $t+1$ consist of the assets carried by households who remained HtM, together with the assets carried by households who were Savers in period t and became HtM.

Notation. Stock holdings are denoted by Θ_t. Let V_t denote the market value of a stock and D_t the real dividend paid per stock. Since households that transition to state H cannot hold stocks, equity is only available to Savers. It should be noted that stocks are presented here to describe the full model, however the consumption functions derivation of the following sections will assume, for simplicity, that $\Theta_t = 0$, as these assets (unlike bonds) cannot be used for Savers' self-insurance against state-transition risk.

Bond holdings depend on household's state $J \in \{S, H\}$ and are denoted by B_t^J and Z_t^J. These variables capture the timing of portfolio decisions: Z_{t+1}^J denotes the bond holdings of a household in state J at the *end of period* t, after the consumption–saving decision has been made but *before* the household's state in period $t+1$ is revealed. Instead, B_{t+1}^J denotes the bond holdings of a household in state J at the *beginning of period* $t+1$, after the state realization has occurred. Therefore the distinction between Z_{t+1}^J and B_{t+1}^J reflects the pooling and redistribution of assets after the state transitions have happened.

Asset Redistribution Across States. Given the transition probabilities across states, the aggregate bond holdings received by each group at the beginning of period $t+1$ reflect the flows of households between the two states. Hence, from accounting, the aggregate bond holdings of the Savers group must satisfy

$$(1-\lambda)B^S_{t+1} = (1-\lambda)sZ^S_{t+1} + \lambda(1-h)Z^H_{t+1}. \tag{18}$$

Namely, $(1-\lambda)B^S_{t+1}$ are the bond holdings available to the Savers at the beginning of period $t+1$. They consist of $(1-\lambda)sZ^S_{t+1}$, the bonds carried by the Savers in period t who remained Savers, plus $\lambda(1-h)Z^H_{t+1}$, the bonds carried by the HtM in period t who became Savers. Similarly, for the Hand-to-Mouth group

$$\lambda B^H_{t+1} = (1-\lambda)(1-s)Z^S_{t+1} + \lambda hZ^H_{t+1}. \tag{19}$$

has to hold. Using the relationship implied by the stationary distribution $\lambda(1-h) = (1-\lambda)(1-s)$ in (17), the above expressions respectively simplify to

$$B^S_{t+1} = sZ^S_{t+1} + (1-s)Z^H_{t+1}; \tag{20}$$

$$B^H_{t+1} = (1-h)Z^S_{t+1} + hZ^H_{t+1}. \tag{21}$$

Dynamic Programming Problem Households derive utility from consumption C_t and incur disutility from supplying labor N_t. The utility function is denoted $U(C_t, N_t)$ and future utility is discounted at the rate $\beta \in (0,1)$. Let P_t denote the price level of consumption goods at time t, and let W_t denote the real wage.

A crucial assumption, as in Bilbiie (2025), is that households' current states, S or H, determine to which *state-contingent group* they belong. Within each group, the consumption decisions and asset pooling are represented by a family head who assigns equal weight to the utility of each household in that group. These state-contingent groups are therefore akin to "islands", in the sense that resources are pooled and distributed within each island, while insurance across islands remains incomplete.[18]

Under standard compactness assumptions, with a discount factor $\beta \in (0,1)$, the consumption problem of the family head can be represented recursively, where allocations are chosen for the state-contingent groups, assigning weights equal to the stationary population shares given by (17). The program takes the form of a Bellman equation in which the value function depends on the beginning-of-period asset holdings of each household group. Denote by $\Lambda(\cdot)$ the value function and the problem can be written as

$$\Lambda\left(B^H_t, B^S_t, \Theta_t\right) = \max_{\substack{C^H_t, C^S_t, N^H_t, N^S_t \\ Z^H_{t+1}, Z^S_{t+1}, \Theta_{t+1}}} \left\{\lambda U(C^H_t, N^H_t) + (1-\lambda)U\left(C^S_t, N^S_t\right)\right\} + \beta\mathbb{E}_t\left[\Lambda\left(B^H_{t+1}, B^S_{t+1}, \Theta_{t+1}\right)\right]$$

[18] The island analogy refers to the seminal contributions of Phelps (1970) and Lucas (1972), where the economy is populated by distinct islands. Unlike in the Lucas islands model, the role of islands here is not to generate imperfect information, but to describe state-contingent pools of resources.

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t B_t^S + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t B_t^H \\ B_{t+1}^S &= s Z_{t+1}^S + (1-s) Z_{t+1}^H \\ B_{t+1}^H &= (1-h) Z_{t+1}^S + h Z_{t+1}^H \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

Denoting $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$, the resulting KKT condition for the bonds of a Saver yields:

$$\beta \mathbb{E}_t \left[\frac{R_{t+1}}{\Pi_{t+1}} \left(s U_C \left(C_{t+1}^S \right) + (1-s) U_C \left(C_{t+1}^H \right) \right) \right] = U_C \left(C_t^S \right) \quad \text{together with} \quad Z_{t+1}^S > 0 \tag{22}$$

and, focusing on the equilibria where the constraint on the Hand-to-Mouth always binds, the KKT condition for the bonds of a HtM yields:

$$\beta \mathbb{E}_t \left[\frac{R_{t+1}}{\Pi_{t+1}} \left((1-h)\, U_C \left(C_{t+1}^S \right) + h U_C \left(C_{t+1}^H \right) \right) \right] < U_C \left(C_t^H \right) \quad \text{together with} \quad Z_{t+1}^H = 0 \tag{23}$$

The complete derivation of the Euler equations is provided in Appendix C.1.

The Saver Euler equation (22) has a state-contingent interpretation. By purchasing one additional unit of liquid bonds at the end of period t, a Saver gives up current consumption in exchange for a real return R_{t+1}/Π_{t+1} next period. Since the household does not know whether they will be in state S or H in period $t+1$, this return is valued using the probability-weighted marginal utility across future states. Hence, $(1-s)$ measures the idiosyncratic risk faced by the household, and the term $(1-s)U_C(C_{t+1}^H, N_{t+1}^H)$ captures the precautionary saving motive: liquid bonds are valuable because they transfer resources across time and future states, in the event in which the household becomes HtM. In other words, if marginal utility in the future HtM state is high, the value of saving for a current Saver increases.

The HtM condition is instead an inequality since, in the equilibria considered here, the bond constraint binds and $Z_{t+1}^H = 0$. HtM consumption is therefore pinned down by the current budget constraint, which includes the pooled resources brought by Savers who transition into the HtM state.

Thus, the forward-looking intertemporal margin is governed by the Saver Euler equation (22), and the state-contingent consumption functions can be derived in the following subsections.

4.2. Derivation of the Consumption Functions

Instead of deriving the *aggregate* consumption function, I describe the consumption decisions of households in each state. These are obtained by combining the state-contingent Euler equations with the relevant budget constraints, following the household-level derivation in Luetticke et al. (2026).

The key mechanism of the model is that the uncertainty induced by the exogenous transitions between the Saver and HtM states distorts intertemporal consumption. In particular, because Savers may become HtM in the future, two opposed motives arise: on the one hand there is precautionary saving, as Savers may want to self-insure against the risk of becoming constrained HtM; on the other hand, there is front-loading, as the asset structure delivers that bonds can only imperfectly finance future Saver consumption.

I make the following assumptions: there are no stocks ($\Theta_t = 0$), households in the *Saver* state can only save in the form of bonds Z, and the profits from ownership of the firm are distributed in the form of dividends. Total nominal dividends of the firm are denoted $P_t D_t$ and, given that the fraction of Savers is $1-\lambda$, each Saver receives $P_t \frac{D_t}{1-\lambda}$ at time t.

In *real* notation, $\tilde{Z}_t^S = \frac{Z_t^S}{P_{t-1}}$ is the real bond holdings of a Saver, $\tilde{Y}_t^S = \frac{W_t N_t^S}{P_t} + \frac{D_t}{1-\lambda}$ is the real non-interest income of a Saver in period t, $\tilde{Y}_t^H = \frac{W_t N_t^H}{P_t}$ is the real non-interest income of a HtM, and $\tilde{R}_t = \frac{R_t}{\Pi_t}$ is the real interest rate carried by the bonds in the beginning of period t, with $\Pi_t = \frac{P_t}{P_{t-1}}$.

Assuming that the household has CRRA preferences given by the utility function[19]

$$U(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \nu_N \frac{N_t^{1+\psi}}{1+\psi} \tag{24}$$

the Euler equation for bonds (22) then reads

$$\beta \, \mathbb{E}_t \left[\tilde{R}_{t+1} \left(s \left(C_{t+1}^S \right)^{-\sigma} + (1-s) \left(C_{t+1}^H \right)^{-\sigma} \right) \right] = \left(C_t^S \right)^{-\sigma},$$

where the expected future marginal utility is an average of the marginal utility in the Saver state and the marginal utility in the HtM state, both weighted by their respective transition probabilities. That Euler equation can be log-linearized as[20] (see Appendix C.2)

$$c_t^S = \mathbb{E}_t \left[\omega c_{t+1}^S + (1-\omega) c_{t+1}^H - \frac{1}{\sigma} \tilde{r}_{t+1} \right], \tag{25}$$

where the weight ω is

$$\omega \equiv \frac{s}{s + (1-s)\,\zeta^\sigma} \qquad \text{with} \qquad \zeta \equiv \frac{C^S}{C^H}. \tag{26}$$

Hence, ω can be interpreted as the *share* of a Saver's expected future marginal utility derived from remaining a Saver, in steady state. It depends on ζ, the consumption ratio between the two household states in the non-stochastic steady state, raised to the power of σ: a higher curvature of marginal utility amplifies the weight of the HtM state. The optimal consumption choice of a Saver takes into account that bonds can be used to self-insure against the risk of becoming HtM, and this mechanism is captured by $(1-\omega)\mathbb{E}_t \left[c_{t+1}^H \right]$ in the log-linearized Euler equation.

[19] Here, σ is the coefficient of relative risk aversion and hence $1/\sigma$ denotes the intertemporal elasticity of substitution. This is the reciprocal of the convention used in Bilbiie (2025).

[20] All log-linearizations in this paper proceed, for a given variable X_t, by defining $X_t = X e^{x_t}$ and taking the first order Taylor approximation around X, the non-stochastic steady state of X_t.

In steady state, the Euler equation implies (see Appendix C.2)

$$s\tilde{R} = \frac{\omega}{\beta}, \tag{27}$$

so that, if the steady-state consumption of a Saver is higher than that of a HtM, $\omega < s$ and the real interest rate is lower than $\frac{1}{\beta}$: the precautionary savings motive is amplified because of the risk of transitioning to a state where steady-state consumption is lower.

Similarly, the log-linearized flow budget constraints are

$$c_t^S = \frac{\tilde{Y}^S}{C^S}\tilde{y}_t^S + \frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t + \tilde{z}_t^S\right) - \frac{\tilde{Z}^S}{C^S}\tilde{z}_{t+1}^S, \quad \text{and} \quad c_t^H = \frac{\tilde{Y}^H}{C^H}\tilde{y}_t^H + \frac{C^H - \tilde{Y}^H}{C^H}\left(\tilde{r}_t + \tilde{z}_t^S\right)$$

for a Saver and a HtM, respectively. Combining these equations with the log-linearized Euler equation for a Saver yields an expression summarizing the dynamics of bonds:

$$\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \mathcal{A}_\zeta\,\tilde{z}_{t+1}^S + \beta^{-1}\tilde{z}_t^S = F_t, \tag{28}$$

where the term $\mathcal{A}_\zeta$ is a variable combining primitives s, h, β, and steady-state objects ω and ζ, while F_t is a forcing term containing incomes and interest rates. The characteristic equation of (28) is given by

$$\mu^2 - \mathcal{A}_\zeta\mu + \beta^{-1} = 0. \tag{29}$$

Its roots, denoted μ_1 and μ_2, satisfy $\mu_1 + \mu_2 = \mathcal{A}_\zeta$ as well as $\mu_1\mu_2 = \beta^{-1}$, and Appendix C.3 shows that the saddle configuration $\mu_1 < 1 < \mu_2$ is verified if and only if

$$\frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega} \tag{30}$$

which is satisfied by standard calibrations of the model, where $\omega < \beta$ holds.

Factoring the equation for bond dynamics (28) using the roots μ_1 and μ_2, and imposing the transversality condition $\lim_{j\to\infty}\mu_2^{-j}\mathbb{E}_t\left[\tilde{z}_{t+j+1}^S - \mu_1\tilde{z}_{t+j}^S\right] = 0$ yields the forward solution

$$\tilde{z}_{t+1}^S = \mu_1\tilde{z}_t^S - \sum_{j=0}^{\infty}\mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right] \tag{31}$$

that clarifies the economic interpretation of the roots: $\mu_1 < 1$ captures the persistence of bonds in the Saver state, and $\mu_2 > 1$ their future discounting. A representative agent model, without idiosyncratic state-transition risk, is then nested with $\mu_1 = 1$, and $\mu_2 = \frac{1}{\beta} = \tilde{R}$.

The Consumption Functions The consumption function of a Saver is obtained by combining the forward solution for bonds (31) together with the budget constraint of a Saver, and rearranging. The consumption function of a HtM is their log-linearized budget constraint. That leads to the following proposition.

Proposition 1 (The Consumption Functions)**.** *In the infinite-horizon THANK model, the consumption functions in period t, log-linearized around the non-stochastic steady-state, are, if the household is in the Saver state:*

$$\begin{aligned} c_t^S &= \left(1-\frac{\mu_1}{\omega}\beta\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^S\right] \\ &+\left[\frac{\tilde{Z}^S}{C^S}\left(1-\frac{\mu_1}{\omega}\beta\right)\mu_2-\frac{1}{\omega\sigma}\right]\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\ &+\left(1-\frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t+\tilde{z}_t^S\right) \\ &+\frac{\tilde{Y}^H}{C^H}\frac{1-\omega}{\omega}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^H\right] \end{aligned} \tag{32}$$

and if the household is in the Hand-to-Mouth state:

$$c_t^H=\frac{\tilde{Y}^H}{C^H}\tilde{y}_t^H+\frac{C^H-\tilde{Y}^H}{C^H}\left(\tilde{r}_t+\tilde{z}_t^S\right). \tag{33}$$

Proof. See Appendix C.3. □

The consumption function of a HtM (33) is their log-linearized budget constraint: the HtM household consumes out of their income $\tilde{y}_t^H$ and out of the beginning-of-period wealth, $\tilde{r}_t+\tilde{z}_t^S$, consisting of the bonds brought by the Savers who transition into the HtM state. It contains no expectations of future variables.

The consumption function of a Saver (32) depends on the entire sequences of expected real incomes and real rate. It has four components: the first line captures consumption's response to changes in the sequence of the Saver real (non-interest) income, the second line to changes in the sequence of the real interest rate, the third line to changes in the beginning-of-period t bonds, and the fourth line to changes in the sequence of the expected HtM real income. All expected future variables are discounted by the same root, $\mu_2>1$.

Savers' response to the expected path of real interest rates. The second line of (32) combines two channels. The first is the intertemporal-substitution channel, captured by $-\frac{1}{\sigma\omega}$: a higher expected future real rate makes saving more attractive and reduces current consumption. In a representative-agent model, the strength of this channel is governed by the intertemporal elasticity of substitution, $1/\sigma$. In THANK, it is scaled by $1/\omega$, because the Euler equation weights future marginal utility across both possible states. The second is a wealth-revaluation channel: if Savers hold liquid assets in steady state, a higher future real rate raises the return on the bonds that reach that date, which tends to increase current consumption. With zero liquidity, which is assumed from Section 4.3 onwards, the second channel vanishes, and the expected path of real rates affects Saver consumption only through intertemporal substitution. This is the channel through which inflation expectations enter aggregate demand.

Intertemporal MPCs. The consumption function (32) also characterize the intertemporal marginal propensities to consume of Savers. Appendix C.4 shows three properties. First, a Saver consumes a fraction $1 - \frac{\mu_1}{\omega}\beta$ of a current income shock on impact, which exceeds its representative-agent counterpart, $1 - \beta$, if and only if $\mu_1 < \omega$, a condition satisfied under standard calibrations. Second, the consumption response to income at horizon k relative to current income is $\mu_2^{-k} = (\beta\mu_1)^k$, so that the consumption profile tilts towards the present. Third, Savers respond more strongly to news about future HtM income than to news about their own future income at the baseline calibration, because higher future HtM income weakens the self-insurance motive.

4.3. Subjective Inflation Expectations

The consumption functions of Proposition 1 are derived under full-information rational expectations (FIRE). This subsection shows that they also hold under subjective expectations. Throughout, $\mathbb{E}_t$ denotes FIRE expectations and $\hat{E}_t$ the subjective expectations of Savers.

Replacing the expectation operator. Proposition 1 is derived from the Saver Euler equation, the flow budget constraints, and the transversality condition (see Appendix C.3). In that derivation, the expectation operator needs to be linear and to satisfy the law of iterated expectations, in order to solve forward the difference equation for bonds. Hence, if the expectations of a Saver are described by any operator $\hat{E}_t$ with these two properties, every step of the derivation goes through with $\hat{E}_t$ in place of $\mathbb{E}_t$, and as a result Proposition 1 holds with $\mathbb{E}_t$ replaced by $\hat{E}_t$.

Linearity holds for any expectation taken with respect to a probability distribution. The law of iterated expectations holds whenever the expectations of a Saver are conditional expectations under a subjective model of the economy with the information set updated over time. This does not require $\hat{E}_t = \mathbb{E}_t$, since the subjective model can differ from the actual one.

Which expectations are subjective. The consumption function of a Saver (32) depends on three discounted sums of expected future variables: their own income, the real interest rates, and the HtM income. It is assumed here that Savers forecast their own income and the income they would receive in the HtM state rationally, hence:

$$\hat{E}_t\left[\tilde{y}^S_{t+k}\right] = \mathbb{E}_t\left[\tilde{y}^S_{t+k}\right] \quad \text{and} \quad \hat{E}_t\left[\tilde{y}^H_{t+k}\right] = \mathbb{E}_t\left[\tilde{y}^H_{t+k}\right].$$

Since the consumption function is linear in the three discounted sums, the effect of departures from FIRE on Saver consumption is the sum of their effects through each of them. Holding FIRE on income expectations therefore isolates the channel through which inflation expectations affect demand: the expected path of real interest rates. In equilibrium, the paths of income respond to inflation beliefs, and Savers anticipate this response rationally. FIRE on income thus keeps the general-equilibrium structure of THANK in Bilbiie (2025) unchanged, with future income discounted at μ_2 and compounded through the income of the HtM state, so that the comparison with FIRE varies inflation expectations only.

The real return $\tilde{R}_t = R_t/\Pi_t$ is carried by the bonds into period t, where the gross nominal rate R_t is set in period $t-1$. Let i_t denote the log deviation of the nominal rate set in period

t, that is of R_{t+1}, and similarly π_t the log deviation of Π_t. Then $\tilde{r}_{t+k} = i_{t+k-1} - \pi_{t+k}$, and the discounted sum of expected real rates in the consumption function splits into expected nominal rates and expected inflation:

$$\sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[\tilde{r}_{t+k}] = \sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[i_{t+k-1}] - \sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[\pi_{t+k}]. \tag{34}$$

Savers therefore need beliefs about inflation and about future nominal rates at all horizons, not only one period ahead. This is why, under subjective expectations, as in Preston (2005), the Euler equation alone does not pin down consumption.

The Perceived Law of Motion (PLM) of Inflation. Savers' inflation expectations follow the noisy-information model of Section 3. The Saver's perceived law of motion (PLM) of inflation is then an AR(1) process with persistence ρ^S, Savers observe a noisy signal of current inflation, and update their nowcast with the steady-state Kalman gain $\mathbf{K}^S$. In line with the estimation, assume that all Savers share the same beliefs and that the idiosyncratic signal noise averages out in the Savers group. Given that the perceived signal bias ξ simply adds a constant to the nowcast but does not affect the dynamics, it is not considered in this exercise.

To lighten the notation, since only the beliefs of Savers enter the forward-looking margins of consumption, the superscript $j = S$ for Savers is dropped, and the nowcast of inflation is simply denoted $\hat{\pi}_t \equiv \hat{\pi}_{t|t}$. Also, the expectations conditional on the information in period t are written $\hat{E}_t[\,\cdot\,] \equiv \hat{E}[\,\cdot \mid \mathcal{I}_t]$.

Accordingly, the nowcast of inflation evolves as

$$\hat{\pi}_t = \rho(1 - \mathbf{K})\,\hat{\pi}_{t-1} + \mathbf{K}\,\pi_t \tag{35}$$

and the forecasts of future inflation under the PLM satisfy

$$\hat{E}_t[\pi_{t+k}] = \hat{\pi}_{t+k|t} = \rho^k\,\hat{\pi}_t, \qquad k \geq 1. \tag{36}$$

These beliefs satisfy the two properties above, as the Kalman filter yields that the forecasts are conditional expectations under the PLM.

Then, by (36), the inflation sum in (34) is a geometric series in ρ/μ_2, that yields

$$\sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[\pi_{t+k}] = \frac{\rho}{\mu_2 - \rho}\,\hat{\pi}_t, \tag{37}$$

which converges since $\rho < \mu_2$ is satisfied. In fact, the PLM is assumed to be stationary hence $\rho < 1$, which is satisfied for every Savers' estimation in Section 3.3, and the saddle condition ensures $1 < \mu_2$.

Nominal-rate expectations. The Saver's consumption function requires an assumption on how expectations of the nominal rate are formed. The next subsection considers the monetary policy rule (50): $i_t = \phi_\pi \pi_t$. In the spirit of Farhi and Werning (2019), I assume that current

inflation and the current nominal rate are treated asymmetrically: the current nominal rate is observed by Savers (as in the baseline of Gabaix, 2020), while current inflation is known only through the nowcast under the PLM.[21] Concerning the future nominal rates, following Eusepi and Preston (2010), I assume that the monetary policy rule is communicated to Savers, who apply it to their own inflation forecasts. I refer to that as "rule-aware" beliefs.[22]

Therefore one has: $\hat{E}_t[i_{t+j}] = \phi_\pi \hat{E}_t[\pi_{t+j}] = \phi_\pi \rho^j \hat{\pi}_t$ for $j \geq 1$. Separating the term $k = 1$, the nominal-rate sum is then

$$\sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[i_{t+k-1}] = \frac{i_t}{\mu_2} + \frac{\phi_\pi}{\mu_2} \frac{\rho}{\mu_2 - \rho} \hat{\pi}_t. \tag{38}$$

Combining (37) and (38), the discounted sum of expected real rates, which I refer to as the perceived real-rate path, is

$$\sum_{k=1}^{\infty} \mu_2^{-k} \hat{E}_t[\tilde{r}_{t+k}] = \frac{i_t}{\mu_2} + \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} \hat{\pi}_t. \tag{39}$$

A higher nowcast raises the perceived real-rate path if $\phi_\pi > \mu_2$, because Savers then expect the policy response to be stronger than the higher inflation.

Zero liquidity. Bonds are in zero net supply and, in the steady state considered here, as in the baseline of Bilbiie (2025), there is zero liquidity, $\tilde{Z}^S = 0$. The steady-state budget constraints then give $C^S = \tilde{Y}^S$ and $C^H = \tilde{Y}^H$. In the consumption functions (32) and (33), the income shares are then equal to one, the beginning-of-period wealth terms vanish, and the coefficient on expected real rates reduces to $-1/(\omega\sigma)$.[23]

Overall, substituting the perceived real-rate path (39) and FIRE income expectations into the consumption functions of Proposition 1 with zero liquidity gives the following result.

Corollary 1 (Consumption functions under the perceived law of motion)**.** *With zero liquidity, FIRE income expectations, the PLM* (35)–(36) *with* $\rho < \mu_2$*, and rule-aware nominal-rate beliefs, the consumption functions are, if the household is in the Saver state:*

$$c_t^S = \left(1 - \frac{\mu_1}{\omega}\beta\right) \sum_{k=0}^{\infty} \mu_2^{-k} \mathbb{E}_t\left[\tilde{y}_{t+k}^S\right] + \frac{1-\omega}{\omega} \sum_{k=1}^{\infty} \mu_2^{-k} \mathbb{E}_t\left[\tilde{y}_{t+k}^H\right] - \frac{1}{\omega\sigma}\left[\frac{i_t}{\mu_2} + \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} \hat{\pi}_t\right], \tag{40}$$

[21] In other words, Savers do not back out current inflation from the observed nominal rate through the policy rule. Doing so would make their nowcast equal to current inflation, with a Kalman gain of one, whereas the estimated gain of Savers is $\mathbf{K} = 0.48$, indicating that the information households effectively process about current inflation is far from complete. This is also consistent with the evidence that household expectations of interest rates respond to monetary policy announcements, while their inflation expectations barely do (De Fiore et al., 2022).

[22] Some survey evidence suggests that households' expectations of interest rates co-move with their inflation expectations in the direction implied by a Taylor rule, see for example Carvalho and Nechio (2014), Dräger et al. (2016), and D'Acunto et al. (2025), as well as Bauer et al. (2024) for professional forecasters.

[23] The roots μ_1 and μ_2 do not depend on the level of liquidity, since $\mathcal{A}_\zeta$ in (29) contains only s, h, β, ζ, and ω (Appendix C.3).

with $\hat{\pi}_t$ given by (35), and if the household is in the Hand-to-Mouth state:

$$c_t^H = \tilde{y}_t^H. \tag{41}$$

The belief parameters appear in two places only. The perceived persistence ρ sets the loading of Saver consumption on the nowcast, $\frac{\rho\,(\phi_\pi - \mu_2)}{\omega\sigma\,\mu_2(\mu_2-\rho)}$, whose absolute value increases with ρ through $\frac{\rho}{\mu_2-\rho}$: the more persistent Savers perceive inflation to be, the further a given nowcast is projected into the future path of real rates. In addition, ρ and $\mathbf{K}$ together govern how the nowcast moves with actual inflation, through (35): on impact it moves by $\mathbf{K}$, and it then decays at rate $\rho(1-\mathbf{K})$ in the absence of further news. Both are recovered from the estimation of Section 3.3.

4.4. Income Closure and Aggregate Demand

The functions (40) and (41) give consumption for given sequences of income and interest rates, under assumptions on beliefs. The closure adopted here, that corresponds to thebaseline of Bilbiie (2025), determines income in equilibrium under three additional assumptions.

1. Equal steady-state consumption. The benchmark is the symmetric steady-state, where $\zeta = C^S/C^H = 1$, so that $\omega = s$. Together with zero liquidity, that yields that the two groups also have the same steady-state income, $C^S = \tilde{Y}^S$ and $C^H = \tilde{Y}^H$, and that

$$c_t^S = \tilde{y}_t^S, \qquad c_t^H = \tilde{y}_t^H. \tag{42}$$

2. Goods market clearing. Aggregate output equals aggregate consumption, and since the steady-state levels of the two groups are equal, that yields:

$$y_t = c_t = \lambda\, c_t^H + (1-\lambda)\, c_t^S, \tag{43}$$

3. Income distribution. HtM income is related to aggregate income with elasticity χ:

$$\tilde{y}_t^H = \chi\, y_t. \tag{44}$$

Solving the income identity in (43) for Saver income yields then

$$\tilde{y}_t^S = \frac{y_t - \lambda\chi\, y_t}{1-\lambda} = \gamma_S\, y_t, \qquad \gamma_S \equiv \frac{1-\lambda\chi}{1-\lambda}. \tag{45}$$

With $\chi > 1$, HtM income rises more than one-for-one with output, and income inequality is countercyclical (i.e. $\partial\,(\tilde{y}_t^S - \tilde{y}_t^H)/\partial\, y_t < 0$). By (42), $c_t^S = \gamma_S y_t$ and $c_t^H = \chi y_t$, and goods market clearing holds since $\lambda\chi + (1-\lambda)\gamma_S = 1$.

Substitution. As a result, replacing ω by s, c_t^S and $\tilde{y}_{t+k}^S$ by $\gamma_S y_t$ and $\gamma_S y_{t+k}$, and $\tilde{y}_{t+k}^H$ by χy_{t+k}, in the consumption function of a Saver (40) gives

$$\gamma_S\, y_t = \left(1 - \frac{\mu_1}{s}\beta\right)\gamma_S \sum_{k=0}^{\infty} \mu_2^{-k}\, \mathbb{E}_t[y_{t+k}] + \frac{1-s}{s}\,\chi \sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_t[y_{t+k}] - \frac{1}{s\sigma}\left[\frac{i_t}{\mu_2} + \frac{\rho\,(\phi_\pi - \mu_2)}{\mu_2\,(\mu_2 - \rho)}\,\hat{\pi}_t\right], \tag{46}$$

and rearranging gives the following result.

Corollary 2 (Aggregate demand)**.** *Under the closure 1–3, the consumption function of a Saver* (40) *holds in equilibrium if and only if aggregate demand satisfies*

$$\gamma_S\,\frac{\mu_1\beta}{s}\,y_t = \left[\left(1 - \frac{\mu_1\beta}{s}\right)\gamma_S + \frac{1-s}{s}\,\chi\right]\sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_t[y_{t+k}] - \frac{1}{s\sigma}\left[\frac{i_t}{\mu_2} + \frac{\rho\,(\phi_\pi - \mu_2)}{\mu_2\,(\mu_2 - \rho)}\,\hat{\pi}_t\right]. \tag{47}$$

Where current output depends on the expected path of future output, discounted at μ_2, under FIRE. Its weight has two parts: $\left(1 - \frac{\mu_1\beta}{s}\right)\gamma_S$ comes from the future income of the Saver, and $\frac{1-s}{s}\chi$ from the income the Saver would receive if the transition in the HtM state happens. With $\chi > 1$, higher expected output raises expected HtM income more than proportionally, weakens the precautionary-saving motive, and raises demand today.[24]

Current output also depends on the perceived path of real rates, that enters through the current nominal rate i_t and the nowcast of Savers $\hat{\pi}_t$. This is the only instance in which the inflation beliefs of Section 3 appear: ρ in the loading of the nowcast, and ρ and $\mathbf{K}$ in the law of motion of the nowcast (35).

Remark: Relation to THANK with FIRE. It is shown in Appendix D that quasi-differencing the aggregate-demand equation (47) gives the aggregate Euler equation under subjective expectations:

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left(i_t - \hat{E}_t[\pi_{t+1}] + \left(\hat{E}_t - \mathbb{E}_t\right)\left[\sum_{k=1}^{\infty} \mu_2^{-k}\,\hat{E}_{t+1}[i_{t+k} - \pi_{t+1+k}]\right]\right). \tag{48}$$

If Savers have rational expectations about the real rate, then $\hat{E} = \mathbb{E}$ yields

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left(i_t - \mathbb{E}_t[\pi_{t+1}]\right), \qquad \delta \equiv 1 + (\chi - 1)\frac{1-s}{1-\lambda\chi}. \tag{49}$$

That is exactly the Euler equation of Bilbiie (2025). Equation (48) illustrates the point made by Preston (2005), that even in the equilibrium with zero liquidity, the long-horizon expectations matter for aggregate dynamics, and the effects of deviations from FIRE cannot be characterized simply by the one-period-ahead expectation operator in the Euler equation.

[24] This is the compounding mechanism of THANK that drives aggregate demand amplification.

Supply and policy. Overall, the economy is described by the aggregate-demand equation (47) and the law of motion of the nowcast (35). Output, inflation, and the nominal rate are determined with a New Keynesian Phillips curve and a simple Taylor rule of the form

$$\pi_t = \beta\, \mathbb{E}_t[\pi_{t+1}] + \kappa\, y_t + u_t, \qquad i_t = \phi_\pi\, \pi_t, \tag{50}$$

respectively, where the cost-push shock u_t follows an AR(1) process

$$u_t = \rho_u\, u_{t-1} + \varepsilon_t^u, \qquad \varepsilon_t^u \sim \mathcal{N}(0, \sigma_u^2). \tag{51}$$

In order to isolate the effects of the inflation expectations of the Savers in aggregate demand (47), I assume that the expectations in the Phillips curve are those of price setters, for whom I retain FIRE.[25]

4.5. Equilibrium in THANK under Subjective Inflation Expectations

Definition 1 (Equilibrium)**.** Assume that Savers hold (i) the PLM of inflation characterized by the perceived persistence ρ together with the Kalman gain $\mathbf{K}$, (ii) FIRE about income, and (iii) rule-aware beliefs about the nominal rate, and that price setters have FIRE. Given the initial nowcast $\hat{\pi}_{-1}$ and the process of the cost-push shock (51), an equilibrium is a collection of stochastic processes for consumption, $\{c_t^S\}$ and $\{c_t^H\}$, incomes, $\{\tilde{y}_t^S\}$ and $\{\tilde{y}_t^H\}$, output, $\{y_t\}$, inflation, $\{\pi_t\}$, the nominal rate, $\{i_t\}$, and the inflation nowcast of Savers, $\{\hat{\pi}_t\}$, such that:

1. given $\{\tilde{y}_t^S\}$, $\{\tilde{y}_t^H\}$, $\{i_t\}$ and $\{\pi_t\}$, Savers update their nowcast according to (35) and choose consumption optimally given their beliefs, so that by Corollary 1 their consumption is given by (40), and HtM households consume according to (41), and

2. in every period $t = 0, 1, 2, \ldots$ and in every state of the world, the bond market clears with zero bonds, so that each group consumes its income (42), the goods market clears (43), incomes are given by (44) and (45), inflation is given by the Phillips curve and the nominal rate by the Taylor rule (50).

By Corollary 2, under the closure 1–3 the consumption function of a Saver holds if and only if aggregate demand satisfies (47). An equilibrium is therefore characterized by stochastic processes $\{y_t\}$, $\{\pi_t\}$, $\{i_t\}$ and $\{\hat{\pi}_t\}$ that satisfy (35), (47), and (50).

Solution. The equilibrium is solved by undetermined coefficients in Appendix E, after aggregate demand (47) is rewritten as the aggregate Euler equation (48). The state in period t consists of the nowcast of the previous period, $\hat{\pi}_{t-1}$, and the cost-push shock, u_t. Output y_t and inflation π_t are guessed to be linear in both.

Hence the system has two forward-looking variables, output and inflation, and one predetermined variable, the inflation nowcast (apart from the exogenous shock). By Blanchard and Kahn (1980), there is a unique bounded equilibrium if and only if exactly one root lies

[25] For example, in Bilbiie (2025) the Phillips curve follows from firms' optimal pricing under Rotemberg costs.

inside the unit circle. It is verified numerically that this condition is satisfied with a standard $\phi_\pi = 1.5$ in the Taylor rule for every perceived persistence $\rho < 1$, and for every Kalman gain $\mathbf{K} \in (0, 1]$.

The Actual Law of Motion (ALM) of Inflation. The ALM of inflation is obtained by combining the solution for inflation with the law of motion of the nowcast (35). It is shown in Appendix E.3 that the resulting ALM takes the form of the ARMA(2,1) process:

$$\pi_t = \left[\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi + \rho_u\right] \pi_{t-1} - \left[\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi\right] \rho_u \, \pi_{t-2} + \vartheta_u^\pi \, \varepsilon_t^u - \vartheta_u^\pi \, \rho(1-\mathbf{K}) \, \varepsilon_{t-1}^u, \quad (52)$$

where ϑ_π^π and ϑ_u^π are the coefficients of inflation on the lagged nowcast and on the shock (see Appendix E.2).

With $\mathbf{K} = 1$, one has $\vartheta_\pi^\pi = 0$, and the ALM is the AR(1) process of the cost-push shock u_t (see Appendix E.3). If in addition $\rho = \rho_u$, the PLM coincides with the ALM, which is the FIRE equilibrium of THANK. For $\mathbf{K} < 1$, the ALM generally differs from the PLM, and Section 5.1 asks which perceived persistence is consistent with it.

5. Whose Expectations Matter for a Cost-Push Shock

Section 4 solved THANK for given beliefs of Savers, summarized by the parameters $(\rho, \mathbf{K})$. This section uses the parameters values estimated in Section 3 and evaluates their relevance for the propagation of a cost-push shock.

5.1. Calibration and Beliefs

Calibration. Table 5 reports the calibration at quarterly frequency. The Taylor-rule coefficient is $\phi_\pi = 1.5$. The cost-push shock has size 0.25, in quarterly percentage points, and persistence $\rho_u = 0.5$, thus matching the actual persistence of the headline CPI inflation that households forecast in the SCE, 0.48 over the full sample and 0.53 during the inflation surge (see Table 4).

Beliefs. Savers hold the beliefs estimated during the inflation surge, $\rho = 0.843$ and $\mathbf{K} = 0.480$. As a counterfactual, Savers are instead given the beliefs estimated for all households over the same period, $\rho = 0.934$ and $\mathbf{K} = 0.524$, corresponing to the parameters values that a model disciplined by population-wide survey data would use. Hence the comparison measures how much the composition of the surveyed population matters for aggregate dynamics in the THANK model.

Perceived and actual persistence. Savers perceive inflation to be much more persistent than the inflation that the model generates. Given $(\rho, \mathbf{K})$, the ALM of inflation follows the ARMA(2,1) process (52). Its first-order autocorrelation, denoted $T(\rho)$, is the persistence that Savers would estimate from the data generated by the model (see Appendix E.3). With the beliefs of Savers, $T(0.843) = 0.49$, and with the beliefs of all households, $T(0.934) = 0.47$. The gap documented in Table 4 is therefore also present in the model: actual inflation has a persistence of about 0.5, well below the persistence that households perceive.

Table 5: Calibration.

Parameter	Description	Value	Source
s	Probability of remaining a Saver	0.96	Bilbiie (2025)
λ	Share of HtM households	0.33	Bilbiie (2025)
χ	Elasticity of HtM income to output	1.4	Bilbiie (2025)
β	Discount factor	0.99	Quarterly frequency
σ	Inverse of the intertemporal elasticity	1	Bilbiie (2025)
κ	Slope of the Phillips curve	0.02	Bilbiie (2025)
ζ	Steady-state consumption ratio C^S/C^H	1	Closure, Section 4.4
ϕ_π	Taylor-rule coefficient	1.5	
ρ_u	Persistence of the cost-push shock	0.5	Table 4
Implied			
h	Probability of remaining HtM	0.919	Stationarity, (17)
ω	Euler-equation weight of the Saver state	0.96	$\omega = s$ with $\zeta = 1$
μ_1, μ_2	Roots of (29)	0.939, 1.076	
γ_S	Elasticity of Saver income to output	0.803	(45)
δ	Compounding in the aggregate Euler equation	1.030	(49)
Beliefs of Savers, $(\rho, \mathbf{K})$			
	Estimated for Savers	0.843, 0.480	Table 4
	Estimated for all households	0.934, 0.524	Table 4
	Fixed point of the T-map	0.499, 0.480	Appendix E.3

Notes: Quarterly calibration. Bilbiie (2025) calibrates $s = 0.82$ at annual frequency and reports that it corresponds to $s = 0.96$ at quarterly frequency. The values of λ and χ are from the same calibration, and those of σ and κ from his determinacy analysis. Beliefs: OLS estimates for the inflation surge. Fixed point: perceived persistence ρ^* equal to the first-order autocorrelation of the inflation generated by the model, with the Kalman gain of Savers.

The natural benchmark is the perceived persistence that would be consistent with the model, that is the fixed point between the ALM and the PLM, denoted $T(\rho^*) = \rho^*$. With the Kalman gain of Savers, $\rho^* = 0.499$ (see Figure 22).[26] This benchmark keeps the Kalman gain of Savers and removes only their misperception of persistence. As a result, the comparison between FIRE and the fixed point isolates the noisy information of Savers, while the comparison between the fixed point and the estimated beliefs isolates misperceived persistence.

Overall, the next section compares four cases: FIRE, the PLM-ALM fixed point, the beliefs of Savers, and the beliefs of all households, which Table 5 collects.

5.2. The Cost-Push Shock

Figure 5 shows the responses to the cost-push shock, and Table 6 summarizes the output responses.

FIRE and the fixed point. Under FIRE, output falls by 1.15% on impact and by 2.31% cumulatively over twelve quarters. At the PLM-ALM fixed point, the contraction is slightly smaller, 1.05% on impact and 2.16% cumulatively, 6% less than under FIRE. Noisy information alone

[26] The fixed point hardly depends on the Kalman gain. It lies between 0.499 and 0.500 for every $\mathbf{K}$ between 0.1 and 1, and the T-map is almost flat, because the autoregressive and moving-average roots of (52) nearly cancel (0.425 and 0.438 with the beliefs of Savers). Inflation is set by price setters with FIRE along a Phillips curve that is rather flat with $\kappa = 0.02$, so that the beliefs of Savers feed back only weakly into inflation, which stays close to the AR(1) process of the cost-push shock. The fixed point is E-stable, since $T'(\rho^*) = -0.009 < 1$.

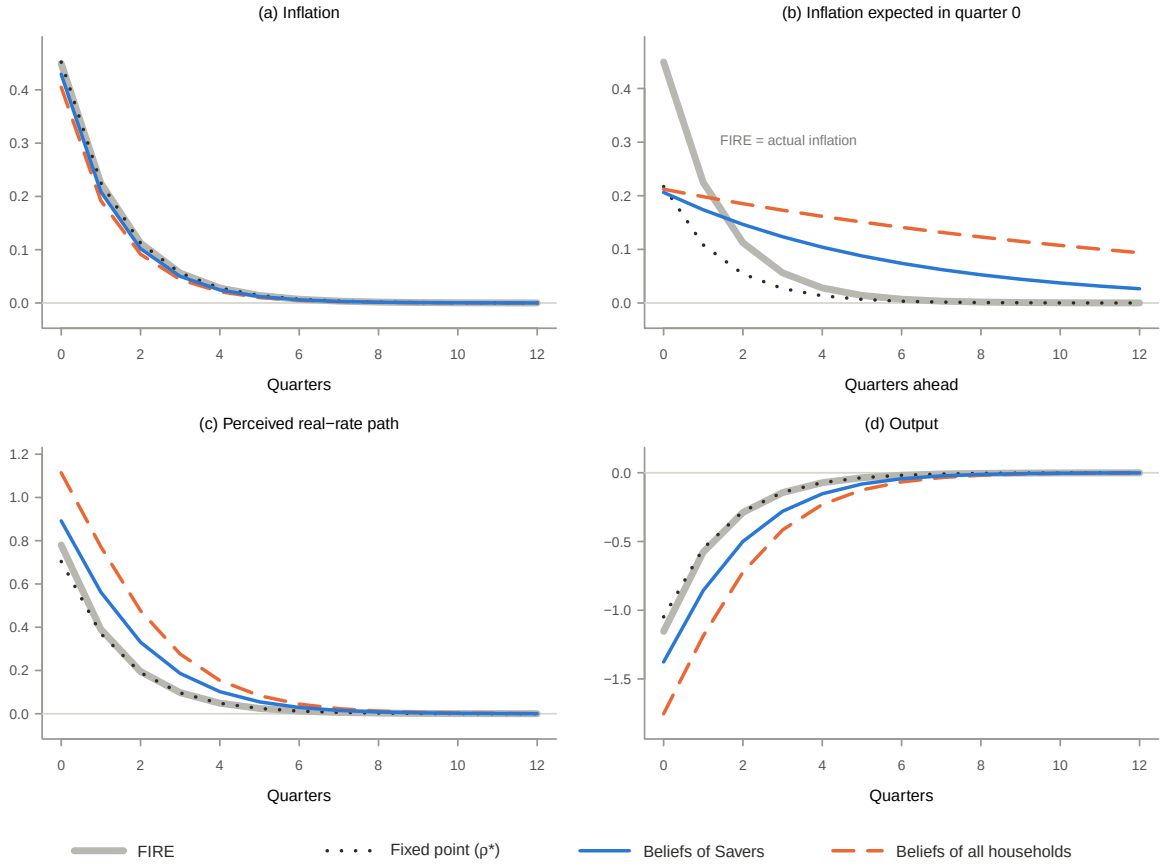


Figure 5: Impulse responses to a cost-push shock.

Notes: Cost-push shock of 0.25 with persistence 0.5. $\phi_\pi = 1.5$, rule-aware nominal-rate beliefs, price setters with FIRE. Beliefs of Savers: $\rho = 0.843$, $\mathbf{K} = 0.480$. Beliefs of all households: $\rho = 0.934$, $\mathbf{K} = 0.524$ (OLS estimates for the inflation surge, Table 4). Fixed point: $\rho = \rho^* = 0.499$ with the Kalman gain of Savers (Appendix E.3). Panel (b): inflation expected by Savers in the quarter of the shock, $\hat{E}_0[\pi_k] = \rho^k \hat{\pi}_0$, for k quarters ahead. Under FIRE, expected inflation equals actual inflation. Panels (a), (c), and (d): responses over time. Output in percent deviations from steady state, inflation and the perceived real-rate path (39) in quarterly percentage points (not annualized). Under FIRE, the perceived real-rate path is computed with rational expectations.

Table 6: Output response to a cost-push shock.

Beliefs	ρ	$\mathbf{K}$	Impact	Cumulative	Channel
FIRE			−1.15	−2.31	
Fixed point	0.499	0.480	−1.05	−2.16	+0.14
Savers	0.843	0.480	−1.38	−3.33	−1.03
All households	0.934	0.524	−1.75	−4.57	−2.26
Savers, with ρ of all households	0.934	0.480	−1.70	−4.54	−2.23
Savers, with $\mathbf{K}$ of all households	0.843	0.524	−1.41	−3.36	−1.06
Savers, with $\mathbf{K} = 1$	0.843	1	−1.77	−3.54	−1.23

Notes: Output response, in percent of steady-state quarterly output, to a cost-push shock of 0.25 (quarterly rate) with persistence 0.5. Impact: response in the quarter of the shock. Cumulative: sum of the responses over twelve quarters, starting with the quarter of the shock. Channel: cumulative response minus the FIRE response. $\phi_\pi = 1.5$, rule-aware nominal-rate beliefs, price setters with FIRE. Beliefs: OLS estimates for the inflation surge (Table 4). Fixed point: $\rho = \rho^*$ with the Kalman gain of Savers (Table 5).

therefore dampens the effect of the shock on demand: as Savers reflect only part of the rise in inflation into their nowcast (Figure 5, panel (b)), they expect a smaller rise in future policy rates, which mitigates the increase in the perceived real-rate path.

The subjective expectations of Savers. With the beliefs of Savers, output falls by 1.38% on impact and by 3.33% cumulatively, and the overall contraction is about 45% larger than under FIRE. Relative to the fixed point, which keeps the Kalman gain of Savers, the cumulative contraction is 54% larger, and this difference comes from misperceived persistence alone. In the quarter of the shock, the nowcast of Savers is 0.21, about half of actual inflation, but Savers perceive inflation as an AR(1) process with persistence 0.84, and four quarters ahead they still expect inflation of 0.10, against 0.03 under FIRE. With rule-aware beliefs, they therefore expect the policy rate to remain high for longer than it actually does. The perceived real-rate path rises to 0.89 on impact, against 0.78 under FIRE and 0.70 at the fixed point, and stays higher in the following quarters (panel (c)), leading to a stronger output contraction.

The subjective expectations of all households. With the beliefs of all households, output falls by 1.75% on impact and by 4.57% cumulatively, almost twice as much as under FIRE. Measured as the gap to FIRE in the cumulative output response, the expectations channel is -2.26 with the beliefs of all households, against -1.03 with the beliefs of Savers. Using population-wide beliefs therefore makes the expectations channel 2.2 times as strong as it is with the beliefs of the households whose expectations enter the intertemporal margin of aggregate demand.

The nowcast responds almost identically with the two sets of beliefs, since their Kalman gains are similar. What differs is how far the nowcast is projected into the future: with the beliefs of all households, the same nowcast implies a larger and more persistent rise in the perceived real-rate path, 1.11 against 0.89 on impact.

Inflation and inflation expectations. The Savers' expectations only affect inflation through aggregate demand. Given a relatively flat Phillips curve, $\kappa = 0.02$, the larger contraction in output translates into only slightly lower inflation (panel (a)). Inflation rises by 0.45 on impact under FIRE, by 0.43 with the beliefs of Savers, and by 0.41 with the beliefs of all households, and in every case it decays at about the rate of the shock.

Panel (b) shows that the model also reproduces the pattern of forecast errors documented in Section 2.2 (Figure 2). In the quarter of the shock, the nowcast of Savers reflects only part of the rise in inflation. Since Savers perceive inflation as more persistent than it is, however, they expect it to remain high after it has fallen, and their forecasts exceed realized inflation for several quarters. With the beliefs of all households, these forecast errors are larger and more persistent, because the population also includes HtM households, whose perceived persistence is higher than that of Savers. As in the data, the expectations of Savers are therefore closer to realized inflation than those of the entire population.

Table 7: The perceived real-rate path on impact.

	Current rate i_0/μ_2	Loading $\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}$	Nowcast $\hat{\pi}_0$	Expected future rates Loading × nowcast	Perceived real-rate path Total
FIRE	0.63	0.34	0.45	0.15	0.78
Fixed point	0.63	0.34	0.22	0.07	0.70
Savers	0.60	1.42	0.21	0.29	0.89
All households	0.56	2.59	0.21	0.55	1.11

Notes: Decomposition of the perceived real-rate path (39) in the quarter of the cost-push shock, in quarterly percentage points. The current nominal rate is $i_0 = \phi_\pi \pi_0$. Under FIRE, the nowcast is actual inflation and the loading is evaluated at the persistence of the shock, $\rho_u = 0.5$. Beliefs as in Table 6.

5.3. Quantitative Mechanisms

By the aggregate-demand equation (47), the inflation beliefs of Savers affect output only through the perceived real-rate path (39). In the quarter of the shock, the path is the sum of two terms, the discounted current nominal rate , i_0/μ_2, and the expected future rates, which are the loading $\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}$ times the nowcast $\hat{\pi}_0$. Table 7 reports these terms in the four cases.

Two opposing forces. The two belief parameters move the perceived real-rate path in opposite directions. The lower Kalman gain dampens it. With noisy information, the nowcast results in only about half of the rise in inflation on impact, 0.21 to 0.22 against 0.45 under FIRE, and the expected future rates are smaller for a given loading. At the fixed point, whose loading is the same as under FIRE, the expected future rates channel then falls from 0.15 to 0.07, and the perceived real-rate path from 0.78 to 0.70.

Over-perceived persistence on the other hand amplifies the real-rate path. With the beliefs of Savers, the loading is 1.42, about four times its value under FIRE, 0.34, which more than offsets the smaller nowcast, and the expected future rates rise to 0.29, against 0.15 under FIRE. With the beliefs of all households, the loading is 2.59, and the expected future rates channel rises to 0.55.

The loading factor, $\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}$, rises steeply with perceived persistence, because the term $\rho/(\mu_2-\rho)$ increases sharply as perceived persistence approaches μ_2. In THANK, the transitions between the Saver and the HtM state make Savers discount the future more than in a representative-agent model, $\mu_2 = 1.076$ against $1/\beta = 1.010$ in the calibration, and this discounting limits how far the nowcast is projected into the future. In a representative-agent or two-agent framework, where $\mu_2 = 1/\beta$, the loading factor would be 2.45 with the beliefs of Savers and 5.97 with those of all households, and perceived persistence would weigh even more on the perceived real-rate path.

The current nominal rate contributes slightly less with subjective expectations: 0.60 and 0.56 against 0.63. Recalling that the current nominal rate follows from the Taylor rule, this difference is due to the fact that inflation is slightly lower in equilibrium (see Figure 5 panel (a)).

Overall, the amplification of the cost-push shock does not come from an underreaction of expectations to inflation news, which on its own dampens it, but from the over-perception of the persistence of inflation.

Attribution and estimation uncertainty. Table 6 also reports two hybrid cases that change one belief parameter at a time. Replacing only the perceived persistence of Savers with that of all households gives a cumulative response of −4.54, almost the same as with the beliefs of all households, −4.57. Replacing instead only the Kalman gain with that of all households gives −3.36, almost the same as with the beliefs of Savers alone, −3.33. Thus confirming that difference between the two sets of beliefs comes almost entirely from perceived persistence.

These results rest on point estimates of the expectation parameters. Appendix F.2 draws the parameters of Savers and of all households jointly from their estimated distribution and recomputes the impulse responses for each draw. In 97% of the draws, the expectations channel is stronger with the beliefs of all households than with those of Savers, and the ratio of the two channels has a median of 2.2. The results also hold with the beliefs estimated over the full sample, 2015–2025, with which the expectations channel of all households is 2.7 times that of Savers (see Appendix F.1).

6. Conclusion

Whose inflation expectations drive aggregate demand? In a heterogeneous-agent setting, the answer is the expectations of the unconstrained households that can smooth consumption over time. This paper shows that this answer matters quantitatively, because the inflation expectations of these households are different from those of the entire population.

Using the Survey of Consumer Expectations, I document that unconstrained households (Savers) have lower inflation expectations than constrained households (Hand-to-Mouth, HtM), that they make smaller forecast errors, and that their expectations are less volatile and less dispersed in the cross-section. A noisy-information model estimated on the panel dimension of the survey shows that, during the 2021–2023 inflation surge, Savers perceive inflation as less persistent than HtM households, and have a somewhat smaller Kalman gain, so that they react less to inflation news.

Introducing these beliefs into a tractable heterogeneous-agent New Keynesian (THANK) model quantifies their aggregate implications. With the beliefs of Savers, the cumulative output contraction after a cost-push shock is 45% larger than under FIRE. With the beliefs of all households, which a model disciplined by population-wide survey data would use, the contraction is almost twice as large as under FIRE, and the expectations channel is 2.2 times as strong as with the beliefs of Savers alone. The amplification comes from the over-perception of the persistence of inflation rather than from the underreaction of expectations to inflation news, which on its own dampens the effect of the shock. Since Savers expect the central bank to respond to inflation, the more persistent they perceive inflation to be, the higher the path of real rates they expect, and the lower their demand.

These results suggest that whose expectations are used to discipline aggregate demand matters. For monetary policy, models with FIRE may understate the effect of a cost-push shock on demand, while models that calibrate household expectations with population-wide survey data may overstate it. Distinguishing between the expectations of Savers and HtM households could then also be informative for central banks that monitor household inflation expectations.

Several interesting questions remain open for future research. The model considered in this paper keeps price setters at FIRE in order to isolate the expectations of Savers in aggregate demand, and only considers a cost-push shock. A natural next step is to ask whose expectations enter the Phillips curve, and how the composition of beliefs on the supply side interacts with the one documented here for demand. Extending the analysis to other shocks is also important, since for demand shocks the same beliefs would stabilize output rather than amplify its response. In addition, the paper takes the Saver or HtM status of households as given, and measures expectations conditional on it. It will be interesting to study the connection between the heterogeneity in expectations and the source of heterogeneity in households' constrained status.

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Appendix A. Facts: Robustness

A.1. Figures

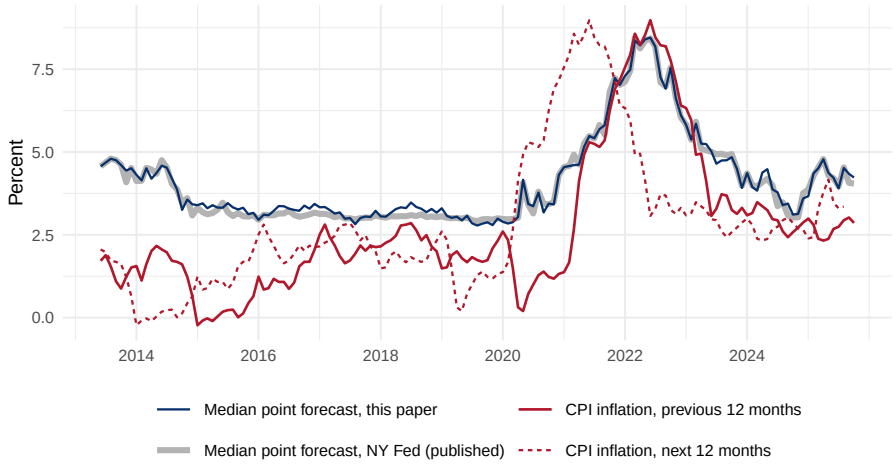


Figure 6: Average inflation expectations for the whole sample.

Notes: The average inflation expectation is derived by treating each point prediction as grouped data and taking a linearly interpolated median with survey weights, as described in the main text (dark blue). The median point prediction published by the Federal Reserve Bank of New York is the thick grey line. CPI inflation over the twelve months before the survey month is the solid red line, and realized CPI inflation over the twelve months that follow it is the dashed red line. All series are plotted at the survey month.

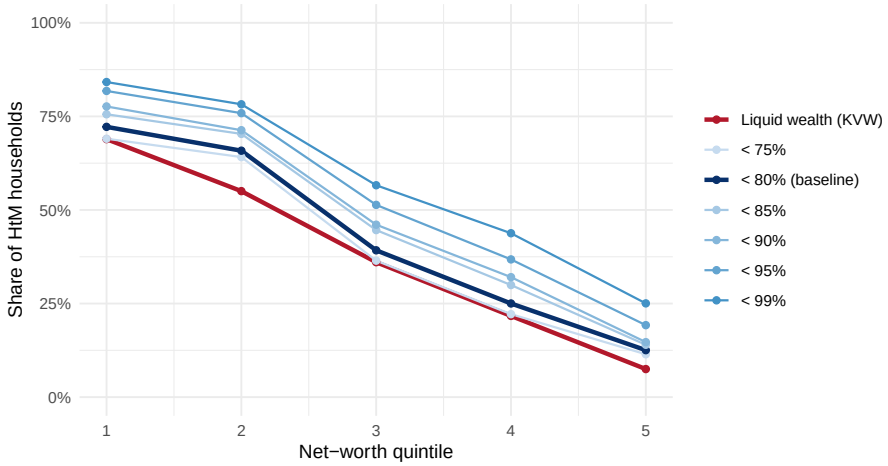


Figure 7: Distribution of Hand-to-Mouth households across wealth quintiles.

Notes: Share of HtM households by quintile of net worth, for households that answer both the Household Finance module (2015–2019) and, within six months of it, the Credit Access module; quintiles within each wave, with survey weights. HtM households identified with their liquid balance, following Kaplan and Violante (2014), and with the perceived ability to come up with $\$2,000$ if an unexpected need materializes within the following month, for thresholds between 75% and 99%; the baseline threshold is 80%.

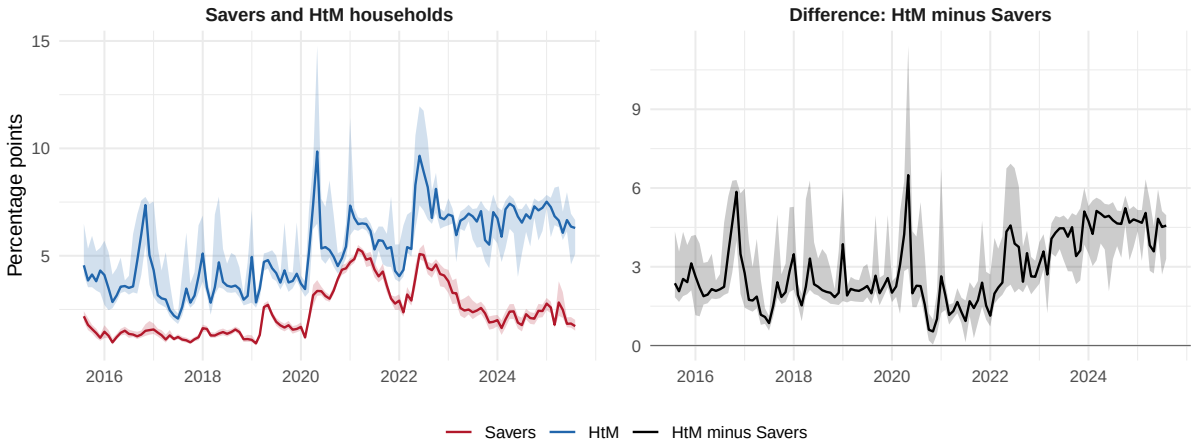


Figure 8: Inflation expectations absolute forecast errors of Savers and HtM households.

Notes: In the left panel, the Savers' absolute forecast errors are in red and HtM absolute forecast errors are in blue; the right panel shows the difference. Absolute forecast errors are the interpolated medians of the absolute difference between individual expectations and realized inflation over the following twelve months. The bands represent the 90% confidence interval.

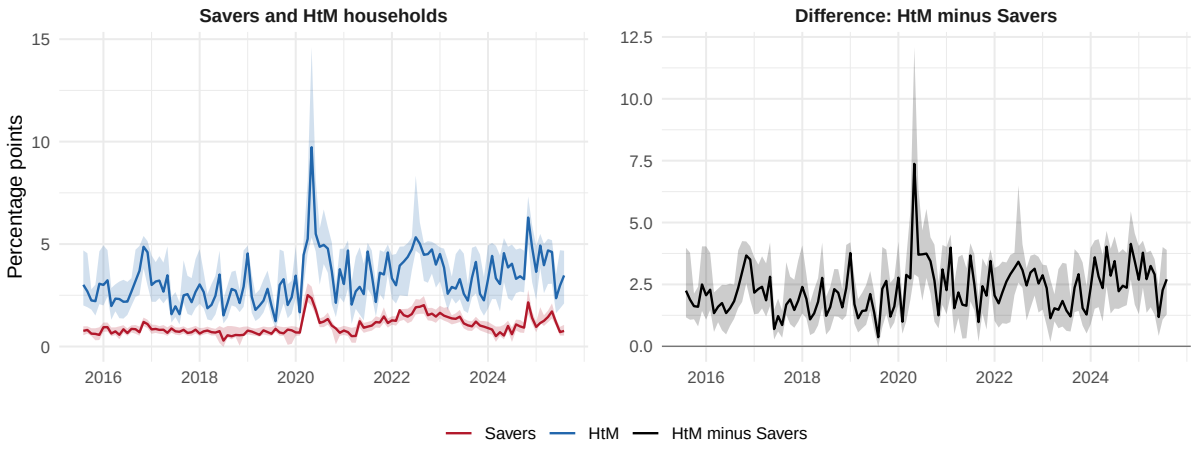


Figure 9: Month-to-month revisions of individual inflation expectations.

Notes: Interpolated median of the absolute change in a household's one-year-ahead inflation expectation from the previous month, for households that answer in two consecutive months. In the left panel, Savers are in red and HtM households in blue; the right panel shows the difference. The bands represent the 90% confidence interval.

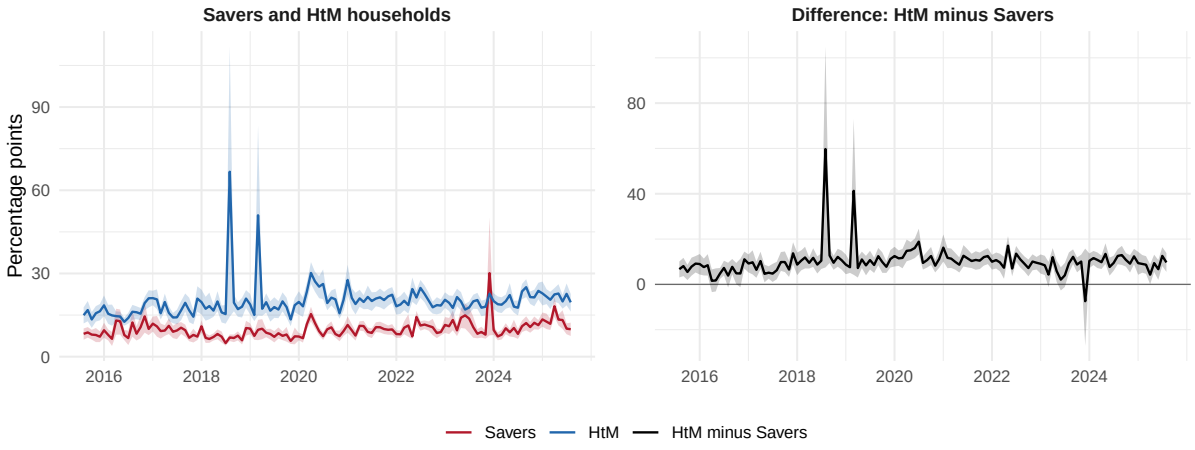


Figure 10: Dispersion of inflation expectations without winsorization.

Notes: Root-mean-square deviation from the group mean, computed on all point predictions (no winsorization). In the left panel, Saver dispersion is in red and HtM dispersion is in blue; the right panel shows the difference. The bands represent the 90% confidence interval.

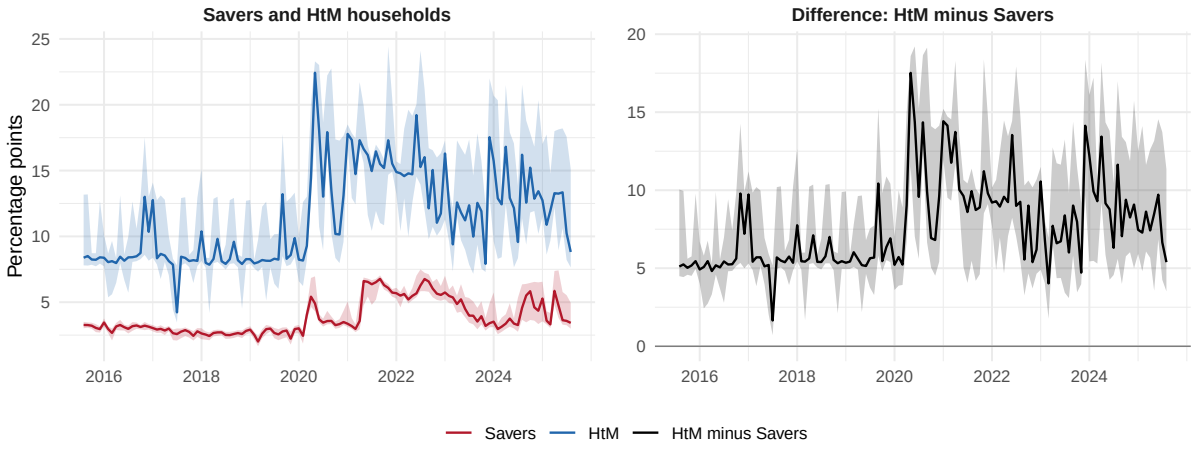


Figure 11: Interquartile range of inflation expectations.

Notes: Interquartile range of point predictions within each group, from interpolated quantiles. In the left panel, Savers are in red and HtM households in blue; the right panel shows the difference. The bands represent the 90% confidence interval.

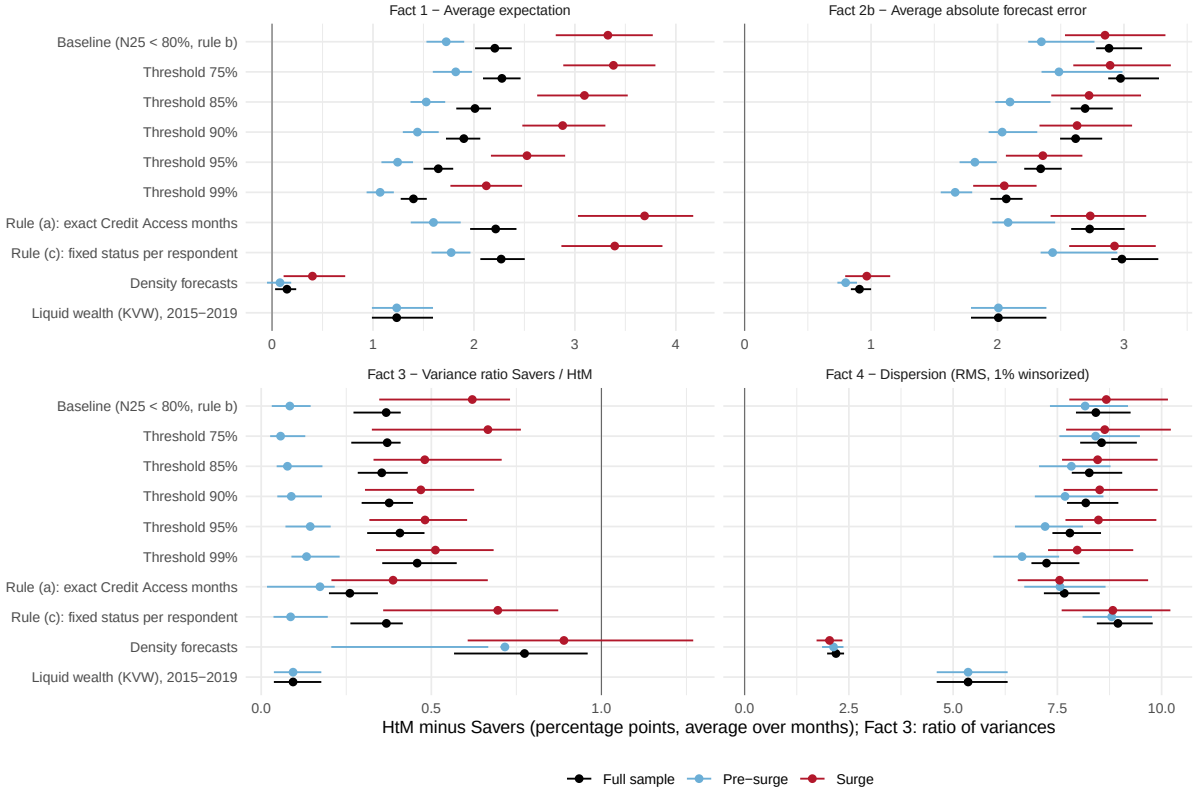


Figure 12: Robustness of the four facts to the Saver-HtM classification.

Notes: Differences between HtM households and Savers, averaged over the full sample (black), the pre-surge period (light blue) and the surge (red); for Fact 3, the ratio of the variances of the average expectation, Savers over HtM. Baseline: threshold of 80%, nearest wave of the Credit Access module within two months. Alternatives: thresholds between 75% and 99%; only the months in which the module is fielded; each household's most frequent status; the liquid-balance definition over 2015–2019; and, for comparison, the means of the probabilistic forecasts instead of point predictions. The bars represent the 90% confidence interval.

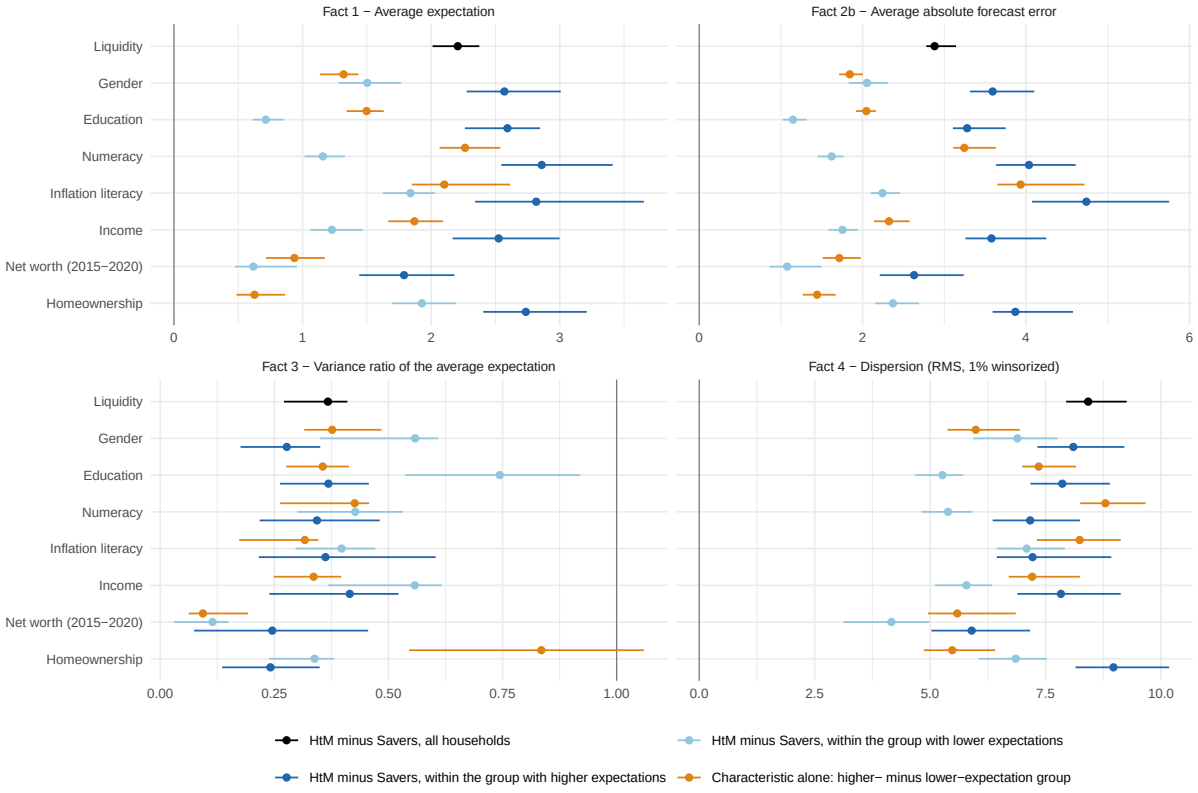


Figure 13: The four facts within groups of households with the same characteristics.

Notes: Differences averaged over the full sample, August 2015 to August 2025 (net worth: August 2015 to February 2020). Black: HtM households minus Savers, all households. Dark and light blue: HtM households minus Savers within the group with higher and the group with lower average inflation expectations of each characteristic (women and men; no college degree and college degree; low and high numeracy; without and with inflation literacy; household income below and above $\$50,000$; net worth below and above the median; not homeowners and homeowners). Orange: group with higher minus group with lower average expectations, regardless of HtM status. For Fact 3, ratios of the variances of the average expectation: Savers over HtM households (black and blue), and the group with lower over the group with higher average expectations (orange). The bars represent the 90% confidence interval.

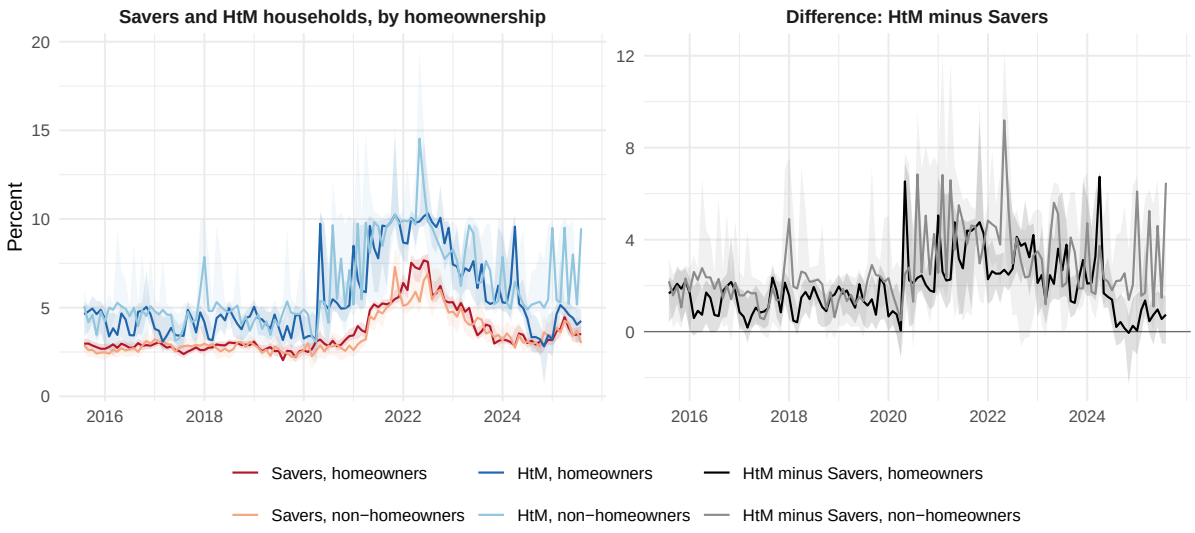


Figure 14: Fact 1 by homeownership: average inflation expectations.

Notes: In the left panel, Savers are in red and HtM households in blue, with darker colors for homeowners and lighter colors for households that do not own their primary residence. The right panel shows the difference between HtM households and Savers among homeowners (black) and among non-homeowners (grey). The bands represent the 90% confidence interval.

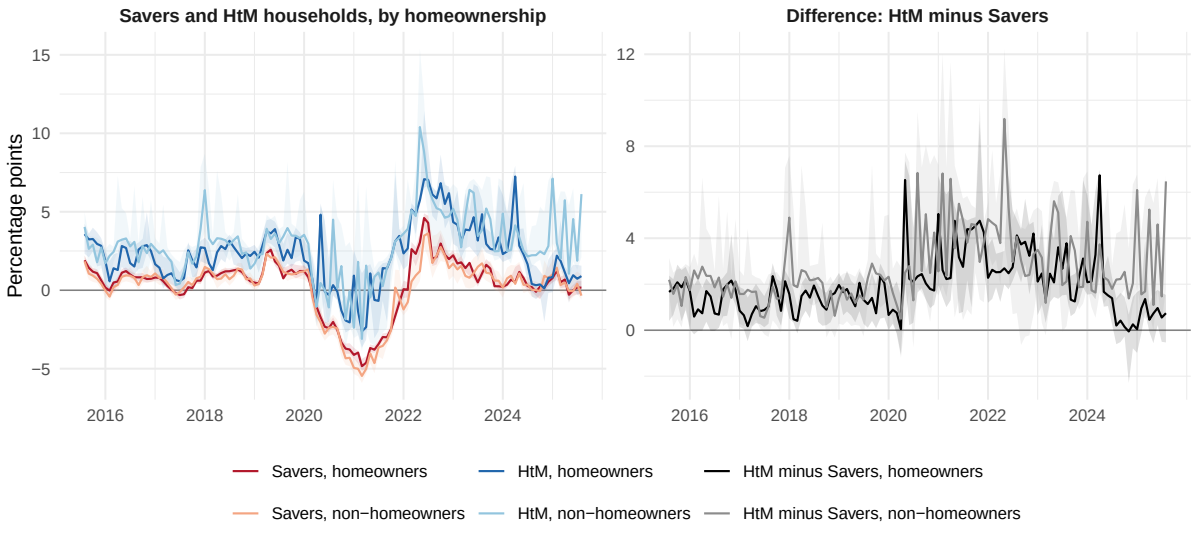


Figure 15: Fact 2a by homeownership: forecast errors.

Notes: In the left panel, Savers are in red and HtM households in blue, with darker colors for homeowners and lighter colors for households that do not own their primary residence. The right panel shows the difference between HtM households and Savers among homeowners (black) and among non-homeowners (grey). The bands represent the 90% confidence interval.

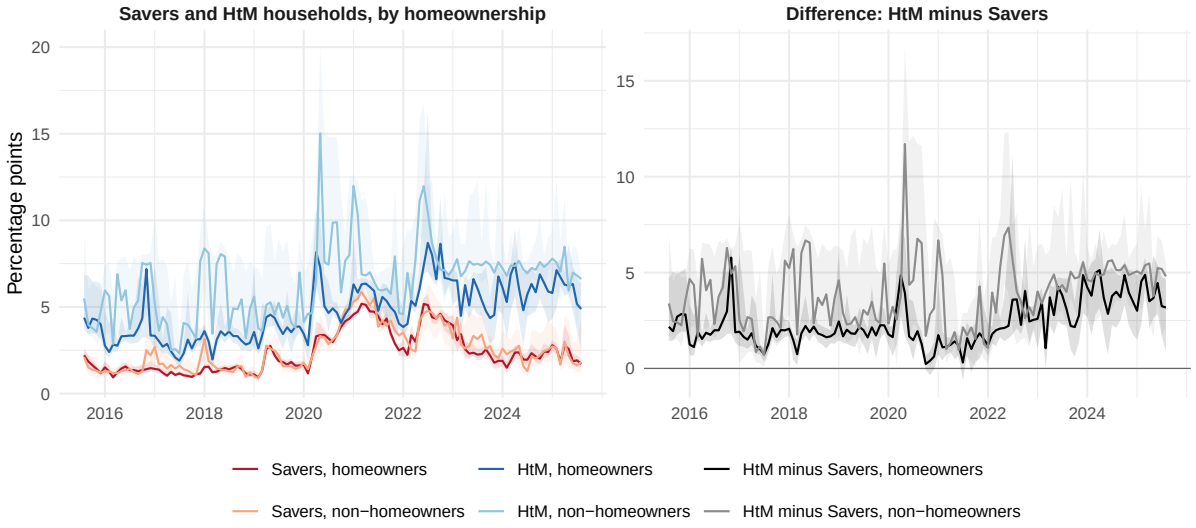


Figure 16: Fact 2b by homeownership: absolute forecast errors.

Notes: In the left panel, Savers are in red and HtM households in blue, with darker colors for homeowners and lighter colors for households that do not own their primary residence. The right panel shows the difference between HtM households and Savers among homeowners (black) and among non-homeowners (grey). The bands represent the 90% confidence interval.

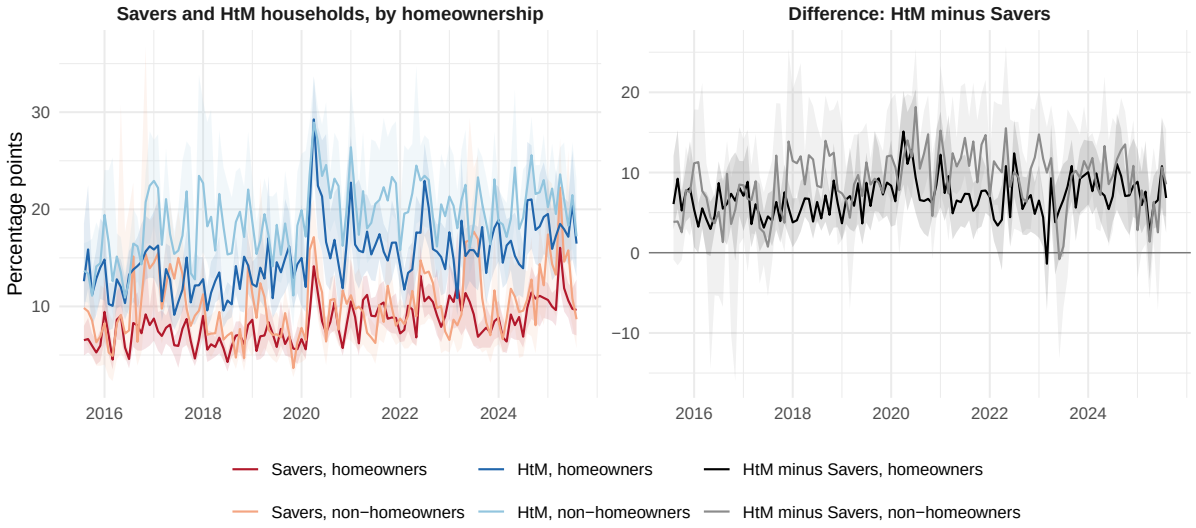


Figure 17: Fact 4 by homeownership: dispersion of inflation expectations.

Notes: Dispersion is the root-mean-square deviation from the group mean, after winsorizing expectations at the 1st and 99th percentiles of each month. In the left panel, Savers are in red and HtM households in blue, with darker colors for homeowners and lighter colors for households that do not own their primary residence. The right panel shows the difference between HtM households and Savers among homeowners (black) and among non-homeowners (grey). The bands represent the 90% confidence interval.

A.2. Tables

Table 8: Liquid wealth and Net Worth Statistics in the Sample

		Kaplan and Violante (2014)	SCE
Liquid wealth	Mean	\$45,327	\$111,829
	Median	\$3,844	\$1,529
	Fraction positive	0.77	0.54
Net worth	Mean	\$219,919	\$425,985
	Median	\$91,298	\$107,167
	Fraction positive	0.90	0.72
HtM Households	Fraction	0.35	0.42

Notes: The table reports summary statistics on the net worth and the liquid balance of the households in the SCE, as well as the fraction of HtM households in the sample. For reference, the same figures from the Survey of Consumer Finances reported in Kaplan and Violante (2014) are also shown. All amounts are in 2020 dollars. Kaplan and Violante (2014): 2001 wave of the Survey of Consumer Finances, excluding the top 5% of households by net worth, converted from 2001 dollars with the Consumer Price Index. SCE: liquid wealth and net worth are measured only in the Household Finance module, whose waves of August 2015 to August 2019 are used; the statistics refer to the households of these waves that are in the sample used for the facts and the estimation (no trimming), with survey weights; each wave is converted with the Consumer Price Index of its survey month. Liquid wealth is the value of checking and savings accounts and directly held financial assets outside retirement accounts, net of credit-card debt. The fraction of HtM households in the SCE refers to the baseline classification in the main-module sample, August 2015 to August 2025; the fraction reported by Kaplan and Violante (2014) is the upper end of their estimated range of 17.5% to 35%. Because the SCE statistics include the top of the wealth distribution, which Kaplan and Violante (2014) exclude, means are not comparable across the two columns, while medians and fractions positive are much less sensitive to it: excluding the top 5% of households by net worth in each wave, the SCE means are \$39,274 for liquid wealth and \$222,630 for net worth (medians \$514 and \$91,218). Credit-card debt in the SCE is not restricted to revolving balances: it is computed as total non-housing debt times the share of it that respondents report as credit-card debt, and may therefore include balances that are repaid in full at the end of the billing cycle. Kaplan and Violante (2014) subtract only revolving balances. SCE liquid wealth is therefore likely to be understated, which lowers its median and the fraction of households with positive liquid wealth.

Table 9: Characteristics of Savers, poor HtM households, and wealthy HtM households.

	Savers	Poor HtM (not homeowners)	Wealthy HtM (homeowners)
Share of respondents	57%	20%	23%
Respondents	9,087	2,329	2,905
Observations (respondent-months)	78,921	19,178	24,131
Household income: median bracket	\$75,000–\$99,999	\$20,000–\$29,999	\$40,000–\$49,999
Household income: mean (\$)	\$99,100	\$37,060	\$61,720
Household income below \$50,000	27%	76%	50%
Net worth: mean (\$)	\$487,830	−\$10,190	\$111,720
Net worth: median (\$)	\$269,030	−\$5,380	\$80,710
Liquid wealth: mean (\$)	\$94,690	−\$4,090	\$6,890
Liquid wealth: median (\$)	\$23,230	−\$160	−\$360
Respondents with wealth data	3,081	764	962
College degree	45%	17%	23%
Inflation literate	91%	62%	75%
Numeracy score (0 to 5), mean	4.12	3.04	3.39
High numeracy	77%	38%	50%
Age (years), mean	52.9	44.1	51.9
Male	58%	32%	44%
Homeowner	80%	0%	100%

Notes: Respondents of the sample used for the four facts, August 2015 to August 2025. Each respondent is assigned to the Saver or HtM group in which it is observed most often, and HtM households are divided into homeowners (wealthy HtM) and households that do not own their primary residence (poor HtM). Characteristics refer to the first month in which the respondent is observed, and statistics are weighted with the respondent's average survey weight. Household income is reported in brackets, and its mean is computed from the midpoints of the brackets (\$250,000 for the top bracket). Net worth and liquid wealth are observed only in the Household Finance waves of 2015 to 2019, and are expressed in 2020 dollars. Their means exclude the top 1% of each variable across all respondents, while medians use all respondents. Education, numeracy, and inflation literacy are defined as in Section 2.3.

Table 10: Fact 3 by homeownership: variance of the average expectation over time.

	Full sample 2015–2025	Pre-surge 2015–2020	Surge 2021–2023	Post-surge 2024–2025
Homeowners				
Savers	1.61	0.05	1.66	0.16
HtM	4.77	0.90	2.76	2.19
Ratio, Savers / HtM	0.34	0.06	0.60	0.07
	[0.24, 0.38]	[0.03, 0.11]	[0.31, 0.75]	[0.03, 0.21]
Non-homeowners				
Savers	1.21	0.05	1.24	0.17
HtM	5.03	1.14	3.79	2.75
Ratio, Savers / HtM	0.24	0.05	0.33	0.06
	[0.14, 0.35]	[0.03, 0.16]	[0.12, 0.57]	[0.04, 0.29]

Notes: Variance over the months of each period of the monthly average inflation expectation of each group (weighted interpolated median), as in Table 1. Savers and HtM households are divided into homeowners and households that do not own their primary residence. In brackets, 90% confidence intervals for the ratio from a cluster bootstrap that resamples respondents (500 replications).

Table 11: HtM status and other household characteristics: full specification.

	Inflation expectation	Absolute forecast error	Absolute revision	Absolute deviation from group median
HtM	2.485***	2.970***	2.296***	2.871***
	(0.223)	(0.206)	(0.202)	(0.203)
Age under 20	−2.311	0.047	−0.112	−0.505
	(2.629)	(2.346)	(2.589)	(2.241)
Age 40–60	0.686**	0.411	−0.227	0.232
	(0.290)	(0.275)	(0.245)	(0.267)
Age over 60	0.300	−0.050	−0.233	−0.290
	(0.341)	(0.326)	(0.278)	(0.315)
Female	1.515***	2.264***	1.903***	2.205***
	(0.235)	(0.222)	(0.193)	(0.213)
No college degree	1.472***	1.994***	1.806***	1.965***
	(0.184)	(0.150)	(0.137)	(0.146)
Low numeracy	2.164***	3.508***	3.091***	3.340***
	(0.287)	(0.269)	(0.227)	(0.261)
Not inflation literate	0.682*	3.164***	3.937***	3.204***
	(0.403)	(0.377)	(0.365)	(0.370)
Household income under $50,000	1.948***	1.815***	0.785***	1.715***
	(0.281)	(0.265)	(0.240)	(0.258)
Household income $50,000–$100,000	0.247	0.066	−0.112	0.124
	(0.196)	(0.183)	(0.188)	(0.181)
Northeast	0.427	0.267	0.085	0.244
	(0.350)	(0.323)	(0.266)	(0.311)
South	0.785**	0.943***	0.898***	0.789***
	(0.314)	(0.291)	(0.247)	(0.282)
West	0.310	−0.179	−0.092	−0.223
	(0.337)	(0.297)	(0.247)	(0.287)
Not a homeowner	0.222	1.063***	0.920***	1.039***
	(0.318)	(0.286)	(0.252)	(0.278)
Month fixed effects	Yes	Yes	Yes	Yes
Observations	119,092	119,092	107,481	119,092
Respondents	13,727	13,727	13,536	13,727

Notes: Specification (5) of Table 2. Reference groups: age 20 to 40, male, college degree, high numeracy, inflation literate, household income above $100,000, Midwest, homeowner. The group of respondents aged under 20 includes only 20 respondents. Standard errors clustered by respondent and month in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Appendix B. The Expectation Model: Derivations and Robustness

B.1. Derivation of the Inflation Forecast

The perceived law of motion for inflation reads:

$$\pi_{t+1} = \rho^j \pi_t + \eta_{t+1}$$

The actual inflation signal reads

$$s_t^A = \pi_t + \varepsilon_t^A$$

with noise being generated as $\varepsilon_t^A \sim \mathcal{N}(0, \sigma_A^2)$. The perceived signal instead is generated from the observation equation:

$$s_t^j = \pi_t - \xi^j + \varepsilon_t^j \tag{B.1}$$

where ξ^j captures the bias and $\varepsilon_t^j \sim \mathcal{N}(0, \sigma_j^2)$, with σ_j^2 inversely related to the household's confidence in the perceived signal. Also assume

$$\mathbb{E}[\eta_t \eta_\tau] = \begin{cases} \sigma_\eta^2 & \text{for } t = \tau \\ 0 & \text{otherwise} \end{cases} \qquad \text{and} \qquad \mathbb{E}[\varepsilon_t^j \varepsilon_\tau^j] = \begin{cases} \sigma_j^2 & \text{for } t = \tau \\ 0 & \text{otherwise} \end{cases}$$

The forecast is defined as:

$$\hat{\pi}_{t+1|t}^j \equiv \hat{E}\left[\pi_{t+1} \mid \mathcal{I}_t^j\right]$$

where:

$$\mathcal{I}_t^j = \left(\rho^j, s_t^j, s_{t-1}^j, ..., s_1^j\right)$$

Associated with each forecast, the mean square error is defined as

$$\mathbf{P}_{t+1|t}^j \equiv \mathbb{E}\left[\left(\pi_{t+1} - \hat{\pi}_{t+1|t}^j\right)^2\right]$$

The forecast of s_t^j is defined as:

$$\hat{s}_{t|t-1}^j \equiv \hat{E}\left[s_t^j \mid \mathcal{I}_{t-1}^j\right]$$

From the observation equation (B.1), that yields:

$$\hat{s}_{t|t-1}^j = \hat{\pi}_{t|t-1}^j - \xi^j$$

And the MSE of this forecast is:

$$\begin{aligned}\mathbb{E}\left[(s_t^j - \hat{s}_{t|t-1}^j)^2\right] &= \mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j + \varepsilon_t^j\right)^2\right] \\ &= \mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j\right)^2\right] + \mathbb{E}\left[\left(\varepsilon_t^j\right)^2\right] \quad \text{, given that} \quad \mathbb{E}\left[(\pi_t - \hat{\pi}_{t|t-1}^j)\varepsilon_t^j\right] = 0 \\ &= \mathbf{P}_{t|t-1}^j + \sigma_j^2\end{aligned}$$

Let's now consider the nowcast of inflation, denoted $\hat{\pi}_{t|t}^j$ from iterated projections, that takes the form

$$\hat{\pi}_{t|t}^j = \hat{\pi}_{t|t-1}^j + \frac{\mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j\right)\left(s_t^j - \hat{s}_{t|t-1}^j\right)\right]}{\mathbb{E}\left[\left(s_t^j - \hat{s}_{t|t-1}^j\right)^2\right]}\left(s_t^A - \hat{s}_{t|t-1}^j\right)$$

Focusing on the fraction, the denominator reduces to the MSE of $\hat{s}_{t|t-1}^j$ and for the numerator one has:

$$\mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j\right)\left(s_t^j - \hat{s}_{t|t-1}^j\right)\right] = \mathbb{E}\left[\left(\pi_t - \hat{\pi}_{t|t-1}^j\right)\left(\pi_t - \hat{\pi}_{t|t-1}^j + \varepsilon_t^j\right)\right] = \mathbf{P}_{t|t-1}^j$$

given that $\mathbb{E}\left[(\pi_t - \hat{\pi}_{t|t-1}^j)\varepsilon_t^j\right] = 0$.

Hence the nowcast simplifies to:

$$\begin{aligned}\hat{\pi}_{t|t}^j &= \hat{\pi}_{t|t-1}^j + \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(s_t^A - \hat{s}_{t|t-1}^j\right) \\ &= \hat{\pi}_{t|t-1}^j + \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(\pi_t + \varepsilon_t^A - \left(\hat{\pi}_{t|t-1}^j - \xi^j\right)\right) \\ &= \hat{\pi}_{t|t-1}^j + \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(\pi_t - \hat{\pi}_{t|t-1}^j + \xi^j + \varepsilon_t^A\right)\end{aligned}$$

Which in turn implies that the forecast reads, from the perceived law of motion:

$$\begin{aligned}\hat{\pi}_{t+1|t}^j &= \rho^j \hat{\pi}_{t|t}^j \\ &= \rho^j \hat{\pi}_{t|t-1}^j + \rho^j \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}\left(\pi_t - \hat{\pi}_{t|t-1}^j + \xi^j + \varepsilon_t^A\right) \\ &= \rho^j \hat{\pi}_{t|t-1}^j + \rho^j \mathbf{K}_t^j\left(\pi_t - \hat{\pi}_{t|t-1}^j + \xi^j + \varepsilon_t^A\right)\end{aligned}$$

where $\mathbf{K}_t^j \equiv \frac{\mathbf{P}_{t|t-1}^j}{\mathbf{P}_{t|t-1}^j + \sigma_j^2}$ is the Kalman gain. Further, one can evaluate the MSE of this forecast

$$\mathbf{P}_{t+1|t}^j = \mathbb{E}\left[\left(\pi_{t+1} - \hat{\pi}_{t+1|t}^j\right)^2\right]$$

$$= \mathbb{E}\left[\left(\rho^j \pi_t - \rho^j \hat{\pi}^j_{t|t} + \eta_{t+1}\right)^2\right]$$
$$= \left(\rho^j\right)^2 \mathbf{P}^j_{t|t} + \sigma^2_\eta$$

noting that by assumption $\mathbb{E}\left[(\pi_t - \hat{\pi}^j_{t|t-1})\eta_t\right] = 0$.

B.2. From Quarterly to Year-Ahead Inflation Expectations

This appendix shows how the quarterly inflation-forecasting equation derived in Equation (11) can be mapped into the one-year-ahead inflation expectations elicited in the SCE.

For a household in state $j \in \{\text{Saver}, \text{HtM}\}$, the quarterly forecasting equation is

$$\hat{\pi}^j_{t+1|t} = \rho^j \hat{\pi}^j_{t|t-1} + \rho^j \mathbf{K}^j_t \left(\pi_t - \hat{\pi}^j_{t|t-1} + \xi^j + \varepsilon^A_t\right). \tag{B.2}$$

Equivalently,

$$\hat{\pi}^j_{t+1|t} = \rho^j \left[\left(1 - \mathbf{K}^j_t\right) \hat{\pi}^j_{t|t-1} + \mathbf{K}^j_t \left(\pi_t + \xi^j + \varepsilon^A_t\right)\right]. \tag{B.3}$$

The SCE elicits expectations about inflation over the next twelve months. In the survey data, a model period corresponds to three survey months: if t denotes a survey month, $t-1$ denotes the survey month three months earlier, and $t-2$ the survey month six months earlier. Since the model is written at quarterly frequency, define year-ahead expected inflation as the sum of expected quarterly inflation over the next four quarters:

$$\hat{\Pi}^j_{t+4|t} \equiv \sum_{s=1}^{4} \hat{\pi}^j_{t+s|t}. \tag{B.4}$$

Using the perceived law of motion for inflation,

$$\begin{aligned}
\hat{\Pi}^j_{t+4|t} &= \hat{\pi}^j_{t+1|t} + \hat{\pi}^j_{t+2|t} + \hat{\pi}^j_{t+3|t} + \hat{\pi}^j_{t+4|t} \\
&= \left[1 + \rho^j + (\rho^j)^2 + (\rho^j)^3\right] \hat{\pi}^j_{t+1|t} \\
&= \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \hat{\pi}^j_{t|t}.
\end{aligned} \tag{B.5}$$

Similarly, the year-ahead forecast formed in the previous quarter satisfies

$$\hat{\Pi}^j_{t+3|t-1} = \left[1 + \rho^j + (\rho^j)^2 + (\rho^j)^3\right] \hat{\pi}^j_{t|t-1}. \tag{B.6}$$

Substituting these two expressions into the quarterly updating equation gives

$$\frac{\hat{\Pi}^j_{t+4|t}}{1 + \rho^j + (\rho^j)^2 + (\rho^j)^3} = \rho^j \left[\left(1 - \mathbf{K}^j_t\right) \frac{\hat{\Pi}^j_{t+3|t-1}}{1 + \rho^j + (\rho^j)^2 + (\rho^j)^3} + \mathbf{K}^j_t \left(\pi_t + \xi^j + \varepsilon^A_t\right)\right]. \tag{B.7}$$

Multiplying both sides by $1 + \rho^j + (\rho^j)^2 + (\rho^j)^3$ yields

$$\hat{\Pi}^j_{t+4|t} = \rho^j \left(1 - \mathbf{K}^j_t\right) \hat{\Pi}^j_{t+3|t-1} + \rho^j \mathbf{K}^j_t \left[1 + \rho^j + (\rho^j)^2 + (\rho^j)^3\right] \left(\pi_t + \xi^j + \varepsilon^A_t\right). \tag{B.8}$$

Since

$$\rho^j \left[1 + \rho^j + (\rho^j)^2 + (\rho^j)^3\right] = \rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4, \tag{B.9}$$

the year-ahead updating equation can be written as

$$\hat{\Pi}^j_{t+4|t} = \rho^j \left(1 - \mathbf{K}^j_t\right) \hat{\Pi}^j_{t+3|t-1} + \mathbf{K}^j_t \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \left(\pi_t + \xi^j + \varepsilon^A_t\right). \tag{B.10}$$

Under the steady-state Kalman gain, $\mathbf{K}^j_t = \mathbf{K}^j$, this expression motivates the empirical specification

$$\hat{\Pi}^j_{t+4|t} = \beta^j_0 + \beta^j_1 \hat{\Pi}^j_{t+3|t-1} + \beta^j_2 \pi_t + \nu^j_t, \tag{B.11}$$

where

$$\beta^j_0 = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \xi^j, \tag{B.12}$$

$$\beta^j_1 = \rho^j \left(1 - \mathbf{K}^j\right), \tag{B.13}$$

$$\beta^j_2 = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right], \tag{B.14}$$

$$\nu^j_t = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right] \varepsilon^A_t. \tag{B.15}$$

The regression coefficients recover the population coefficients of the model under the maintained orthogonality assumptions. In particular, assume that the actual signal noise satisfies

$$\mathbb{E}\left[\varepsilon^A_t\right] = 0, \qquad \mathbb{E}\left[\varepsilon^A_t \mid \mathcal{I}^j_{t-1}\right] = 0, \qquad \mathbb{E}\left[\varepsilon^A_t \pi_t\right] = 0. \tag{B.16}$$

Then the regression error satisfies

$$\mathbb{E}\left[\nu^j_t\right] = 0, \qquad \mathbb{E}\left[\nu^j_t \hat{\Pi}^j_{t+3|t-1}\right] = 0, \qquad \mathbb{E}\left[\nu^j_t \pi_t\right] = 0. \tag{B.17}$$

Therefore, the population projection of $\hat{\Pi}^j_{t+4|t}$ on a constant, $\hat{\Pi}^j_{t+3|t-1}$, and π_t recovers the coefficients β^j_0, β^j_1, and β^j_2.

Finally, these coefficients can be mapped back into the structural parameters of the expectation-formation process. The coefficients β^j_1 and β^j_2 identify ρ^j and $\mathbf{K}^j$ through

$$\begin{cases} \beta^j_1 = \rho^j \left(1 - \mathbf{K}^j\right), \\ \beta^j_2 = \mathbf{K}^j \left[\rho^j + (\rho^j)^2 + (\rho^j)^3 + (\rho^j)^4\right]. \end{cases} \tag{B.18}$$

Once ρ^j and $\mathbf{K}^j$ are recovered, the perceived signal bias is obtained from

$$\xi^j = \frac{\beta_0^j}{\beta_2^j}. \tag{B.19}$$

B.3. Additional Estimates and Illustrations

Table 12: Co-movement of three-year and one-year inflation expectations.

	Full sample, 2015–2025			Surge, 2021–2023		
	HtM	Savers	Difference	HtM	Savers	Difference
Levels	0.29^{***}	0.19^{***}	0.10	0.38^{***}	0.12	0.26^{**}
	(0.04)	(0.07)	(0.07)	(0.09)	(0.08)	(0.11)

Notes: Slope of the three-year inflation expectation (inflation over the twelve months between 24 and 36 months ahead) on the one-year inflation expectation, separately for HtM households and Savers. In the noisy-information model, the slope equals $(\rho^j)^8$. Monthly average expectations of each group (weighted interpolated medians), Driscoll–Kraay standard errors. Difference: HtM minus Savers. ***, **, *: different from zero at the 1%, 5%, and 10% levels.

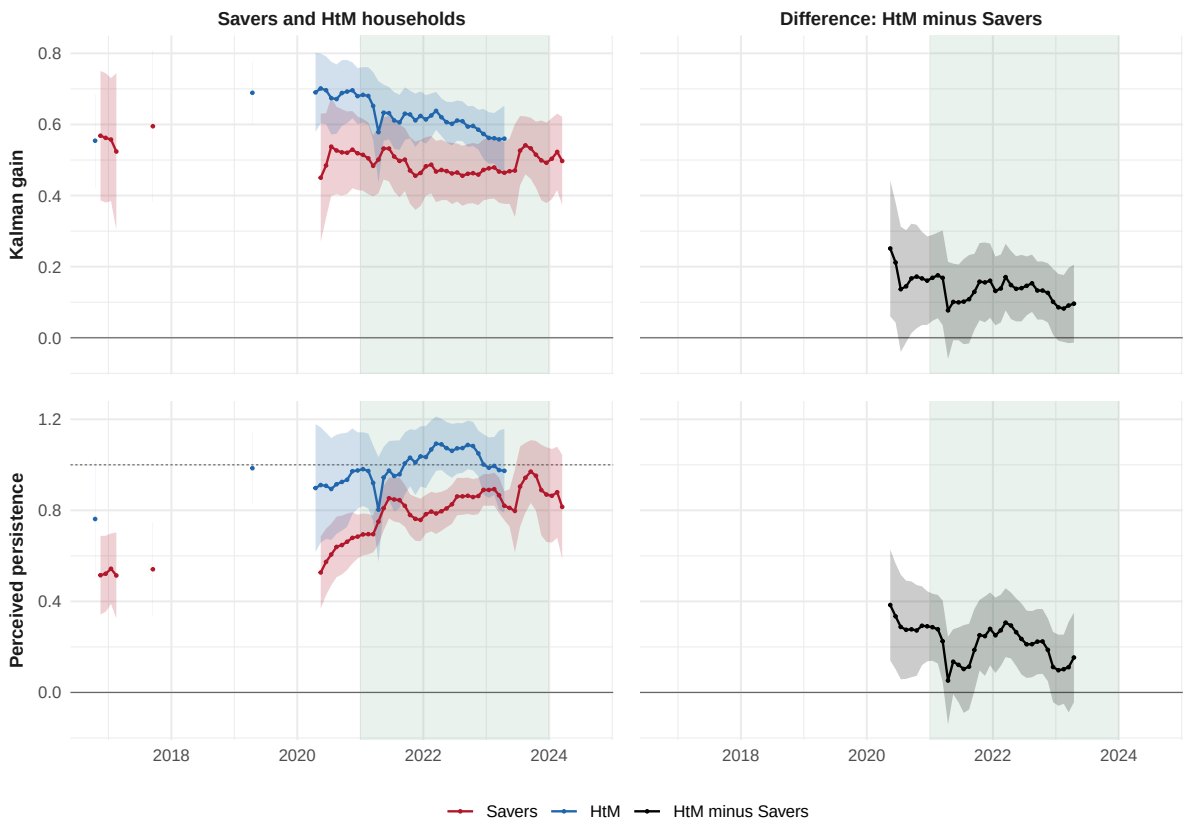


Figure 18: Kalman gains and perceived persistence in rolling windows: 24-month windows.

Notes: Estimates of the quarterly noisy-information model by weighted OLS in 24-month windows moved by one month, plotted at the centre of each window. The top panels show the Kalman gains and the bottom panels the perceived persistence of inflation. In the left panels, Savers are in red and HtM households in blue, and the right panels show the difference between HtM households and Savers. The estimates are shown only in windows where the response to inflation, $\hat{\beta}_2^j$, is statistically significant at the 10% level (for both groups, in the case of the difference). The bands represent the 90% confidence interval, and the dashed line marks a perceived persistence of one. The shaded area marks the inflation surge, 2021 to 2023.

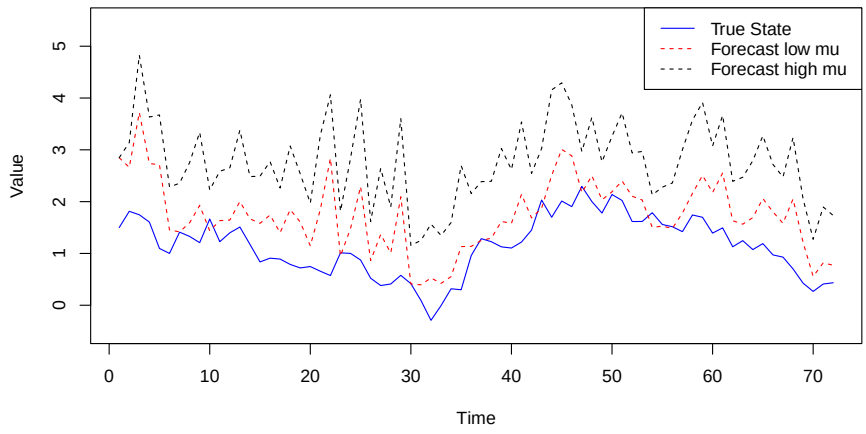


Figure 19: Simulation of the time series of Inflation Expectations with high and low ξ^j.

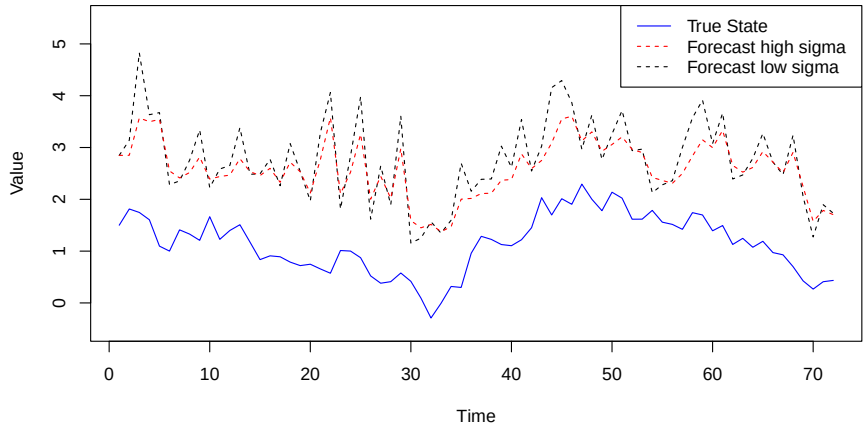
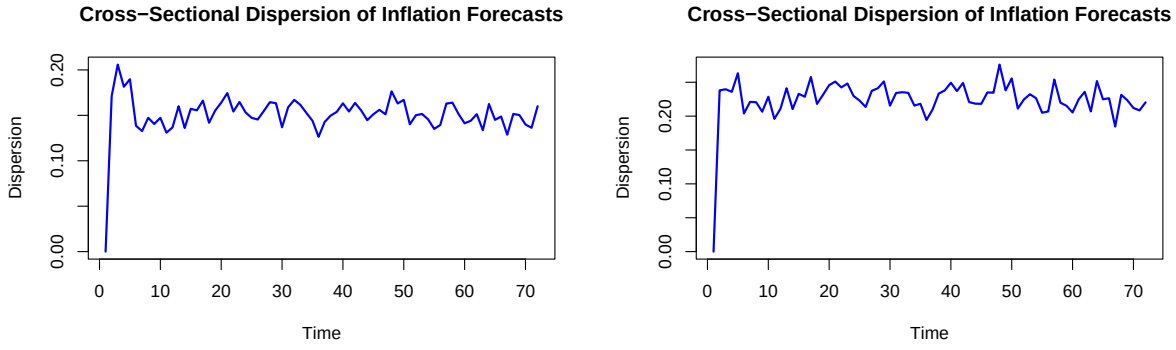


Figure 20: Simulation of the time series of Inflation Expectations with high and low σ_j^2.



(a) $\sigma_j^2 = \sigma_A^2$ **(b)** $\sigma_j^2 > \sigma_A^2$

Figure 21: Simulation of the dispersion of Inflation Expectations with high and low σ_j^2.

Appendix C. THANK: Derivations

C.1. Derivation of the Euler Equations in THANK

The problem takes the form of the Bellman equation:

$$\Lambda\left(B_t^H, B_t^S, \Theta_t\right) = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} + \beta \mathbb{E}\left[\Lambda\left(B_{t+1}^H, B_{t+1}^S, \Theta_{t+1}\right)\right]$$

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t B_t^S + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t B_t^H \\ B_{t+1}^S &= s Z_{t+1}^S + (1-s) Z_{t+1}^H \\ B_{t+1}^H &= (1-h) Z_{t+1}^S + h Z_{t+1}^H \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

This can be rewritten more succinctly, by substituting the different B assets as:

$$\Lambda\left((1-h)Z_t^S + hZ_t^H, sZ_t^S + (1-s)Z_t^H, \Theta_t\right) = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} + \beta \mathbb{E}\left[\Lambda\left((1-h)Z_{t+1}^S + hZ_{t+1}^H, sZ_{t+1}^S + (1-s)Z_{t+1}^H, \Theta_{t+1}\right)\right]$$

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t\left(sZ_t^S + (1-s)Z_t^H\right) + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t\left((1-h)Z_t^S + hZ_t^H\right) \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

The dynamic programming problem can be written as the following Lagrangian:

$$\begin{aligned} &\Lambda\left((1-h)Z_t^S + hZ_t^H, sZ_t^S + (1-s)Z_t^H, \Theta_t\right) \\ &\quad = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} \\ &\quad + \beta \mathbb{E}\left[\Lambda\left((1-h)Z_{t+1}^S + hZ_{t+1}^H, sZ_{t+1}^S + (1-s)Z_{t+1}^H, \Theta_{t+1}\right)\right] \\ &\quad + \ell_t^S\left[W_t N_t^S + R_t\left(sZ_t^S + (1-s)Z_t^H\right) + (V_t + P_t D_t)\Theta_t - P_t C_t^S - Z_{t+1}^S - V_t \Theta_{t+1}\right] \\ &\quad + \ell_t^H\left[W_t N_t^H + R_t\left((1-h)Z_t^S + hZ_t^H\right) - P_t C_t^H - Z_{t+1}^H\right] \\ &\quad + m_t^S Z_{t+1}^S + m_t^H Z_{t+1}^H + n_t \Theta_{t+1} \end{aligned}$$

where ℓ_t^S, ℓ_t^H, m_t^S, m_t^H, and n_t are Lagrangian multipliers. The corresponding KKT conditions are then, with complementary slackness (omitted):[27]

$$\begin{aligned}
\frac{\partial \Lambda(\cdot)}{\partial C_t^H} = 0 \quad &\iff \quad \lambda U_C(C_t^H, N_t^H) - \ell_t^H P_t = 0 \\
&\qquad \frac{\lambda U_C(C_t^H, N_t^H)}{P_t} = \ell_t^H \\
\frac{\partial \Lambda(\cdot)}{\partial C_t^S} = 0 \quad &\iff \quad (1-\lambda) U_C(C_t^S, N_t^S) - \ell_t^S P_t = 0 \\
&\qquad \frac{(1-\lambda) U_C(C_t^S, N_t^S)}{P_t} = \ell_t^S \\
\frac{\partial \Lambda(\cdot)}{\partial N_t^H} = 0 \quad &\iff \quad \lambda U_N(C_t^H, N_t^H) + \ell_t^H W_t = 0 \\
&\qquad \frac{-\lambda U_N(C_t^H, N_t^H)}{W_t} = \ell_t^H \\
\frac{\partial \Lambda(\cdot)}{\partial N_t^S} = 0 \quad &\iff \quad (1-\lambda) U_N(C_t^S, N_t^S) + \ell_t^S W_t = 0 \\
&\qquad \frac{-(1-\lambda) U_N(C_t^S, N_t^S)}{W_t} = \ell_t^S \\
\frac{\partial \Lambda(\cdot)}{\partial Z_{t+1}^H} = 0 \quad &\iff \quad \beta h \mathbb{E}[\Lambda_{B^H}(\cdot_{t+1})] + \beta(1-s)\mathbb{E}[\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H = 0 \\
&\qquad \beta \mathbb{E}[h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H = 0 \\
\frac{\partial \Lambda(\cdot)}{\partial Z_{t+1}^S} = 0 \quad &\iff \quad \beta(1-h)\mathbb{E}[\Lambda_{B^H}(\cdot_{t+1})] + \beta s \mathbb{E}[\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S = 0 \\
&\qquad \beta \mathbb{E}[(1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S = 0 \\
\frac{\partial \Lambda(\cdot)}{\partial \Theta_{t+1}} = 0 \quad &\iff \quad \beta \mathbb{E}[\Lambda_\Theta(\cdot_{t+1})] - \ell_t^S V_t + n_t = 0
\end{aligned}$$

and the Envelope Conditions are:

$$\begin{aligned}
\frac{\partial \Lambda(\cdot)}{\partial Z_t^H} = 0 \quad &\iff \quad h\Lambda_{B^H}(\cdot_t) + (1-s)\Lambda_{B^S}(\cdot_t) = \ell_t^S R_t(1-s) + \ell_t^H R_t h \\
&\text{iterated:} \quad h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1}) = \ell_{t+1}^S R_{t+1}(1-s) + \ell_{t+1}^H R_{t+1} h \\
\frac{\partial \Lambda(\cdot)}{\partial Z_t^S} = 0 \quad &\iff \quad (1-h)\Lambda_{B^H}(\cdot_t) + s\Lambda_{B^S}(\cdot_t) = \ell_t^S R_t s + \ell_t^H R_t(1-h) \\
&\text{iterated:} \quad (1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1}) = \ell_{t+1}^S R_{t+1} s + \ell_{t+1}^H R_{t+1}(1-h) \\
\frac{\partial \Lambda(\cdot)}{\partial \Theta_t} = 0 \quad &\iff \quad \Lambda_\Theta(\cdot_t) = \ell_t^S(V_t + P_t D_t) \\
&\text{iterated:} \quad \Lambda_\Theta(\cdot_{t+1}) = \ell_{t+1}^S(V_{t+1} + P_{t+1} D_{t+1})
\end{aligned}$$

[27] The following notation is used consistently: for a function $F(X, Y)$, the partial derivative $F_X \equiv \frac{\partial F(X,Y)}{\partial X}$.

Hence, denoting $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$, the Euler equations are:

$$\begin{aligned}
\beta\mathbb{E}[(1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S R_{t+1} s + \ell_{t+1}^H R_{t+1}(1-h)\right] &\leq \ell_t^S \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}} R_{t+1} s + \frac{\lambda U_C(C_{t+1}^H, N_{t+1}^H)}{P_{t+1}} R_{t+1}(1-h)\right] &\leq \frac{(1-\lambda)U_C(C_t^S, N_t^S)}{P_t} \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + \frac{\lambda}{1-\lambda}(1-h)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^S, N_t^S) \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + (1-s)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^S, N_t^S) \qquad \text{(C.1)}
\end{aligned}$$

together with

$$0 = Z_{t+1}^S \left\{U_C(C_t^S, N_t^S) - \beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + (1-s)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right]\right\} \qquad \text{(C.2)}$$

and

$$\begin{aligned}
\beta\mathbb{E}[h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S R_{t+1}(1-s) + \ell_{t+1}^H R_{t+1} h\right] &\leq \ell_t^H \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}} R_{t+1}(1-s) + \frac{\lambda U_C(C_{t+1}^H, N_{t+1}^H)}{P_{t+1}} R_{t+1} h\right] &\leq \frac{\lambda U_C(C_t^H, N_t^H)}{P_t} \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(\frac{1-\lambda}{\lambda}(1-s)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^H, N_t^H) \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left((1-h)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^H, N_t^H) \qquad \text{(C.3)}
\end{aligned}$$

together with

$$0 = Z_{t+1}^H \left\{U_C(C_t^H, N_t^H) - \beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left((1-h)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right]\right\} \qquad \text{(C.4)}$$

and

$$\begin{aligned}
\beta\mathbb{E}[\Lambda_\Theta(\cdot_{t+1})] - \ell_t^S V_t + n_t &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S(V_{t+1} + P_{t+1}D_{t+1})\right] &\leq \ell_t^S V_t \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}}(V_{t+1} + P_{t+1}D_{t+1})\right] &\leq \frac{(1-\lambda)U_C(C_t^S, N_t^S)}{P_t} V_t \\
\beta\mathbb{E}\left[U_C(C_{t+1}^S, N_{t+1}^S)\frac{P_t}{P_{t+1}}\frac{V_{t+1} + P_{t+1}D_{t+1}}{V_t}\right] &\leq U_C(C_t^S, N_t^S) \qquad \text{(C.5)}
\end{aligned}$$

together with

$$0 = \Theta_{t+1}\left\{U_C(C_t^S, N_t^S) - \beta\mathbb{E}\left[U_C(C_{t+1}^S, N_{t+1}^S)\frac{P_t}{P_{t+1}}\frac{V_{t+1} + P_{t+1}D_{t+1}}{V_t}\right]\right\} \qquad \text{(C.6)}$$

and at time t it also has to be that

$$U_C(C_t^H, N_t^H) = -U_N(C_t^H, N_t^H)\frac{P_t}{W_t} \tag{C.7}$$

as well as

$$U_C(C_t^S, N_t^S) = -U_N(C_t^S, N_t^S)\frac{P_t}{W_t} \tag{C.8}$$

C.2. The Log-Linearized Euler Equation of a Saver

The derivation is written between periods 0 and 1. The same steps apply between any two periods t and $t+1$, with $\mathbb{E}$ replaced by $\mathbb{E}_t$.

With CRRA preferences, the Euler equation for bonds of a Saver reads

$$\beta\,\mathbb{E}\Big[\tilde{R}_1\left(s\left(C_1^S\right)^{-\sigma} + (1-s)\left(C_1^H\right)^{-\sigma}\right)\Big] \;=\; \left(C_0^S\right)^{-\sigma} \tag{C.9}$$

expected future marginal utility averages the two states, weighted by the transition probabilities. Define

$$\Omega \;\equiv\; s\left(C^S\right)^{-\sigma} + (1-s)\left(C^H\right)^{-\sigma}, \qquad \omega \;\equiv\; \frac{s\left(C^S\right)^{-\sigma}}{\Omega} \;=\; \frac{s}{s+(1-s)\,\zeta^{\sigma}}. \tag{C.10}$$

Then ω is the share of the Saver's expected future marginal utility attributable to remaining a Saver: the transition probability s, adjusted by the relative marginal utility of the two states, $\zeta^{\sigma} = \left(\frac{C^S}{C^H}\right)^{\sigma}$.

To log-linearize (C.9), write $C_1^S = C^S e^{c_1^S}$, $C_1^H = C^H e^{c_1^H}$, $C_0^S = C^S e^{c_0^S}$ and $\tilde{R}_1 = \tilde{R}e^{\tilde{r}_1}$, and expand to first order ($e^{-\sigma c} \approx 1-\sigma c$, $e^{\tilde{r}} \approx 1+\tilde{r}$):

$$\beta\tilde{R}\,\mathbb{E}\Big[(1+\tilde{r}_1)\left(s\left(C^S\right)^{-\sigma}\left(1-\sigma c_1^S\right) + (1-s)\left(C^H\right)^{-\sigma}\left(1-\sigma c_1^H\right)\right)\Big] \;=\; \left(C^S\right)^{-\sigma}\left(1-\sigma c_0^S\right).$$

Using the definitions in (C.10), the inner bracket equals $\Omega\left[1-\sigma\omega\,c_1^S - \sigma(1-\omega)\,c_1^H\right]$, so that, dropping second-order cross products,

$$\beta\tilde{R}\,\Omega\,\mathbb{E}\left[1+\tilde{r}_1 - \sigma\omega\,c_1^S - \sigma(1-\omega)\,c_1^H\right] \;=\; \left(C^S\right)^{-\sigma}\left(1-\sigma c_0^S\right).$$

Evaluating at the steady state gives $\beta\tilde{R}\,\Omega = \left(C^S\right)^{-\sigma}$; dividing both sides by this relation and rearranging yields

$$c_0^S \;=\; \mathbb{E}\left[\omega\,c_1^S + (1-\omega)\,c_1^H - \frac{1}{\sigma}\,\tilde{r}_1\right], \tag{C.11}$$

or, solved for the expected future Saver consumption,

$$\mathbb{E}\left[c_1^S\right] \;=\; \frac{1}{\omega}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right) - \frac{1-\omega}{\omega}\,\mathbb{E}\left[c_1^H\right]. \tag{C.12}$$

The steady-state return

As established above, the steady-state version of (C.9) is $\beta\tilde{R}\,\Omega = \left(C^S\right)^{-\sigma}$, that is

$$\beta\tilde{R}\left[s + (1-s)\,\zeta^{\sigma}\right] \;=\; 1 \qquad \Longleftrightarrow \qquad s\tilde{R} \;=\; \frac{\omega}{\beta}, \tag{C.13}$$

where the implication follows from multiplying both sides of the first equality by s and using the definition of ω in (C.10). Since $\zeta > 1$ implies $\omega < s$, the steady-state return satisfies $\tilde{R} < \frac{1}{\beta}$: the risk of transitioning into the high-marginal-utility HtM state makes households willing to hold bonds at a depressed return. Equation (C.13) is used repeatedly below: the effective gross return on carried wealth, $s\tilde{R}$, equals $\frac{\omega}{\beta}$.

C.3. Derivation of the Saver's Consumption Function in Infinite-Horizon THANK

This appendix derives the infinite-horizon Saver consumption function in a THANK economy with positive steady-state liquidity, $\tilde{Z}^S > 0$, and unequal steady-state consumption, $C^S > C^H$, by the characteristic-root method. Since liquidity is positive, all variables — including per-Saver bonds — are log deviations from the steady state:

$$C_t^j = C^j e^{c_t^j}, \qquad \tilde{Y}_t^j = \tilde{Y}^j e^{\tilde{y}_t^j}, \qquad \tilde{R}_t = \tilde{R}e^{\tilde{r}_t}, \qquad \tilde{Z}_t^S = \tilde{Z}^S e^{\tilde{z}_t^S}, \qquad j \in \{S, H\}. \tag{C.14}$$

Aggregate liquid assets are $B_t = (1-\lambda)\tilde{Z}_t^S$, so $\tilde{z}_t^S$ is also the log deviation of aggregate liquidity.

The marginal-utility weight and the steady-state Euler equation are, as derived in Appendix C.2:

$$\omega = \frac{s}{s + (1-s)\zeta^{\sigma}}, \qquad \beta\tilde{R} = \frac{1}{s + (1-s)\zeta^{\sigma}}, \qquad s\tilde{R} = \frac{\omega}{\beta}, \qquad \zeta \equiv \frac{C^S}{C^H} > 1. \tag{C.15}$$

C.3.1. *Budget Constraints*

As in Section 4.1, bonds are pooled within groups: a continuing Saver retains the fraction s of per-Saver bonds, and the HtM group receives, per member, $\frac{(1-\lambda)(1-s)}{\lambda}\tilde{Z}_t^S = (1-h)\tilde{Z}_t^S$ by the stationarity of the population shares, $(1-\lambda)(1-s) = \lambda(1-h)$. The level budget constraints are therefore

$$C_t^H = \tilde{Y}_t^H + (1-h)\,\tilde{R}_t\,\tilde{Z}_t^S, \tag{C.16}$$

$$C_t^S + \tilde{Z}_{t+1}^S = \tilde{Y}_t^S + s\,\tilde{R}_t\,\tilde{Z}_t^S. \tag{C.17}$$

Evaluated at the steady state:

$$C^H - \tilde{Y}^H = (1-h)\,\tilde{R}\tilde{Z}^S, \qquad C^S - \tilde{Y}^S = \left(s\tilde{R} - 1\right)\tilde{Z}^S. \tag{C.18}$$

Log-linearizing (C.16) and (C.17) around the steady state:

$$c_t^H = \frac{\tilde{Y}^H}{C^H}\,\tilde{y}_t^H + \alpha_H\left(\tilde{r}_t + \tilde{z}_t^S\right), \qquad \alpha_H \equiv \frac{C^H - \tilde{Y}^H}{C^H}, \tag{C.19}$$

$$c_t^S = \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_t^S + \alpha_S\left(\tilde{r}_t + \tilde{z}_t^S\right) - \alpha_B\,\tilde{z}_{t+1}^S, \qquad \alpha_S \equiv \frac{s\tilde{R}\tilde{Z}^S}{C^S}, \quad \alpha_B \equiv \frac{\tilde{Z}^S}{C^S}, \tag{C.20}$$

with $\alpha_S = s\tilde{R}\,\alpha_B$. In contrast with the zero-liquidity case, the interest rate now enters the budget constraints at first order: $\tilde{r}_t$ revalues the outstanding stock.

C.3.2. *Euler Equation and Liquid-Asset Dynamics*

The log-linearized CRRA Euler equation of a Saver is

$$c_t^S = \mathbb{E}_t\left[\omega c_{t+1}^S + (1-\omega)c_{t+1}^H - \frac{1}{\sigma}\tilde{r}_{t+1}\right]. \tag{C.21}$$

Substituting (C.20) and (C.19) into (C.21), and collecting the asset terms on the left-hand side:

$$\omega\alpha_B\,\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \left[\alpha_B + \omega\alpha_S + (1-\omega)\alpha_H\right]\tilde{z}_{t+1}^S + \alpha_S\,\tilde{z}_t^S = F_t^0, \tag{C.22}$$

where the forcing term now contains current and expected interest rates:

$$F_t^0 \equiv \omega\,\frac{\tilde{Y}^S}{C^S}\,\mathbb{E}_t\left[\tilde{y}_{t+1}^S\right] + (1-\omega)\,\frac{\tilde{Y}^H}{C^H}\,\mathbb{E}_t\left[\tilde{y}_{t+1}^H\right] - \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_t^S + \left[\omega\alpha_S + (1-\omega)\alpha_H - \frac{1}{\sigma}\right]\mathbb{E}_t[\tilde{r}_{t+1}] - \alpha_S\,\tilde{r}_t. \tag{C.23}$$

Dividing (C.22) by $\omega\alpha_B$:

$$\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \mathcal{A}_\zeta\,\tilde{z}_{t+1}^S + \beta^{-1}\tilde{z}_t^S = F_t, \qquad F_t \equiv \frac{F_t^0}{\omega\alpha_B}, \tag{C.24}$$

since $\frac{\alpha_S}{\omega\alpha_B} = \frac{s\tilde{R}}{\omega} = \beta^{-1}$ by (C.15), and

$$\mathcal{A}_\zeta \equiv \frac{\alpha_B + \omega\alpha_S + (1-\omega)\alpha_H}{\omega\alpha_B} = \frac{1}{\omega} + \frac{\omega}{\beta} + \frac{(1-\omega)}{\beta}\,\frac{(1-h)\,\zeta}{s}, \tag{C.25}$$

where the second equality uses $\alpha_S/\alpha_B = s\tilde{R} = \omega/\beta$ and $\alpha_H/\alpha_B = \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{\tilde{Z}^S} = (1-h)\tilde{R}\,\zeta$ from (C.18).

The level of liquidity has cancelled from the homogeneous dynamics, and appears only in the forcing term (C.23), through the interest-rate revaluation.

C.3.3. *Characteristic Equation*

The characteristic equation is obtained from the homogeneous part of (C.24). Then, asset dynamics satisfy

$$\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \mathcal{A}_\zeta\tilde{z}_{t+1}^S + \beta^{-1}\tilde{z}_t^S = 0. \tag{C.26}$$

Guessing a solution of the form $\tilde{z}_t^S = K\mu^t$ gives

$$K\mu^{t+2} - \mathcal{A}_\zeta K\mu^{t+1} + \beta^{-1}K\mu^t = 0. \tag{C.27}$$

Dividing by $K\mu^t$ gives the characteristic equation

$$\mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1} = 0. \tag{C.28}$$

Let μ_1 and μ_2 denote the two roots of (C.28), ordered so that $\mu_1 < \mu_2$. By the quadratic formula,

$$\mu_1 = \frac{\mathcal{A}_\zeta - \sqrt{\mathcal{A}_\zeta^2 - 4\beta^{-1}}}{2}, \qquad \mu_2 = \frac{\mathcal{A}_\zeta + \sqrt{\mathcal{A}_\zeta^2 - 4\beta^{-1}}}{2} \tag{C.29}$$

and the quadratic polynomial can be written equivalently as

$$\mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1} = (\mu - \mu_1)(\mu - \mu_2) = \mu^2 - (\mu_1 + \mu_2)\mu + \mu_1\mu_2 \tag{C.30}$$

therefore matching coefficients gives

$$\mu_1 + \mu_2 = \mathcal{A}_\zeta, \qquad \mu_1\mu_2 = \beta^{-1}. \tag{C.31}$$

The first identity says that $\mathcal{A}_\zeta$ governs the sum of the two roots. The second identity says that intertemporal discounting governs their product through β^{-1}.

The saddle configuration $\mu_1 < 1 < \mu_2$ is satisfied if and only if

$$\frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega}. \tag{C.32}$$

In standard calibrations this condition is typically satisfied because $\omega < \beta$. In that case, the right-hand side of (C.32) is negative, while the left-hand side is clearly always positive: $h, s \in (0,1)$ and $\zeta > 0$.

To see why (C.32) holds, one has from the characteristic equation (C.29) that $p(\mu) \equiv \mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1}$ and then $p(0) = \beta^{-1} > 0$. Hence, a root lies between zero and one if and only if $p(1) < 0$, then:

$$\begin{aligned} p(1) < 0 \quad &\iff \quad \mathcal{A}_\zeta > 1 + \beta^{-1} \\ &\iff \quad \frac{1}{\omega} + \frac{\omega}{\beta} + \frac{1-\omega}{\beta}\frac{(1-h)\zeta}{s} > 1 + \frac{1}{\beta} \\ &\iff \quad \frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega}. \end{aligned}$$

Since $\mu_1\mu_2 = \beta^{-1} > 1$, if $\mu_1 < 1$ then the other root must satisfy $\mu_2 > 1$.

C.3.4. *Forward Solution*

Using the two identities (C.31) that hold for the roots, the equation for asset dynamics (C.24) can be factored as

$$\mathbb{E}_t\left[\tilde{z}^S_{t+2} - \mu_1 \tilde{z}^S_{t+1}\right] - \mu_2\left(\tilde{z}^S_{t+1} - \mu_1 \tilde{z}^S_t\right) = F_t. \tag{C.33}$$

Defining $q_t \equiv \tilde{z}^S_{t+1} - \mu_1 \tilde{z}^S_t$ gives the first-order forward equation

$$\mathbb{E}_t\left[q_{t+1}\right] - \mu_2 q_t = F_t. \tag{C.34}$$

Rearranging:

$$q_t = \mu_2^{-1}\mathbb{E}_t\left[q_{t+1}\right] - \mu_2^{-1} F_t. \tag{C.35}$$

Iterating this equation forward for $J+1$ periods gives

$$q_t = -\sum_{j=0}^{J} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right] + \mu_2^{-(J+1)}\mathbb{E}_t\left[q_{t+J+1}\right]. \tag{C.36}$$

The final term is the terminal assets component associated with the unstable root $\mu_2 > 1$. The forward equation alone would then allow this component to survive, generating explosive paths for liquid assets. The transversality condition must then require that

$$\lim_{J\to\infty} \mu_2^{-(J+1)}\mathbb{E}_t\left[q_{t+J+1}\right] = 0 \tag{C.37}$$

or, equivalently, that $\lim_{j\to\infty} \mu_2^{-j}\mathbb{E}_t\left[q_{t+j}\right] = 0$. Taking the limit of (C.36) then yields

$$q_t = -\sum_{j=0}^{\infty} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right]. \tag{C.38}$$

Using the definition of q_t, the forward solution for liquid assets is

$$\tilde{z}^S_{t+1} = \mu_1 \tilde{z}^S_t - \sum_{j=0}^{\infty} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right]. \tag{C.39}$$

This shows that the root μ_1 governs the persistence of bonds, while μ_2 governs their future discounting.

C.3.5. *The Saver Consumption Function*

The log-linear Saver budget constraint (C.20) can be written as

$$c^S_t = \frac{\tilde{Y}^S}{C^S}\tilde{y}^S_t + \alpha_S(\tilde{r}_t + \tilde{z}^S_t) - \alpha_B \tilde{z}^S_{t+1}. \tag{C.40}$$

The only object that is not dated t is next-period liquid assets, $\tilde{z}^S_{t+1}$. Using the forward solution (C.39) that reads

$$\tilde{z}^S_{t+1} = \mu_1 \tilde{z}^S_t - \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right],$$

the last term of the budget constraint (C.40) becomes

$$-\alpha_B \tilde{z}^S_{t+1} = -\alpha_B \mu_1 \tilde{z}^S_t + \alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right]. \tag{C.41}$$

Substituting this expression back into the budget constraint gives

$$c^S_t = \frac{\tilde{Y}^S}{C^S} \tilde{y}^S_t + \alpha_S \tilde{r}_t + \left(\alpha_S - \alpha_B \mu_1 \right) \tilde{z}^S_t + \alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right]. \tag{C.42}$$

To make the next step explicit, use $F_t = F^0_t / (\omega \alpha_B)$, so that $\alpha_B F_t = F^0_t / \omega$. Therefore,

$$\alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right] = \frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F^0_{t+j} \right]. \tag{C.43}$$

Using the definition of F^0_t in (C.23) and the law of iterated expectations, this last term becomes

$$\begin{aligned} \frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \Bigg\{ & \omega \frac{\tilde{Y}^S}{C^S} \mathbb{E}_t \left[\tilde{y}^S_{t+j+1} \right] + (1-\omega) \frac{\tilde{Y}^H}{C^H} \mathbb{E}_t \left[\tilde{y}^H_{t+j+1} \right] \\ & - \frac{\tilde{Y}^S}{C^S} \mathbb{E}_t \left[\tilde{y}^S_{t+j} \right] + \left[\omega \alpha_S + (1-\omega) \alpha_H - \frac{1}{\sigma} \right] \mathbb{E}_t \left[\tilde{r}_{t+j+1} \right] \\ & - \alpha_S \mathbb{E}_t \left[\tilde{r}_{t+j} \right] \Bigg\}. \end{aligned} \tag{C.44}$$

Substituting (C.44) into (C.42) gives the consumption function before collecting terms. Grouping together the variables that will enter the same line of the final expression:

$$\begin{aligned} c^S_t = & \underbrace{\frac{\tilde{Y}^S}{C^S} \tilde{y}^S_t + \frac{\tilde{Y}^S}{C^S} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{y}^S_{t+j+1} \right] - \frac{1}{\omega} \frac{\tilde{Y}^S}{C^S} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{y}^S_{t+j} \right]}_{\text{Saver income}} \\ & + \underbrace{\frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \left[\omega \alpha_S + (1-\omega) \alpha_H - \frac{1}{\sigma} \right] \mathbb{E}_t \left[\tilde{r}_{t+j+1} \right] - \frac{\alpha_S}{\omega} \sum_{j=1}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{r}_{t+j} \right]}_{\text{Expected future real rates}} \\ & + \underbrace{\left(\alpha_S - \frac{\alpha_S}{\omega \mu_2} \right) \tilde{r}_t + \left(\alpha_S - \alpha_B \mu_1 \right) \tilde{z}^S_t}_{\text{Beginning-of-period wealth}} \end{aligned}$$

$$+\underbrace{\frac{1-\omega}{\omega}\frac{\tilde{Y}^H}{C^H}\sum_{j=0}^{\infty}\mu_2^{-(j+1)}\mathbb{E}_t\left[\tilde{y}^H_{t+j+1}\right]}_{\text{Future HtM income}}. \tag{C.45}$$

Equation (C.45) contains the direct budget effects and the full present-value forcing term, grouped according to the four economic blocks of the final consumption function. Rearranging it requires only shifting the indices inside the sums and simplifying the coefficients, then:

$$\begin{aligned} c_t^S \quad = \quad & \left(1-\frac{1}{\omega\mu_2}\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^S_{t+k}\right] \\ & + \left\{\frac{1}{\omega}\left[\omega\alpha_S+(1-\omega)\alpha_H-\frac{1}{\sigma}\right]-\frac{\alpha_S}{\omega\mu_2}\right\}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\ & + \alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\tilde{r}_t+\left(\alpha_S-\alpha_B\mu_1\right)\tilde{z}^S_t \\ & + \frac{1-\omega}{\omega}\frac{\tilde{Y}^H}{C^H}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^H_{t+k}\right]. \end{aligned} \tag{C.46}$$

The following paragraphs explain the last rearrangements needed to obtain the final formulation of the consumption function that appears in the body of the paper.

The income block. The Saver-income line in (C.46) has the coefficient

$$1-\frac{1}{\omega\mu_2}.$$

Using the root product $\mu_1\mu_2=\beta^{-1}$, one has $\mu_2^{-1}=\beta\mu_1$. Hence

$$1-\frac{1}{\omega\mu_2}=1-\frac{\mu_1}{\omega}\beta.$$

This is the prefactor that appears in the first line of (C.49). The HtM-income block does not contain this prefactor because it is dated only from $t+1$ onward in the forcing term, and therefore enters directly as the future-income present value.

The expected future real-rate block. The coefficient on expected future real rates in (C.46) is

$$\frac{1}{\omega}\left[\omega\alpha_S+(1-\omega)\alpha_H-\frac{1}{\sigma}\right]-\frac{\alpha_S}{\omega\mu_2}.$$

Using $\omega\alpha_S+(1-\omega)\alpha_H=\alpha_B(\omega\mathcal{A}_\zeta-1)$, $\mathcal{A}_\zeta=\mu_1+\mu_2$, and $\alpha_S/\mu_2=\omega\alpha_B\mu_1$, this coefficient becomes

$$\frac{1}{\omega}\left[\alpha_B\left(\omega\mu_1+\omega\mu_2-1\right)-\frac{1}{\sigma}-\omega\alpha_B\mu_1\right]=\alpha_B\left(\mu_2-\frac{1}{\omega}\right)-\frac{1}{\omega\sigma}.$$

Finally,

$$\alpha_B\left(\mu_2-\frac{1}{\omega}\right)-\frac{1}{\omega\sigma}=\frac{\tilde{Z}^S}{C^S}\left(1-\frac{1}{\omega\mu_2}\right)\mu_2-\frac{1}{\omega\sigma}=\frac{\tilde{Z}^S}{C^S}\left(1-\frac{\mu_1}{\omega}\beta\right)\mu_2-\frac{1}{\omega\sigma}, \tag{C.47}$$

where the last equality uses $\mu_2^{-1}=\beta\mu_1$ again. This is the coefficient in the second line of (C.49).

The beginning-of-period wealth block. The beginning-of-period wealth block in (C.46) is

$$\alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\tilde{r}_t+\left(\alpha_S-\alpha_B\mu_1\right)\tilde{z}_t^S.$$

Since $\alpha_S=\omega\alpha_B\mu_1\mu_2$, one has $\alpha_S/(\omega\mu_2)=\alpha_B\mu_1$. Therefore,

$$\alpha_S-\alpha_B\mu_1=\alpha_S-\frac{\alpha_S}{\omega\mu_2}=\alpha_S\left(1-\frac{1}{\omega\mu_2}\right),$$

and the current return and beginning-of-period bond stock have the same coefficient:

$$\alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\left(\tilde{r}_t+\tilde{z}_t^S\right).$$

Using $\mu_2^{-1}=\beta\mu_1$ and $\alpha_S=s\tilde{R}\tilde{Z}^S/C^S$ gives

$$\left(1-\frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t+\tilde{z}_t^S\right), \tag{C.48}$$

which is the third line of (C.49).

Rearranging (C.46) after the simplifications described in the last three paragraphs delivers that the consumption function of a Saver can be written as

$$\begin{aligned}
c_t^S \quad = \quad & \left(1-\frac{\mu_1}{\omega}\beta\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^S\right] \\
& +\left[\frac{\tilde{Z}^S}{C^S}\left(1-\frac{\mu_1}{\omega}\beta\right)\mu_2-\frac{1}{\omega\sigma}\right]\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\
& +\left(1-\frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t+\tilde{z}_t^S\right) \\
& +\frac{\tilde{Y}^H}{C^H}\frac{1-\omega}{\omega}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^H\right].
\end{aligned} \tag{C.49}$$

C.3.6. *Intertemporal MPC Profiles*

This section derives the iMPC profile out of a one-time income shock. That corresponds to the first column of the intertemporal MPC matrix in Auclert et al. (2024).

Saver iMPC profile out of own income. Assume a one-time shock to current Saver income, $\tilde{y}_0^S = 1$. At $t = 0$, the consumption function (C.49) then simplifies to

$$c_0^S = \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{\tilde{Y}^S}{C^S} \tag{C.50}$$

and from the budget constraint (C.20), the beginning-of-next-period bond position (i.e. the liquid assets carried into period $t = 1$) is

$$\tilde{z}_1^S = \frac{\tilde{Y}^S}{\tilde{Z}^S} \frac{\mu_1}{\omega}\beta. \tag{C.51}$$

From $t \geq 1$ there is no income and no rate movement, so consumption from (C.49) reduces to

$$c_t^S = \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s\tilde{R}\tilde{Z}^S}{C^S} \tilde{z}_t^S \tag{C.52}$$

and recalling that asset persistence is governed by μ_1, one has

$$\tilde{z}_{t+1}^S = \mu_1 \tilde{z}_t^S \quad \text{therefore:} \quad \tilde{z}_t^S = \mu_1^{t-1} \tilde{z}_1^S. \tag{C.53}$$

Combining (C.52) together with (C.53), and using $s\tilde{R} = \frac{\omega}{\beta}$ from (C.15) one has

$$c_t^S \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s\tilde{R}\tilde{Z}^S}{C^S} \mu_1^{t-1} \frac{\tilde{Y}^S}{\tilde{Z}^S} \frac{\mu_1}{\omega}\beta \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{\tilde{Y}^S}{C^S} \mu_1^t, \qquad t \geq 1, \tag{C.54}$$

which also reproduces (C.50) at $t = 0$. Converting to level responses, the Saver's iMPC path is then given by

$$\frac{\partial C_t^S}{\partial \tilde{Y}_0^S} \quad = \quad \frac{C^S}{\tilde{Y}^S} \frac{\partial c_t^S}{\partial \tilde{y}_0^S} \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \mu_1^t, \qquad t \geq 0. \tag{C.55}$$

On impact, the Saver consumes a fraction $\left(1 - \frac{\mu_1}{\omega}\beta\right)$ of the shock, the remaining fraction $\frac{\mu_1}{\omega}\beta$ goes into savings, through liquid bonds, which then decay at the stable root $\mu_1 \in (0, 1)$, and so does consumption. Since (C.55) involves only μ_1, ω and β, the consumption path is independent of the level of steady-state liquidity.

Aggregate iMPC profile out of aggregate income. Similarly to the previous paragraph, assume a one-time aggregate income shock with equal incidence across states, that is

$$\tilde{y}_0^S = \tilde{y}_0^H = 1.$$

The HtM income shock affects neither the Saver's consumption at $t = 0$, nor aggregate liquid assets since HtM are constrained. Hence the Saver's path is unchanged from the one described in (C.55).

The HtM on the other hand consume their income on impact, intuitively from the budget constraint (C.19):

$$\frac{\partial c_0^H}{\partial \tilde{y}_0^H} = \frac{\tilde{Y}^H}{C^H} \quad \text{and} \quad \frac{\partial C_0^H}{\partial \tilde{Y}_0^H} = 1 \tag{C.56}$$

Then, from $t \geq 1$, the HtM consume the gross payoff on the pooled bonds that are brought by transitioning Savers. From the HtM flow budget constraint (C.19) combined with bond persistence (C.53) one then has:

$$c_t^H = \frac{C^H - \tilde{Y}^H}{C^H} \mu_1^{t-1} \tilde{z}_1^S, \qquad t \geq 1. \tag{C.57}$$

Concerning aggregate responses, define aggregate consumption and income as

$$C_t = \lambda C_t^H + (1-\lambda) C_t^S \quad \text{and} \quad \tilde{Y}_t = \lambda \tilde{Y}_t^H + (1-\lambda)\tilde{Y}_t^S$$

with steady-state levels

$$C = \lambda C^H + (1-\lambda) C^S \quad \text{and} \quad \tilde{Y} = \lambda \tilde{Y}^H + (1-\lambda)\tilde{Y}^S$$

respectively. The log deviation of aggregate consumption is then the consumption-weighted average of the group log deviations, $c_t = \frac{\lambda C^H c_t^H + (1-\lambda) C^S c_t^S}{C}$, and the equal-incidence shock moves aggregate income by $\tilde{y}_0 = 1$.

On impact at $t = 0$, HtM consume the full income shock and Savers consume the fraction $1 - \frac{\mu_1}{\omega}\beta$ of their own income shock.

For $t \geq 1$, there are no other shocks. Aggregate consumption therefore comes only from the bonds accumulated by Savers at $t = 0$, and then pooled within groups after the transitions. Starting from the definition of aggregate consumption,

$$C c_t = \lambda C^H c_t^H + (1-\lambda) C^S c_t^S.$$

Using the Saver's response (C.52), the bond's persistence (C.53), and the HtM consumption path (C.57), this becomes

$$\begin{aligned} C c_t &= \lambda C^H \left[\frac{C^H - \tilde{Y}^H}{C^H} \mu_1^{t-1} \tilde{z}_1^S \right] + (1-\lambda) C^S \left[\left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s \tilde{R} \tilde{Z}^S}{C^S} \mu_1^{t-1} \tilde{z}_1^S \right] \\ &= \left[\lambda \left(C^H - \tilde{Y}^H\right) + (1-\lambda) \left(1 - \frac{\mu_1}{\omega}\beta\right) s \tilde{R} \tilde{Z}^S \right] \mu_1^{t-1} \tilde{z}_1^S. \end{aligned}$$

The first term in brackets can be rewritten using the HtM steady-state budget constraint and stationarity of population shares:

$$\lambda \left(C^H - \tilde{Y}^H\right) = \lambda (1-h) \tilde{R} \tilde{Z}^S = (1-\lambda)(1-s) \tilde{R} \tilde{Z}^S.$$

Moreover, since $s\tilde{R} = \frac{\omega}{\beta}$, then $\left(1 - \frac{\mu_1}{\omega}\beta\right) s\tilde{R} = s\tilde{R} - \mu_1$. Therefore

$$\begin{aligned} C c_t &= (1-\lambda)\left[(1-s)\tilde{R} + s\tilde{R} - \mu_1\right]\tilde{Z}^S \mu_1^{t-1}\tilde{z}_1^S \\ &= (1-\lambda)\left(\tilde{R} - \mu_1\right)\tilde{Z}^S \mu_1^{t-1}\tilde{z}_1^S. \end{aligned}$$

Finally, substituting (C.51), $\tilde{z}_1^S = \frac{\tilde{Y}^S}{\tilde{Z}^S}\frac{\mu_1}{\omega}\beta$, gives

$$c_t = (1-\lambda)\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{C}\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1}, \qquad t \geq 1,$$

and therefore the aggregate consumption response in log-deviations can be summarized by:

$$c_0 = \frac{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S\left(1 - \frac{\mu_1}{\omega}\beta\right)}{C}, \qquad c_t = (1-\lambda)\,\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{C}\,\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1} \quad t \geq 1. \tag{C.58}$$

Converting to levels, the aggregate iMPC is the log-deviation response rescaled by the aggregate consumption-to-income ratio, $\frac{dC_t}{d\tilde{Y}_0} = \frac{C}{\tilde{Y}}\frac{c_t}{\tilde{y}_0} = \frac{C}{\tilde{Y}}c_t$. The impact aggregate response is then

$$\frac{dC_0}{d\tilde{Y}_0} = \frac{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S\left(1 - \frac{\mu_1}{\omega}\beta\right)}{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S}, \tag{C.59}$$

and for $t \geq 1$

$$\frac{dC_t}{d\tilde{Y}_0} = (1-\lambda)\,\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S}\,\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1}. \tag{C.60}$$

C.3.7. *Comparison with the Representative-Agent Benchmark*

The representative-agent (RA) benchmark is nested at $\omega = 1$: there $\mathcal{A}_\zeta = 1 + \beta^{-1}$, so the characteristic polynomial factors as $P(\mu) = (\mu - 1)\left(\mu - \beta^{-1}\right)$ and

$$\mu_1\big|_{\omega=1} = 1, \qquad \mu_2\big|_{\omega=1} = \beta^{-1}: \tag{C.61}$$

the impact MPC is $1 - \beta$, future income is discounted at β, and the response to the k-period-ahead rate is $-\frac{1}{\sigma}\beta^k$. This section compares the THANK responses with this benchmark, term by term. All three comparisons reduce to the position of the polynomial $P(\mu) = \mu^2 - \mathcal{A}_\zeta\mu + \beta^{-1}$ at a single evaluation point, since $P(\mu) < 0$ exactly for $\mu \in (\mu_1, \mu_2)$.

The impact MPC. The Saver's impact MPC exceeds the RA value if and only if

$$1 - \frac{\mu_1}{\omega}\beta \;>\; 1 - \beta \quad \Longleftrightarrow \quad \mu_1 < \omega \quad \Longleftrightarrow \quad P(\omega) < 0. \tag{C.62}$$

Substituting (C.25) and factoring:

$$P(\omega) = \omega^2 - \mathcal{A}_\zeta\,\omega + \beta^{-1} = (1-\omega)\left[\frac{(1+\omega)(1-\beta)}{\beta} - \frac{\omega}{\beta}\,\frac{(1-h)\,\zeta}{s}\right], \tag{C.63}$$

so that, with $1-\omega>0$ and $1+\frac{1}{\omega}=\frac{2s+(1-s)\zeta^{\sigma}}{s}$,

$$\mu_1<\omega \quad\Longleftrightarrow\quad (1-h)\,\zeta \;>\; (1-\beta)\left[2s+(1-s)\,\zeta^{\sigma}\right]. \tag{C.64}$$

At quarterly discount factors the right-hand side of (C.64) is of order 10^{-2}, so the condition holds except in a negligible neighbourhood of $h=1$: in the infinite-horizon model, front-loading generally prevails and the Saver's impact MPC exceeds the RA value.

The profile of the response to future income. From (C.49), the Saver's responses to income at different horizons satisfy

$$\frac{M_{0,k}}{M_{0,0}}=\mu_2^{-k}=\left(\beta\mu_1\right)^k, \qquad M_{0,k}\equiv\frac{\partial C_0^S}{\partial\,\mathbb{E}_0[\tilde{Y}_k^S]}, \tag{C.65}$$

against β^k in the RA benchmark. The profile tilts towards the present — future income is discounted more steeply than by β — if and only if $\mu_1<1$, which is the saddle condition (C.32) itself, so that the tilt holds whenever the model has a solution.

Intertemporal substitution. The response to the k-period-ahead expected rate is $-\frac{1}{\sigma\omega}\left(\beta\mu_1\right)^k$ (abstracting from the wealth-income terms, which are scaled by $\frac{\tilde{Z}^S}{C^S}$ and vanish at zero liquidity), against $-\frac{1}{\sigma}\beta^k$ in the RA benchmark. The ratio of the two is

$$\frac{\mu_1^k}{\omega}, \tag{C.66}$$

so the substitution response is attenuated at every horizon $k\geq 1$ whenever $\mu_1<\omega$ — the same condition (C.64) that amplifies the impact MPC.

C.3.8. *Relation to Bilbiie (2025) and to Auclert et al. (2024)*

Bilbiie (2025) derives analytical iMPCs for the THANK model at the *symmetric* zero-liquidity steady state ($C^H=C^S$, $\tilde{R}=\beta^{-1}$): his equation (31) is a second-order difference equation in aggregate liquid assets with trace $\Theta\equiv\frac{1}{s}+\beta^{-1}\left[1+\frac{1-s}{s}\left(\frac{1-s}{\lambda}-1\right)\right]$, and his Proposition 8 gives the resulting iMPC profile, decaying geometrically at the stable root x_b. This appendix nests that result. Setting $\zeta=1$ (so $\omega=s$) in (C.25) and using the stationarity of the shares, a direct computation gives

$$\mathcal{A}_{\zeta=1}=\frac{1}{s}+\beta^{-1}\left[1+\frac{1-s}{s}\left(\frac{1-s}{\lambda}-1\right)\right]=\Theta, \tag{C.67}$$

so that $\mu_1|_{\zeta=1}=x_b$ and the iMPCs of Section C.3.6 reproduce his Proposition 8 at the symmetric steady state.

Relative to that benchmark, the derivation here generalizes along three dimensions. First, the steady state is unequal: $\zeta>1$ replaces the transition probability s by the marginal-utility weight $\omega<s$ in the asset dynamics, and depresses the steady-state return, $\tilde{R}<\beta^{-1}$. Since $\mathcal{A}_\zeta$ moves with ζ and σ, steady-state inequality is a determinant of iMPC *persistence* — a

margin absent in the symmetric case, where the stable root depends on (λ, s, β) only; cyclical income inequality, the margin emphasized by Bilbiie (2025), raises the impact response but leaves x_b unchanged. Second, liquidity is positive: the consumption function (C.49) evaluates the full path of real rates (the revaluation of the outstanding stock and the wealth-income effect of expected future rates) channels that vanish at zero liquidity. Third, the object derived is the household-level consumption function in arbitrary income and rate paths, before any equilibrium income closure; the state-by-state iMPCs of Section C.3.6 accordingly sharpen the aggregate profile: the kink between the impact response and the decay phase originates entirely in the HtM spike, while the Saver's own path is geometric from impact.

This last point admits a useful reformulation in the language of the sequence-space literature. The consumption function (C.49) expresses c_t^S as a linear functional of the expected sequences $\{\mathbb{E}_t[\tilde{y}_{t+k}^S]\}$, $\{\mathbb{E}_t[\tilde{y}_{t+k}^H]\}$ and $\{\mathbb{E}_t[\tilde{r}_{t+k}]\}$ and of the bond position; combined with the law of motion (C.39), which solves the position forward in the same sequences. It then delivers the consumption *sequence* as a linear map of the shock sequences and the initial condition. The consumption functions therefore constitute the THANK household block in *sequence space*: its intertemporal Jacobians. The matrices of partial derivatives of consumption at every date with respect to income and rates at every date, which Auclert et al. (2024) compute numerically for quantitative heterogeneous-agent models, are here available in closed form

C.4. Properties of the Consumption Functions

This appendix derives the properties of the consumption function of a Saver (32) summarized in Section 4.2: the intertemporal MPCs out of income, the response to the expected path of real interest rates, the MPC out of beginning-of-period wealth, and the response to news about HtM income.

Savers' iMPC out of their income. From the first line of (32), for a shock that affects the Saver's income in period $t = 0$, a fraction $1 - \frac{\mu_1}{\omega}\beta$ of the additional income is consumed on impact, and the remaining fraction $\frac{\mu_1}{\omega}\beta$ is saved in the form of bonds, whose persistence in the Saver state is captured by the stable root $\mu_1 < 1$. In fact, the intertemporal MPC path can be characterized as follows.

Corollary 3 (iMPC path of a Saver in response to an income shock)**.** *For a shock that only affects the level of income of a Saver at time $t = 0$, the intertemporal MPC path in levels, resulting from the consumption function* (32)*, is given by*

$$\frac{\partial C_t^S}{\partial \tilde{Y}_0^S} = \frac{C^S}{\tilde{Y}^S}\frac{\partial c_t^S}{\partial \tilde{y}_0^S} = \left(1 - \frac{\mu_1}{\omega}\beta\right)\mu_1^t, \qquad t \geq 0. \tag{C.68}$$

Proof. See Appendix C.3. □

It follows that at time $t = 0$ the impact MPC of a Saver, compared to the representative-agent benchmark, $1 - \beta$, is higher if and only if

$$\mu_1 < \omega. \tag{C.69}$$

The stable root μ_1 captures the effective persistence of bonds in the Saver state, while ω captures the valuation of remaining a Saver, from the Euler equation for bonds. When $\mu_1 < \omega$, bonds' persistence is lower than their valuation in the Saver state: saving is a weaker instrument for transferring resources to future Saver consumption, and the Saver front-loads more of the income shock into current consumption. Conversely, when $\mu_1 > \omega$, the self-insurance motive dominates and the impact response is lower than in the representative-agent benchmark. Appendix C.3 shows that the inequality (C.69) can be pinned down in terms of model primitives, and that $\mu_1 < \omega$ is satisfied with standard calibrations.

To evaluate the consumption response to future income, using a notation for iMPCs akin to Auclert et al. (2024), $M_{0,k}^{S,S}$ denotes the *Saver* consumption response in period $t = 0$ to a *Saver* income shock in period $t = k$, that is

$$M_{0,k}^{S,S} \equiv \frac{\partial\, C_0^S}{\partial\, \mathbb{E}_0[\tilde{Y}_k^S]} = \left(1 - \frac{\mu_1}{\omega}\beta\right) \mu_2^{-k}.$$

That, combined with the consumption function (32), yields the following corollary result.

Corollary 4 (iMPC ratios to future income shocks)**.** *The consumption profile in the Saver state, in response to Saver income shocks at different horizons t, satisfies*

$$\frac{M_{0,k}^{S,S}}{M_{0,0}^{S,S}} = \mu_2^{-k} = \left(\beta\mu_1\right)^k. \tag{C.70}$$

Hence, the saddle condition (30), $\mu_1 < 1$, implies that future income in THANK is discounted more than in the representative agent model, and therefore the consumption profile tilts towards the present. Since the saddle condition is necessary for the model to have a solution, the tilt holds whenever the model has a solution.

Overall, the Saver's consumption response to income shocks in the infinite horizon THANK is characterized by front-loading prevailing over self-insurance: the saddle condition $\mu_1 < 1$ tilts the consumption profile towards the present, and under baseline calibration $\mu_1 < \omega$ is satisfied, so that the impact MPC to a current income shock is greater than in a representative agent framework with complete markets.

Savers' response to the expected future real interest rate. The second line of the consumption function (32) describes how current Saver consumption responds to movements in expected future real interest rates. As with future income, the sequence of expected real rates is discounted by μ_2^{-k}.

The coefficient multiplying this sequence combines two channels. The first is the intertemporal-substitution channel, captured by $-\frac{1}{\sigma\omega}$. A higher expected future real rate makes

saving more attractive and tends to reduce current consumption. In a representative-agent model, the strength of this channel is governed by the intertemporal elasticity of substitution, $1/\sigma$. In THANK, the same channel is scaled by $1/\omega$, because the Euler equation weights future marginal utility across both possible states.

The second channel is a wealth-revaluation channel. If Savers hold positive liquid assets in steady state, expected future real rates also affect the value of the asset position carried through the consumption function. A higher future real rate raises the returns on the bonds that reach that future date, and this tends to increase current consumption through the beginning-of-period wealth side of the budget constraint. This channel is proportional to $\left(1 - \frac{\mu_1}{\omega}\beta\right)$, the Saver's impact MPC out of current resources: the more strongly current consumption reacts to liquid resources, the stronger the revaluation effect. Therefore, the sign of the real-rate response depends on the balance between the incentive to postpone consumption when future returns are higher and the increase in resources implied by higher returns on the liquid asset position.

This naturally leads to the next paragraph, which isolates the direct response of current consumption to beginning-of-period wealth.

Savers' MPC out of beginning-of-period wealth. The third line of the consumption function (32) captures the response to changes in beginning-of-period liquid wealth. This includes shocks to the bonds brought into the period, $\tilde{z}_t^S$, and to their real return, $\tilde{r}_t$. The MPC out of gross beginning-of-period wealth, in levels, is

$$\frac{\partial\, C_t^S}{\partial\left(\tilde{R}_t \tilde{Z}_t^S\right)} = s\left(1 - \frac{\mu_1}{\omega}\beta\right) \tag{C.71}$$

and the factor s reflects the state-transition accounting: only the fraction s of Saver bond holdings remains in the Saver state, while the fraction $1 - s$ is pooled into the HtM state. Conditional on this retained liquid wealth, this MPC is the same as the Saver's MPC out of their current non-interest income, $1 - \frac{\mu_1}{\omega}\beta$. It is lower when liquid assets are more persistent in the Saver state, μ_1 is higher, and higher when the Euler-equation valuation of the Saver state, ω, is higher.

Savers' iMPC out of HtM income. From the fourth line of the consumption function (32), the response at time $t = 0$, in levels, to news about HtM income at horizon k is given by

$$M_{0,k}^{S,H} \equiv \frac{\partial\, C_0^S}{\partial\, \mathbb{E}_0\left[\tilde{Y}_k^H\right]} \quad = \quad \frac{C^S}{C^H}\frac{1-\omega}{\omega}\mu_2^{-k} \quad = \quad \zeta^{\sigma+1}\frac{1-s}{s}\left(\beta\mu_1\right)^k, \qquad k \geq 1, \tag{C.72}$$

where the second equality uses the definition of ω. Intuitively, current Saver consumption only responds to news about future HtM income ($k \geq 1$). The term $\frac{1-\omega}{\omega}$ captures the relative Euler-equation weight of the HtM state compared to the Saver state. The term $\zeta = \frac{C^S}{C^H}$ rescales the response across steady-state consumption units. The term $\mu_2^{-k} = (\beta\mu_1)^k$ discounts expected HtM income at horizon k through the same forward-discounting root, $\mu_2 > 1$, that is used for future Saver income.

Future HtM income enters the Saver's consumption function through the Euler equation: a current Saver values this income because it raises consumption in the future HtM state, in which the household would be constrained. A higher expected income in a future HtM state raises future HtM consumption, and thus lowers the marginal-utility value of additional self-insurance against becoming HtM. The self-insurance motive then weakens, and current Saver consumption rises.

Therefore, comparing the Saver's consumption response at $t = 0$ to news about future HtM income, $M_{0,k}^{S,H}$, with the response to news about future Saver income, $M_{0,k}^{S,S}$, yields the following corollary.

Corollary 5 (iMPC comparison: HtM vs. Saver income shocks)**.** *For horizon $k \geq 1$, $M_{0,k}^{S,H} > M_{0,k}^{S,S}$ is verified if and only if*

$$\frac{1-s}{s}\,\zeta^{\sigma+1} > 1 - \frac{\mu_1}{\omega}\,\beta. \tag{C.73}$$

Since both future income sequences are discounted by the same forward-discounting term, μ_2^{-k} drops out of the comparison. The left-hand side measures how strongly current Saver consumption responds to expected future HtM income, while the right-hand side measures the corresponding response to expected future income in the Saver state, which is equal to the Saver impact MPC out of current income. Condition (C.73) is satisfied at the baseline calibration: per unit of income, the Saver's consumption reacts *more* to news about HtM income than to news about Saver income.

Appendix D. Consumption Functions and Closure under Subjective Expectations

This appendix derives equation (48). Before substituting the perceived real-rate path (39), the aggregate-demand equation (47) reads

$$\gamma_S \frac{\mu_1 \beta}{s} y_t = \left[\left(1 - \frac{\mu_1 \beta}{s}\right)\gamma_S + \frac{1-s}{s}\chi\right] \sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_t[y_{t+k}] - \frac{1}{s\sigma} \sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_t[\tilde{r}_{t+k}]. \tag{D.1}$$

Separating the first term of each sum, and using the law of iterated expectations that applies for both expectation operators, that yields

$$\begin{aligned} \sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_t[y_{t+k}] &= \frac{1}{\mu_2}\, \mathbb{E}_t[y_{t+1}] + \frac{1}{\mu_2}\, \mathbb{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_{t+1}[y_{t+1+k}]\right], \\ \sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_t[\tilde{r}_{t+k}] &= \frac{1}{\mu_2}\, \hat{E}_t[\tilde{r}_{t+1}] + \frac{1}{\mu_2}\, \hat{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]. \end{aligned} \tag{D.2}$$

Multiplying (D.1) by μ_2 and using (D.2) gives

$$\begin{aligned} \gamma_S \frac{\mu_1 \mu_2 \beta}{s} y_t = \left[\left(1 - \frac{\mu_1 \beta}{s}\right)\gamma_S + \frac{1-s}{s}\chi\right] &\left(\mathbb{E}_t[y_{t+1}] + \mathbb{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_{t+1}[y_{t+1+k}]\right]\right) \\ - \frac{1}{s\sigma} &\left(\hat{E}_t[\tilde{r}_{t+1}] + \hat{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]\right). \end{aligned}$$

Taking the objective expectation $\mathbb{E}_t$ of (D.1) in period $t+1$ gives

$$\begin{aligned} \gamma_S \frac{\mu_1 \beta}{s}\, \mathbb{E}_t[y_{t+1}] = \left[\left(1 - \frac{\mu_1 \beta}{s}\right)\gamma_S + \frac{1-s}{s}\chi\right] &\mathbb{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \mathbb{E}_{t+1}[y_{t+1+k}]\right] \\ - \frac{1}{s\sigma}\, &\mathbb{E}_t\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]. \end{aligned}$$

Subtracting the second equation from the first eliminates the expected sum of future output, and the coefficients on $\mathbb{E}_t[y_{t+1}]$ add up to $\gamma_S + \frac{1-s}{s}\chi$:

$$\gamma_S \frac{\mu_1 \mu_2 \beta}{s} y_t = \left(\gamma_S + \frac{1-s}{s}\chi\right)\mathbb{E}_t[y_{t+1}] - \frac{1}{s\sigma}\left(\hat{E}_t[\tilde{r}_{t+1}] + \left(\hat{E}_t - \mathbb{E}_t\right)\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]\right).$$

Since $\mu_1 \mu_2 = \beta^{-1}$, the coefficient on y_t is $\gamma_S \frac{\mu_1 \mu_2 \beta}{s} = \frac{\gamma_S}{s}$. Dividing by $\frac{\gamma_S}{s}$, the coefficient on $\mathbb{E}_t[y_{t+1}]$ becomes, recalling $\gamma_S = \frac{1-\lambda\chi}{1-\lambda}$,

$$s + \frac{(1-s)\chi}{\gamma_S} = s + \chi(1-s)\frac{1-\lambda}{1-\lambda\chi} = 1 + (1-s)\frac{\chi(1-\lambda) - (1-\lambda\chi)}{1-\lambda\chi} = 1 + (\chi - 1)\frac{1-s}{1-\lambda\chi} = \delta, \tag{D.3}$$

and the coefficient on the real-rate terms becomes $\frac{1}{s\sigma}\frac{s}{\gamma_S} = \frac{1}{\sigma\gamma_S}$. Hence

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left(\hat{E}_t[\tilde{r}_{t+1}] + \left(\hat{E}_t - \mathbb{E}_t\right)\left[\sum_{k=1}^{\infty}\mu_2^{-k}\,\hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]\right). \tag{D.4}$$

Under subjective beliefs, the aggregate-demand equation therefore implies the aggregate Euler equation of Bilbiie (2025) with the subjective one-period real rate, plus a belief-revision term: the difference between the subjective and the objective expectation of the next-period perceived path of real rates.

Appendix E. Model Solution and the Actual Law of Motion

This appendix solves for the equilibrium defined in Section 4.5 and computes the fixed point between the perceived and the actual law of motion of inflation. Appendix E.1 writes the equilibrium system, Appendix E.2 solves it by undetermined coefficients, and Appendix E.3 derives the actual law of motion of inflation, the T-map, and the fixed point.

E.1. The Equilibrium System

The equilibrium of Section 4.5 is characterized by the aggregate-demand equation (47), the Phillips curve and the Taylor rule (50), the law of motion of the nowcast (35), and the process of the cost-push shock (51).

Aggregate demand in Euler form. Appendix D shows that quasi-differencing the aggregate-demand equation (47) gives the aggregate Euler equation (D.4),

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left(\hat{E}_t[\tilde{r}_{t+1}] + \left(\hat{E}_t - \mathbb{E}_t\right)\left[\sum_{k=1}^{\infty}\mu_2^{-k}\,\hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right]\right). \tag{E.1}$$

This equation holds for any beliefs of Savers about the real rate. The real rate between periods t and $t+1$ is $\tilde{r}_{t+1} = i_t - \pi_{t+1}$. Savers observe the current nominal rate i_t, and, by (36) with $k=1$, they expect next-period inflation to be $\hat{E}_t[\pi_{t+1}] = \rho\,\hat{\pi}_t$. Hence

$$\hat{E}_t[\tilde{r}_{t+1}] = i_t - \rho\,\hat{\pi}_t. \tag{E.2}$$

Equation (39) gives the perceived real-rate path in period t as a function of i_t and $\hat{\pi}_t$. The same equation in period $t+1$ gives

$$\sum_{k=1}^{\infty}\mu_2^{-k}\,\hat{E}_{t+1}[\tilde{r}_{t+1+k}] = \frac{i_{t+1}}{\mu_2} + \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\,\hat{\pi}_{t+1}. \tag{E.3}$$

With rule-aware beliefs, Savers apply the Taylor rule (50) to their own forecast of inflation, so that $\hat{E}_t[i_{t+1}] = \phi_\pi \hat{E}_t[\pi_{t+1}] = \phi_\pi\rho\,\hat{\pi}_t$. Since Savers expect no revision of their nowcast, $\hat{E}_t[\hat{\pi}_{t+1}] = \rho(1-\mathbf{K})\,\hat{\pi}_t + \mathbf{K}\rho\,\hat{\pi}_t = \rho\,\hat{\pi}_t$, and taking the subjective expectation gives

$$\hat{E}_t\left[\sum_{k=1}^{\infty}\mu_2^{-k}\,\hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right] = \frac{\phi_\pi}{\mu_2}\,\rho\,\hat{\pi}_t + \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\,\rho\,\hat{\pi}_t. \tag{E.4}$$

Similarly, the objective expectation of the same path yields

$$\mathbb{E}_t\left[\sum_{k=1}^{\infty}\mu_2^{-k}\,\hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right] = \frac{\phi_\pi}{\mu_2}\,\mathbb{E}_t[\pi_{t+1}] + \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\left(\rho(1-\mathbf{K})\,\hat{\pi}_t + \mathbf{K}\,\mathbb{E}_t[\pi_{t+1}]\right). \tag{E.5}$$

Subtracting the two gives

$$\left(\hat{E}_t - \mathbb{E}_t\right)\left[\sum_{k=1}^{\infty} \mu_2^{-k}\, \hat{E}_{t+1}[\tilde{r}_{t+1+k}]\right] = \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right)\left(\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]\right). \tag{E.6}$$

Substituting into (D.4) gives the aggregate Euler equation under the beliefs of Corollary 1,

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left[i_t - \rho\,\hat{\pi}_t + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right)\left(\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]\right)\right]. \tag{E.7}$$

Since $\mu_2 > 1$, (47) and (E.7) are equivalent among bounded solutions, so that the equilibrium can be computed with (E.7).

Relation to the aggregate Euler equation in THANK under FIRE. Adding and subtracting $\mathbb{E}_t[\pi_{t+1}]$ in the square bracket of (E.7), the bracket equals

$$i_t - \mathbb{E}_t[\pi_{t+1}] + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} - 1\right)\left(\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]\right),$$

where the factor in parentheses simplifies to

$$\frac{\phi_\pi}{\mu_2} - 1 + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} = \frac{(\phi_\pi - \mu_2)(\mu_2 - \rho) + \mathbf{K}\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} = \frac{(\phi_\pi - \mu_2)\left(\mu_2 - \rho(1 - \mathbf{K})\right)}{\mu_2(\mu_2 - \rho)}.$$

Hence (E.7) can be written as

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left(i_t - \mathbb{E}_t[\pi_{t+1}]\right) - \frac{1}{\sigma\gamma_S}\,\frac{(\phi_\pi - \mu_2)\left(\mu_2 - \rho(1 - \mathbf{K})\right)}{\mu_2(\mu_2 - \rho)}\left(\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]\right). \tag{E.8}$$

The first two terms on the RHS are the aggregate Euler equation of Bilbiie (2025), (49). The last term is proportional to $\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]$, the expected error of Savers in forecasting next-period inflation. The aggregate Euler equation of Bilbiie (2025) is therefore recovered when this term is zero, which holds in two cases. The first is when the one-period forecast of Savers is correct, $\rho\,\hat{\pi}_t = \mathbb{E}_t[\pi_{t+1}]$. In equilibrium, this is the case with $\mathbf{K} = 1$ and $\rho = \rho_u$. The second is the knife-edge case $\phi_\pi = \mu_2$, in which forecast errors of Savers do not affect the perceived real-rate path.

The system. With the Taylor rule $i_t = \phi_\pi \pi_t$ of (50), the equilibrium is a solution of three equations in output, inflation, and the nowcast. The first is the aggregate Euler equation (E.7),

$$y_t = \delta\,\mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma\gamma_S}\left[\phi_\pi\pi_t - \rho\,\hat{\pi}_t + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right)\left(\rho\,\hat{\pi}_t - \mathbb{E}_t[\pi_{t+1}]\right)\right].$$

The second is the New Keynesian Phillips curve (50),

$$\pi_t = \beta\,\mathbb{E}_t[\pi_{t+1}] + \kappa\,y_t + u_t.$$

The third is the law of motion of the nowcast (35),

$$\hat{\pi}_t = \rho(1 - \mathbf{K})\,\hat{\pi}_{t-1} + \mathbf{K}\,\pi_t.$$

The cost-push shock follows (51), $u_t = \rho_u u_{t-1} + \varepsilon_t^u$, and the nominal rate follows from inflation through the Taylor rule. The system has two forward-looking variables, y_t and π_t, which depend on expectations of the next period, and one predetermined variable, $\hat{\pi}_{t-1}$.

E.2. Solution by Undetermined Coefficients

This subsection solves the system of Appendix E.1 for given beliefs ρ and $\mathbf{K}$. The solution is the actual law of motion (ALM) of the economy.

The guess. The state of the economy in period t consists of the nowcast of the previous period, $\hat{\pi}_{t-1}$, and the shock, u_t. Output and inflation are therefore guessed to be linear in these two variables,

$$y_t = \vartheta_\pi^y\,\hat{\pi}_{t-1} + \vartheta_u^y\,u_t, \qquad \pi_t = \vartheta_\pi^\pi\,\hat{\pi}_{t-1} + \vartheta_u^\pi\,u_t, \tag{E.9}$$

with four unknown coefficients, ϑ_π^y, ϑ_π^π, ϑ_u^y, and ϑ_u^π. Substituting the guess for inflation into the law of motion of the nowcast (35) gives

$$\hat{\pi}_t = \big(\rho(1 - \mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi\big)\,\hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi\,u_t. \tag{E.10}$$

The coefficients on the nowcast. Without shocks, $u_t = 0$, the guess (E.9) gives $\pi_t = \vartheta_\pi^\pi\hat{\pi}_{t-1}$ and $y_t = \vartheta_\pi^y\hat{\pi}_{t-1}$, and (E.10) gives $\hat{\pi}_t = \big(\rho(1 - \mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi\big)\,\hat{\pi}_{t-1}$. The nowcast, inflation, and output are all proportional to the nowcast of the previous period, and all three decay at the same rate, the persistence of the nowcast in equilibrium, denoted z.

Hence $\hat{\pi}_t = z\,\hat{\pi}_{t-1}$, $\mathbb{E}_t[y_{t+1}] = z\,y_t$, and $\mathbb{E}_t[\pi_{t+1}] = z\,\pi_t$, and each of the three equations of the system gives a relation between $\hat{\pi}_{t-1}$, π_t, and y_t. The law of motion of the nowcast gives

$$\begin{aligned} \hat{\pi}_t = z\,\hat{\pi}_{t-1} = \rho(1 - \mathbf{K})\,\hat{\pi}_{t-1} + \mathbf{K}\,\pi_t \quad &\Longleftrightarrow \quad \big(z - \rho(1 - \mathbf{K})\big)\,\hat{\pi}_{t-1} = \mathbf{K}\pi_t, \\ &\Longleftrightarrow \quad \hat{\pi}_{t-1} = \frac{\mathbf{K}}{z - \rho(1 - \mathbf{K})}\,\pi_t, \end{aligned} \tag{E.11}$$

and the Phillips curve gives

$$\begin{aligned} \pi_t = \beta\,\mathbb{E}_t[\pi_{t+1}] + \kappa\,y_t = \beta z\,\pi_t + \kappa\,y_t \quad &\Longleftrightarrow \quad \kappa\,y_t = (1 - \beta z)\,\pi_t, \\ &\Longleftrightarrow \quad y_t = \frac{1 - \beta z}{\kappa}\,\pi_t. \end{aligned} \tag{E.12}$$

Since $\pi_t = \vartheta_\pi^\pi\hat{\pi}_{t-1}$ and $y_t = \vartheta_\pi^y\hat{\pi}_{t-1}$, relations (E.11) and (E.12) give the coefficients on the nowcast as functions of z,

$$\vartheta_\pi^\pi = \frac{z - \rho(1 - \mathbf{K})}{\mathbf{K}}, \qquad \vartheta_\pi^y = \frac{1 - \beta z}{\kappa}\,\vartheta_\pi^\pi\,, \tag{E.13}$$

and z remains to be determined. It is pinned down by the third equation of the system: the aggregate Euler equation (E.7), with $i_t = \phi_\pi \pi_t$ and $\hat{\pi}_t = z\,\hat{\pi}_{t-1}$. That is

$$y_t = \delta z\, y_t - \frac{1}{\sigma\gamma_S}\left[\phi_\pi \pi_t - \rho z\,\hat{\pi}_{t-1} + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z\left(\rho\,\hat{\pi}_{t-1} - \pi_t\right)\right],$$

or, rearranging,

$$(1 - \delta z)\, y_t + \frac{1}{\sigma\gamma_S}\left[\phi_\pi \pi_t - \rho z\,\hat{\pi}_{t-1} + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z\left(\rho\,\hat{\pi}_{t-1} - \pi_t\right)\right] = 0. \tag{E.14}$$

Substituting (E.11) and (E.12) into (E.14), and dividing by π_t,[28] gives

$$\frac{(1-\delta z)(1-\beta z)}{\kappa} + \frac{1}{\sigma\gamma_S}\left[\phi_\pi - \frac{\rho\mathbf{K}z}{z - \rho(1-\mathbf{K})} + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z\left(\frac{\rho\mathbf{K}}{z - \rho(1-\mathbf{K})} - 1\right)\right] = 0.$$

Expanding the product in the square bracket, the bracket equals

$$\phi_\pi - \frac{\rho\mathbf{K}z}{z - \rho(1-\mathbf{K})} + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right)\frac{\rho\mathbf{K}z}{z - \rho(1-\mathbf{K})} - \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z.$$

The second and the third term both contain $\frac{\rho\mathbf{K}z}{z-\rho(1-\mathbf{K})}$. Collecting them, the bracket equals

$$\phi_\pi - \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z + \frac{\rho\mathbf{K}z}{z - \rho(1-\mathbf{K})}\left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} - 1\right),$$

where the factor in parentheses in the last term simplifies to

$$\frac{\phi_\pi}{\mu_2} - 1 + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} = \frac{(\phi_\pi - \mu_2)(\mu_2 - \rho) + \mathbf{K}\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} = \frac{(\phi_\pi - \mu_2)\left(\mu_2 - \rho(1-\mathbf{K})\right)}{\mu_2(\mu_2 - \rho)}.$$

Hence the condition on z becomes

$$\frac{(1-\delta z)(1-\beta z)}{\kappa} + \frac{1}{\sigma\gamma_S}\left[\phi_\pi - \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right) z + \frac{\rho\mathbf{K}z}{z - \rho(1-\mathbf{K})}\,\frac{(\phi_\pi - \mu_2)\left(\mu_2 - \rho(1-\mathbf{K})\right)}{\mu_2(\mu_2 - \rho)}\right] = 0.$$

Multiplying by $\kappa\left(z - \rho(1-\mathbf{K})\right)$, and expanding $(1-\delta z)(1-\beta z) = \delta\beta z^2 - (\delta+\beta)z + 1$, the condition becomes $P(z) = 0$, where

$$P(z) = \left(z - \rho(1-\mathbf{K})\right)\left[\delta\beta z^2 - \left(\delta + \beta + \frac{\kappa}{\sigma\gamma_S}\left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)}\right)\right) z + 1 + \frac{\kappa}{\sigma\gamma_S}\,\phi_\pi\right] + \frac{\kappa}{\sigma\gamma_S}\,\mathbf{K}\rho\,\frac{(\phi_\pi - \mu_2)\left(\mu_2 - \rho(1-\mathbf{K})\right)}{\mu_2(\mu_2 - \rho)}\, z. \tag{E.15}$$

[28] If $\pi_t = 0$, (E.12) gives $y_t = 0$, so that a nonzero solution has $\hat{\pi}_{t-1} \neq 0$, and (E.11) requires $z = \rho(1-\mathbf{K})$. Relation (E.14) then holds only if $\rho(1-\mathbf{K})\,\rho\left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)} - 1\right) = 0$, in which case $z = \rho(1-\mathbf{K})$ is also a root of the polynomial (E.15) below. Dividing by π_t therefore loses no solution.

Determinacy. The polynomial (E.15) is cubic and has three roots. Apart from the exogenous shock, the system has two forward-looking variables, output and inflation, and one predetermined variable, the nowcast. Hence the equilibrium is determinate if and only if exactly one of these roots lies inside the unit circle (Blanchard and Kahn, 1980). Numerically, with $\phi_\pi = 1.5$, this holds on a grid of $\rho \in (0, 1)$ with step 0.0005 and of $\mathbf{K} \in (0, 1]$. The condition also holds for $1 \leq \rho < \mu_2$ whenever $\mathbf{K} \geq 0.004$.

The coefficients on the shock. Given ϑ_π^π and ϑ_π^y from (E.13), the coefficients ϑ_u^π and ϑ_u^y follow from the terms in u_t on the two sides of the Phillips curve and of the aggregate Euler equation.

By the guess (E.9) in period $t+1$, and by (E.10) for the nowcast $\hat{\pi}_t$,

$$\begin{aligned}
\pi_{t+1} &= \vartheta_\pi^\pi \hat{\pi}_t + \vartheta_u^\pi u_{t+1} = \vartheta_\pi^\pi \left[\left(\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi \right) \hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi \, u_t \right] + \vartheta_u^\pi u_{t+1}, \\
y_{t+1} &= \vartheta_\pi^y \hat{\pi}_t + \vartheta_u^y u_{t+1} = \vartheta_\pi^y \left[\left(\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi \right) \hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi \, u_t \right] + \vartheta_u^y u_{t+1}.
\end{aligned}$$

Taking expectations in period t, with $\mathbb{E}_t[u_{t+1}] = \rho_u u_t$ by (51) gives, respectively,

$$\begin{aligned}
\mathbb{E}_t[\pi_{t+1}] &= \vartheta_\pi^\pi \left(\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi \right) \hat{\pi}_{t-1} + \left(\mathbf{K}\vartheta_\pi^\pi + \rho_u \right) \vartheta_u^\pi \, u_t, \\
\mathbb{E}_t[y_{t+1}] &= \vartheta_\pi^y \left(\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi \right) \hat{\pi}_{t-1} + \left(\mathbf{K}\vartheta_\pi^y \vartheta_u^\pi + \rho_u \vartheta_u^y \right) u_t.
\end{aligned}$$

Substituting the guess (E.9) for π_t and y_t, and the expectation $\mathbb{E}_t[\pi_{t+1}]$ above, into the Phillips curve (50), $\pi_t = \beta \, \mathbb{E}_t[\pi_{t+1}] + \kappa \, y_t + u_t$, gives

$$\begin{aligned}
\vartheta_\pi^\pi \hat{\pi}_{t-1} + \vartheta_u^\pi \, u_t = \beta \Big[& \vartheta_\pi^\pi \left(\rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi \right) \hat{\pi}_{t-1} + \left(\mathbf{K}\vartheta_\pi^\pi + \rho_u \right) \vartheta_u^\pi \, u_t \Big] \\
& + \kappa \left(\vartheta_\pi^y \hat{\pi}_{t-1} + \vartheta_u^y \, u_t \right) + u_t.
\end{aligned}$$

Hence the terms in u_t give

$$\vartheta_u^\pi = \beta \left(\mathbf{K}\vartheta_\pi^\pi + \rho_u \right) \vartheta_u^\pi + \kappa \, \vartheta_u^y + 1, \tag{E.16}$$

and solving for ϑ_u^y,

$$\vartheta_u^y = \frac{\vartheta_u^\pi \left(1 - \beta \left(\mathbf{K}\vartheta_\pi^\pi + \rho_u \right) \right) - 1}{\kappa}. \tag{E.17}$$

Similarly, substituting the guess (E.9) for y_t and π_t, the nowcast (E.10), $\hat{\pi}_t = z \, \hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi \, u_t$ with $z = \rho(1-\mathbf{K}) + \mathbf{K}\vartheta_\pi^\pi$, and the two expectations above into the aggregate Euler equation (E.7), with $i_t = \phi_\pi \pi_t$, gives

$$\begin{aligned}
\vartheta_\pi^y \hat{\pi}_{t-1} + \vartheta_u^y \, u_t = \delta & \left[\vartheta_\pi^y z \, \hat{\pi}_{t-1} + \left(\mathbf{K}\vartheta_\pi^y \vartheta_u^\pi + \rho_u \vartheta_u^y \right) u_t \right] \\
& - \frac{1}{\sigma \gamma_S} \Bigg[\phi_\pi \left(\vartheta_\pi^\pi \hat{\pi}_{t-1} + \vartheta_u^\pi \, u_t \right) - \rho \left(z \, \hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi \, u_t \right) \\
& + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K} \frac{\rho(\phi_\pi - \mu_2)}{\mu_2(\mu_2 - \rho)} \right) \left(\rho \left(z \, \hat{\pi}_{t-1} + \mathbf{K}\vartheta_u^\pi \, u_t \right) \right. \\
& \left. - \vartheta_\pi^\pi z \, \hat{\pi}_{t-1} - \left(\mathbf{K}\vartheta_\pi^\pi + \rho_u \right) \vartheta_u^\pi \, u_t \right) \Bigg].
\end{aligned}$$

Hence the terms in u_t give

$$\vartheta_u^y = \delta\left(\mathbf{K}\vartheta_\pi^y\vartheta_u^\pi + \rho_u\vartheta_u^y\right) - \frac{1}{\sigma\gamma_S}\,\vartheta_u^\pi\left[\phi_\pi - \rho\mathbf{K} + \left(\frac{\phi_\pi}{\mu_2} + \mathbf{K}\,\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}\right)\left(\rho\mathbf{K} - \mathbf{K}\vartheta_\pi^\pi - \rho_u\right)\right].$$

Moving $\delta\rho_u\vartheta_u^y$ to the left-hand side, substituting ϑ_u^y from (E.17), and multiplying by κ gives

$$\begin{aligned}&(1-\delta\rho_u)\left[\vartheta_u^\pi\left(1-\beta\left(\mathbf{K}\vartheta_\pi^\pi+\rho_u\right)\right)-1\right]\\&\quad=\delta\kappa\mathbf{K}\vartheta_\pi^y\vartheta_u^\pi-\frac{\kappa}{\sigma\gamma_S}\,\vartheta_u^\pi\left[\phi_\pi-\rho\mathbf{K}+\left(\frac{\phi_\pi}{\mu_2}+\mathbf{K}\,\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}\right)\left(\rho\mathbf{K}-\mathbf{K}\vartheta_\pi^\pi-\rho_u\right)\right].\end{aligned}$$

Collecting the terms in ϑ_u^π yields

$$\vartheta_u^\pi = \frac{1-\delta\rho_u}{\left(1-\beta(\mathbf{K}\vartheta_\pi^\pi+\rho_u)\right)(1-\delta\rho_u)-\delta\kappa\mathbf{K}\vartheta_\pi^y+\frac{\kappa}{\sigma\gamma_S}\left(\phi_\pi-\rho\mathbf{K}+\left(\frac{\phi_\pi}{\mu_2}+\mathbf{K}\frac{\rho(\phi_\pi-\mu_2)}{\mu_2(\mu_2-\rho)}\right)(\rho\mathbf{K}-\mathbf{K}\vartheta_\pi^\pi-\rho_u)\right)}. \tag{E.18}$$

Given ϑ_π^π and ϑ_π^y from (E.13), equation (E.18) gives ϑ_u^π, and (E.17) then gives ϑ_u^y. All four coefficients of the guess are therefore explicit functions of the parameters and of the root z of (E.15).

E.3. The Actual Law of Motion of Inflation and the T-Map

The guess (E.9) gives inflation as a function of the state of the economy. Combining it with the law of motion of the nowcast gives inflation as a function of past innovations alone, which is the form that can be compared with the AR(1) process of the PLM.

The ALM of inflation as an ARMA process. Let L denote the lag operator, $Lx_t = x_{t-1}$. The law of motion of the nowcast (35) reads $\left(1-\rho(1-\mathbf{K})L\right)\hat{\pi}_t = \mathbf{K}\pi_t$, so that

$$\hat{\pi}_t = \frac{\mathbf{K}}{1-\rho(1-\mathbf{K})L}\,\pi_t, \qquad \hat{\pi}_{t-1} = L\hat{\pi}_t = \frac{\mathbf{K}L}{1-\rho(1-\mathbf{K})L}\,\pi_t.$$

Substituting $\hat{\pi}_{t-1}$ into the guess for inflation (E.9), $\pi_t = \vartheta_\pi^\pi\hat{\pi}_{t-1} + \vartheta_u^\pi\,u_t$, gives

$$\left(1-\frac{\mathbf{K}\vartheta_\pi^\pi L}{1-\rho(1-\mathbf{K})L}\right)\pi_t = \vartheta_u^\pi\,u_t \quad\Longleftrightarrow\quad \frac{1-\left(\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi\right)L}{1-\rho(1-\mathbf{K})L}\,\pi_t = \vartheta_u^\pi\,u_t,$$

and, with $u_t = \frac{\varepsilon_t^u}{1-\rho_u L}$ by (51),

$$\pi_t = \vartheta_u^\pi\,\frac{1-\rho(1-\mathbf{K})L}{\left(1-\left(\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi\right)L\right)\left(1-\rho_u L\right)}\,\varepsilon_t^u. \tag{E.19}$$

Multiplying both sides by the denominator and applying the lag operator yields

$$\pi_t = \left(\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi+\rho_u\right)\pi_{t-1} - \left(\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi\right)\rho_u\,\pi_{t-2} + \vartheta_u^\pi\,\varepsilon_t^u - \vartheta_u^\pi\,\rho(1-\mathbf{K})\,\varepsilon_{t-1}^u.$$

Therefore the actual law of motion of inflation is an ARMA(2,1) process, with autoregressive roots ρ_u and $\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi$, and moving-average root $\rho(1-\mathbf{K})$.

FIRE as a special case where the ALM of inflation is AR(1). Let $\mathbf{K}=1$ and $\rho=\rho_u$. With $\mathbf{K}=1$, $\rho(1-\mathbf{K})=0$, the polynomial (E.15) has the factor z, and its root inside the unit circle is $z=0$, so that (E.13) gives $\vartheta_\pi^\pi=0$ and $\vartheta_\pi^y=0$.

Intuitively, with $\mathbf{K}=1$ the nowcast equals current inflation, $\hat{\pi}_t=\pi_t$. By the guess (E.9), inflation is then $\pi_t=\vartheta_u^\pi u_t$, so that $\mathbb{E}_t[\pi_{t+1}]=\rho_u\pi_t$. With $\rho=\rho_u$, the one-period forecast of Savers is then correct, $\rho\,\hat{\pi}_t=\rho_u\pi_t=\mathbb{E}_t[\pi_{t+1}]$. The belief-revision term in (E.7) is therefore zero, the one-period real rate perceived by Savers, $i_t-\rho\,\hat{\pi}_t$, equals $i_t-\mathbb{E}_t[\pi_{t+1}]$, and (E.7) becomes the baseline aggregate Euler equation of Bilbiie (2025), (49). Since the FIRE equilibrium is unique for $\phi_\pi>1.012$, the two equilibria coincide.

In addition, if $\vartheta_\pi^\pi=0$, then (E.19) reduces to

$$\pi_t=\frac{\vartheta_u^\pi}{1-\rho_u L}\,\varepsilon_t^u \quad\Longleftrightarrow\quad \pi_t=\rho_u\,\pi_{t-1}+\vartheta_u^\pi\,\varepsilon_t^u. \tag{E.20}$$

Given the assumptions on the shock u_t, inflation is then an AR(1) process with the persistence of the shock, ρ_u.

The T-map. Savers' PLM of inflation is an AR(1) process with persistence ρ, that delivers an ALM of inflation that is the ARMA(2,1) process of (E.19).

The AR(1) coefficient that best fits actual inflation, in the sense of a population least-squares regression of π_t on π_{t-1}, is its first-order autocorrelation. Let T denote the function that maps the PLM parameter ρ into the first-order autocorrelation of inflation under the ALM, that is

$$T(\rho)=\frac{\mathrm{Cov}(\pi_t,\pi_{t-1})}{\mathrm{Var}(\pi_t)}.$$

The T-map is computed numerically. For given ρ and $\mathbf{K}$, the solution of Appendix E.2, together with the law of motion of the nowcast and the process of the shock, gives the autocovariances of inflation, and hence $T(\rho)$.

As seen in the FIRE paragraph above, if $\vartheta_\pi^\pi=0$, inflation is an AR(1) process with the persistence of the shock, and $T(\rho)=\rho_u$. That is satisfied with $\mathbf{K}=1$, implying $\vartheta_\pi^\pi=0$ for any ρ, so that $T(\rho)=\rho_u$.

With $0<\mathbf{K}<1$ and $\rho>0$, T depends on ρ, since $\vartheta_\pi^\pi\neq 0$: by (E.13), $\vartheta_\pi^\pi=0$ would require $z=\rho(1-\mathbf{K})$, where (E.15) reduces to its last term, which is nonzero for $\phi_\pi\neq\mu_2$. As ρ goes to zero, $\rho(1-\mathbf{K})$ and the last term of (E.15) go to zero, so that (E.15) tends to z times the bracket, and its root inside the unit circle goes to zero. By (E.13), ϑ_π^π then goes to zero, and $T(\rho)$ goes to ρ_u.

At the calibration, both with the Kalman gains of Savers and with the one of all households, $T(\rho)$ is non-increasing in ρ and does not exceed $\rho_u=0.5$ on the whole interval $0<\rho<\mu_2$. At the estimated persistence of Savers, $T(0.843)=0.487$. The T-map is mostly flat because, under this calibration, the autoregressive root $\rho(1-\mathbf{K})+\mathbf{K}\vartheta_\pi^\pi$ and the moving-average root $\rho(1-\mathbf{K})$

of (E.19) nearly cancel (0.425 and 0.438 with the beliefs of Savers), so that actual inflation stays close to an AR(1) process with the persistence of the shock, as shown in Figure 22.

Fixed point and E-stability. A restricted perceptions equilibrium is a perceived persistence ρ^* such that the fixed point

$$T(\rho^*) = \rho^* \tag{E.21}$$

is satisfied.

At ρ^*, the AR(1) process of Savers is the best AR(1) process for the inflation that their beliefs generate, and the data do not contradict it. The fixed point is E-stable if Savers who revise their perceived persistence towards the persistence they observe, $\frac{d\rho}{d\tau} = T(\rho) - \rho$ in notional time τ, converge to it, which holds if and only if $T'(\rho^*) < 1$ (Evans and Honkapohja, 2001).

With $\mathbf{K} = 1$, $T(\rho) = \rho_u$ for every ρ, so that the unique fixed point is $\rho^* = \rho_u$, with $T'(\rho^*) = 0 < 1$. The restricted perceptions equilibrium with $\mathbf{K} = 1$ is therefore the FIRE equilibrium of THANK.

With $0 < \mathbf{K} < 1$, the fixed point exists and is unique at the calibration.The fixed point is $\rho^* = 0.4987$ with the Kalman gain of Savers and 0.4988 with that of all households, with $T'(\rho^*) = -0.009$ and -0.008, so that both are E-stable. Numerically, for Kalman gains between 0.1 and 1, the fixed point lies between 0.4987 and 0.5000, close to the persistence of the shock.

Figure 22 plots the T-map, the fixed point, and the estimated beliefs of Savers and of all households.

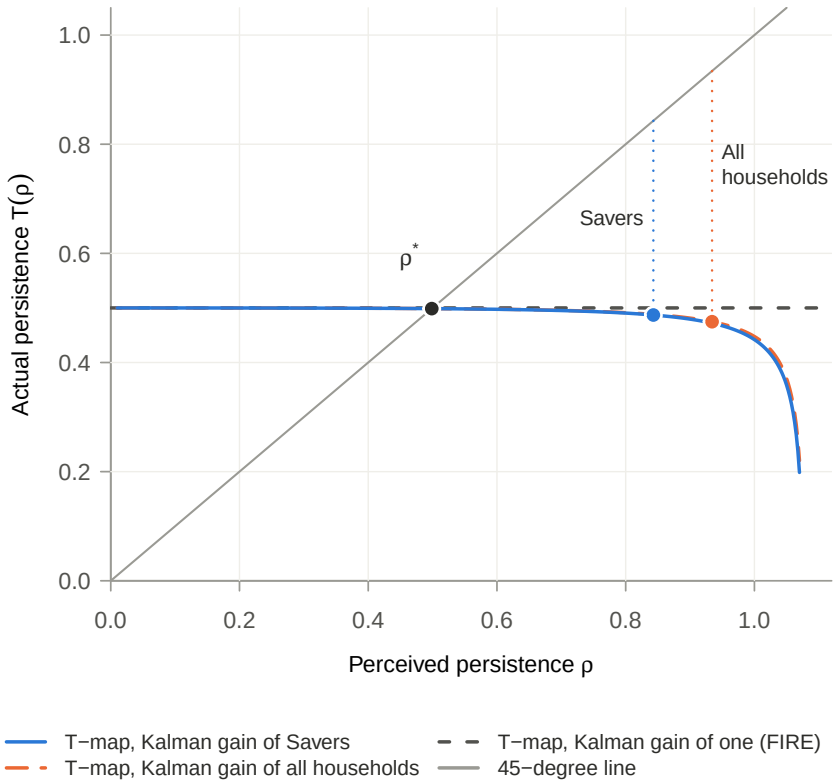


Figure 22: The T-map.

Notes: $T(\rho)$ is the first-order autocorrelation of inflation under the actual law of motion generated by a PLM with perceived persistence ρ, for the Kalman gain of Savers ($\mathbf{K} = 0.480$), of all households ($\mathbf{K} = 0.524$), and for $\mathbf{K} = 1$. The fixed point ρ^* is where the T-map crosses the 45-degree line. Dots: estimated perceived persistence of Savers ($\rho = 0.843$, $T(\rho) = 0.487$) and of all households ($\rho = 0.934$, $T(\rho) = 0.475$). Dotted segments: gap between perceived and actual persistence. The T-map falls as ρ approaches $\mu_2 = 1.076$. $\phi_\pi = 1.5$, rule-aware nominal-rate beliefs, firms with rational expectations, cost-push shock with persistence 0.5.

Appendix F. Quantitative Results: Robustness

This appendix complements Section 5. Unless stated otherwise, the calibration is that of Table 5, $\phi_\pi = 1.5$, nominal-rate beliefs are rule-aware, and the beliefs are the OLS estimates for the inflation surge.

F.1. Full-Sample Beliefs

Section 5 uses the beliefs estimated for the inflation surge. Table 13 repeats the exercise with the OLS estimates for the full sample, 2015–2025, $\rho = 0.764$ and $\mathbf{K} = 0.549$ for Savers, and $\rho = 0.887$ and $\mathbf{K} = 0.568$ for all households (Table 4). Over the full sample, both groups perceive inflation as less persistent than during the surge, but still as much more persistent than actual inflation, whose persistence is 0.48. The cumulative contraction after the cost-push shock is 24 percent larger than under FIRE with the beliefs of Savers, and 67 percent larger with the beliefs of all households, against 45 and 98 percent with the surge estimates. The expectations channel of all households is 2.7 times that of Savers, against 2.2 with the surge estimates. The amplification relative to FIRE, and the overstatement implied by population-wide beliefs, are therefore not specific to the surge. The equilibrium is determinate in both cases.

F.2. Estimation Uncertainty

The regression coefficients of Savers and of all households are estimated on overlapping observations, so that their estimates are correlated. Their joint covariance matrix is obtained by stacking the scores of the two regressions observation by observation and applying the two-way clustered covariance estimator of Section 3.3, by respondent and by survey month. From this joint distribution, 2,000 draws of the six coefficients are taken. For each draw, ρ and $\mathbf{K}$ are recovered for Savers and for all households from the mapping (16), and the impulse responses are recomputed. Table 14 reports the results.

Two results stand out. First, the expectations channel is clearly different from zero for both sets of beliefs: the beliefs of Savers alone make the cumulative contraction after a cost-push shock significantly larger than under FIRE. Second, the difference between the two sets of beliefs is statistically significant, but not by a wide margin. The 90 percent bands of the difference in perceived persistence, $[0.007, 0.168]$, and of the difference between the two channels, $[-3.13, -0.09]$, exclude zero, and the channel of all households is the larger one in 97 percent of the draws. The ratio of the two channels is centered on 2.2, with a wide band, from 1.07 to 4.57.

Table 13: Output response to a cost-push shock: full-sample beliefs.

Beliefs	ρ	$\mathbf{K}$	Impact	Cumulative	Channel	Relative to FIRE
FIRE			−1.15	−2.31		
Savers	0.764	0.549	−1.28	−2.87	−0.56	+24%
All households	0.887	0.568	−1.58	−3.85	−1.54	+67%

Notes: As in Table 6, with the OLS estimates for the full sample, 2015–2025 (Table 4). Relative to FIRE: percentage by which the cumulative contraction exceeds the one under FIRE. The ratio of the channels of all households and of Savers is 2.74.

Table 14: Estimation uncertainty.

	Median	5%	95%
Beliefs			
ρ, Savers	0.843	0.769	0.910
ρ, all households	0.934	0.854	1.000
ρ, all households minus Savers	0.089	0.007	0.168
K, Savers	0.480	0.402	0.552
K, all households	0.525	0.469	0.574
Cost-push shock: cumulative output response			
Channel, Savers	−1.03	−1.81	−0.53
Channel, all households	−2.26	−4.43	−1.13
Channel, all households minus Savers	−1.22	−3.13	−0.09
Ratio of the channels	2.20	1.07	4.57

Notes: 2,000 joint draws of the regression coefficients of Savers and of all households (OLS, inflation surge), from their joint two-way clustered covariance matrix. For each draw, ρ and **K** are recovered from the coefficients and the impulse responses are recomputed. Channel: cumulative output response over twelve quarters to a cost-push shock of 0.25 with persistence 0.5, minus the FIRE response. The channel of all households is the larger one in 97 percent of the draws.